

OBLIGATORY UNIFORM HYPERGRAPHS: CANONICAL ATOMS AND EXACT SPECTRA

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ABSTRACT. Erdős Problem 593 asks which finite triple systems occur in every triple system of uncountable chromatic number. We give an independent proof of the classification and develop its finite structural consequences. After isolated vertices are removed, a system is obligatory if and only if it is linear, every hyperedge-node of its Levi graph is incident with a bridge, and every Berge cycle has even length. Its canonical atoms are single triples and private-vertex expansions of finite 2-connected bipartite graphs. We extend this classification and canonical description to every fixed finite uniformity $r \geq 3$, with at least $r - 2$ bridge incidences required at each hyperedge-node. The proof iterates a finite linear trace lemma over hypergraph bases. For reduced obligatory systems with at least one hyperedge, we determine the exact parameter and atom-count spectra, including the parity obstruction at cycle rank one. The supported one-point factorizations form a product of partition lattices: every attachment excess profile can be realized with the atoms and component count fixed, and the lattice type recovers that profile. For connected systems of positive cycle rank, an exact budget relates cyclic-rank splitting and excess core order to the available factorization rank. At every prescribed original-edge piece count we obtain sharp binomial and Stirling-number bounds, with binary joints and a common joint as the interior equality cases. The two extremal assemblies can have the same atoms and global parameters. These results separate the constraints imposed by cyclic content from the freedom of assembly. The finite structural spectra extend to all finite uniformities under a corresponding change of parameters. Every non-obligatory finite system can also be avoided at each prescribed uncountable chromatic cardinal. We prove simultaneous avoidance for finite linear families and exact chromatic calibration with bounded edge intersections, together with explicit bounds on host order. We characterize universal pair-marginal realization by the matching polytope and prove density estimates for all finite linear patterns as the uniformity increases. Under regularity, the shortest Berge cycle determines the optimal decay rate, including for canonical forest assemblies. Conditional on a recent rank-layer estimate of OpenAI, we also construct a single obligatory atom in every uniformity that violates the Sidorenko density bound. The same graph has no Sidorenko expansion, giving a counterexample to the expansion conjecture of Nie–Spiro under that input. The triple-system classification and specified structural extensions are verified in Lean 4; profile realization, full lattice spectra, the new counting endpoints, the higher-uniformity extension, chromatic spectra and density transfer are not yet fully formalized. All arguments are in ZFC.

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INTRODUCTION

Although the host systems are infinite, the classification is determined by finite structure. A finite triple system is *obligatory* if it occurs in every triple system of uncountable chromatic number. Erdős asked for a characterisation of these finite systems [1]; the question is catalogued as Erdős Problem 593 [2]. We give both constructive and intrinsic descriptions of the answer.

For a finite graph J , write J^+ for its private-vertex expansion. Let $\mathcal{B}$ be the smallest class containing J^+ for every finite bipartite graph J and every finite edgeless triple system, and closed under finite disjoint unions and one-point amalgamations. Membership in $\mathcal{B}$ is understood up to triple-system isomorphism. If F is a triple system, let F° denote the system obtained by deleting its isolated vertices.

Theorem A. *For every finite triple system F , the following are equivalent:*

- (1) F is obligatory;
- (2) $F \in \mathcal{B}$;
- (3) F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with a bridge, and every Berge cycle of F° has even length.

The classification first appeared in Li’s preprint, posted on 23 June 2026 [3, Theorem 1.1]. The present proof and Lean implementation were developed independently. Both proofs use one-apex sequence lifts and Levi bridges, but their finite trace arguments differ: Li uses a bridge-trace formulation and graph derivatives [3, Sections 3–4 and Theorem 4.6], whereas we decompose traces into expansion fibres and a support-incidence forest. We cite Li’s work for chronology and comparison; none of its results is used below as a black box.

Extension to all finite uniformities. The finite linear trace argument can be iterated beyond triples. For a finite graph J and an integer $r \geq 3$, write $J^{[r]}$ for the hypergraph obtained by adding $r - 2$ distinct private vertices to every edge of J . Let $\mathcal{B}_r$ be the class generated by these expansions of finite bipartite graphs and by finite edgeless systems, under finite disjoint unions and one-point amalgamations. As throughout the paper, embeddings are injective and non-induced, and proper vertex colourings have no monochromatic edge.

Theorem B. *For every fixed integer $r \geq 3$ and every finite simple r -uniform hypergraph F , the following are equivalent:*

- (1) F occurs in every r -uniform hypergraph of uncountable chromatic number;
- (2) $F \in \mathcal{B}_r$;
- (3) after isolated vertices are removed, F is linear, each hyperedge-node of its Levi graph is incident with at least $r - 2$ bridges, and every Berge cycle has even length.

Theorem 11.1 proves this extension. Reiher’s expansion theorem already supplies the positive generators for every uniformity [4, Theorem 1.2]. The

additional step is the finite linear trace lemma with a hypergraph base: projection preserves linearity, and one-private-vertex inflation commutes with the forest assembly. Iterating the lift therefore reduces the negative direction to the same high-odd-girth graph input used for triples. The bridge requirement increases with r ; the example in Remark 11.8 shows why one bridge no longer suffices at uniformity four.

The canonical atoms in this generality are single r -edges and $J^{[r]}$ with J a finite 2-connected bipartite graph. Writing $s = n - (r - 2)m$ and $\beta = (r - 1)m - n + c$ transports the finite spectra, factorization products and fixed-atom profiles below to every finite uniformity (Corollary 11.9). The detailed triple-system argument and its finite structure remain in Sections 1–10.

Prescribing the chromatic cardinal. The structural answer does not depend on which uncountable chromatic cardinal is prescribed. The next theorem strengthens the negative direction from an avoiding host to a host of any specified uncountable chromatic number.

Theorem C. *Fix a finite integer $r \geq 3$ and a finite simple r -uniform hypergraph F . If $F \notin \mathcal{B}_r$, then for every uncountable cardinal κ there is an F -free r -uniform hypergraph H with $\chi(H) = \kappa$. If F is linear, one may also require $|V(H)| \leq 2^{<\kappa}$. If $F \in \mathcal{B}_r$, no such host exists.*

Here $2^{<\kappa} = \sup_{\nu < \kappa} 2^\nu$, with the supremum taken over cardinals. Theorem 12.7 proves the assertion, including singular κ . Li established the triple-system spectrum [3, Theorem 7.1]. For linear forbidden systems in higher uniformity, the key identity is $\chi(\mathcal{L}_\kappa(H)) = \min\{\chi(H), \kappa\}$ for the sequence lift of height κ . It combines exact chromatic control with the finite trace theorem already used for the classification.

Nonlinear forbidden systems require a different construction. We extend Li's reservoir method [3, Section 6] to add several old vertices to a labelled base edge. With the simultaneous-colour systems of Erdős–Galvin–Hajnal [5, Corollary 12.3], this gives, for $1 \leq i \leq r - 2$, an r -uniform host with pairwise edge intersections at most i , chromatic number exactly μ^+ , and order at most $\beth_{\lceil (r-1)/i \rceil}(\mu)$. Here $\beth_0(\mu) = \mu$ and $\beth_{j+1}(\mu) = 2^{\beth_j(\mu)}$. The order bounds are in ZFC; the corresponding optimal GCH thresholds are classical consequences of Erdős–Galvin–Hajnal.

The same lift avoids any finite family of linear non-obligatory systems simultaneously. This family theorem has a real boundary: for every uniformity there is a finite mixed family whose members are individually avoidable at every uncountable cardinal, while their simultaneous omission forces countable chromatic number (Proposition 12.8). The new results in Sections 11 and 12 have manuscript proofs; they are not additional Lean-verified endpoints.

Obligatory atoms and finite density inequalities. The atom description leads to a further question: does obligatoriness impose the lower bound on copies

predicted by a random host of the same density? The next result separates these properties within a single cyclic atom.

Theorem D. *Assume the rank-layer limit in OpenAI’s A counterexample to Sidorenko’s conjecture [6]. There is an explicit 2-connected bipartite graph J with 35 vertices and 66 edges such that its private expansion $J^{[r]}$ is not Sidorenko for any finite $r \geq 2$, with $J^{[2]} = J$. For every $r \geq 3$, the expansion is one canonical obligatory atom with 66 edges, $66r - 97$ vertices and cyclic rank 32.*

Here a finite r -uniform pattern F is Sidorenko if every finite host G satisfies $t(F, G) \geq p_r(G)^{|E(F)|}$, where t counts all homomorphisms and $p_r(G) = r!|E(G)|/|V(G)|^r$. Nie–Spiro conjectured that every bipartite graph admits a Sidorenko expansion in some uniformity [7, Conjecture 5.1]. Theorem D supplies a counterexample under the stated rank-layer input.

The graph and its density deficit originate in OpenAI [6]. The new transfer uses more than that deficit: in the source’s mean-one kernel family, every proper edge-subgraph of J has density tending to one. We prove this additional limit and use an average over matchings to realize the required graph marginals by symmetric r -kernels. Section 13 gives these proofs and a finite-density separation witnessed by $C_4^{[r]}$. The long finite-field estimate remains a cited preprint input; the transfer has not been formalized in Lean.

The unconditional results develop this framework further. Theorem 13.1 characterizes the universal realization of arbitrary collections of pair kernels by the matching polytope, and proves that the symmetric transfer coefficient is optimal. For any graph F with m edges, Theorem 13.5 gives the explicit two-sided estimate

$$|t(F^{[r]}, U) - p^m| \leq m(m-1)p(1-p)/r$$

for every symmetric $[0, 1]$ -valued r -kernel U of mean p . For regular kernels, Theorem 13.6 improves the error to $O_F(r^{-g})$ when the core has girth g , and gives exact equality for forests. Explicit kernels of density $1/2$ show that these powers of r are optimal. These results concern quantitative homomorphism densities and require no rank-layer hypothesis.

Theorem 13.9 extends these estimates to every finite linear r -uniform pattern H , including arbitrary one-point assemblies of the canonical atoms. If H has m edges and at most b shared vertices on each edge, then

$$|t(H, U) - p^m| \leq \binom{m}{2} \frac{b}{r} p(1-p).$$

Under regularity the error has order r^{-g} , where g is the shortest Berge-cycle length; every fixed cyclic incidence pattern attains this rate. Consequently every obligatory pattern with a fixed edge count has error $O_m(r^{-4})$ in regular hosts, with exact equality for Berge forests. A further estimate separates the contribution of degree variance from the cyclic error. Worked examples show

both how to realize a marginal profile and how the choice of attachment vertices changes the density of assemblies with the same atoms and numerical parameters.

Finite structure beyond the classification. Theorem A answers an embedding question about infinite hosts. The further results concern a different question: how much finite structure is fixed by cyclic content, and how much remains free under one-point assembly? The finite systems considered in this overview are reduced (they have no isolated vertices), obligatory, and have at least one hyperedge. Write $n = |V(F)|$, $m = |E(F)|$, and c for their number of components; put $s = n - m$ and $\beta = 2m - n + c$, the Levi cycle rank. The canonical normal form identifies the atoms as single triples and expansions of finite 2-connected bipartite graphs (Theorem 5.3). Block decomposition itself is classical [8, 9]; the point here is the exact list of allowable block types and what it implies about the systems assembled from them.

The finite results give two complementary classifications. The numerical spectra determine which global parameters and canonical atom counts are possible. The local partition product determines which decompositions remain available once the atoms have been fixed. Neither classification alone captures the other: even assemblies with identical atoms, order, size and cycle rank can have different factorization lattices and different numbers of decompositions.

The numerical consequences are exact existence statements, not only bounds. Theorem 10.3 determines every feasible order, size and component count. Theorem 10.9 and Proposition 10.12 then determine all possible canonical atom counts k at fixed s, β, c , including the parity obstruction when $\beta = 1$. For connected systems, the indecomposability phase diagram separates parameter ranges where every system is decomposable, where both behaviours occur, and where every system is indecomposable (Theorem 10.10). Its boundary cases identify cycle, even-theta and complete-bipartite expansions, and the extremal decomposable assembly (Corollary 10.11). These conclusions do not say that numerical parameters determine every core up to isomorphism.

The remaining freedom is described by a lattice, not another numerical invariant. At each point p shared by $\mu(p)$ canonical atoms, one may choose a partition of those atoms independently. The supported one-point forest decompositions of the original hyperedge set, ordered by refinement, therefore form $\mathcal{D}(F) \cong \prod_{p \in S(F)} \Pi_{\mu(p)}$, where $S(F)$ is the set of shared points and Π_r is the partition lattice on r elements (Lemma 10.15). Its rank is $N = k - c$. Every integer partition $\lambda \vdash N$ is realized as an attachment excess profile by reassembling the same atoms with the same component count, order, size and cycle rank (Lemma 10.16). This gives the exact factorization-lattice spectrum: its types are $\prod_i \Pi_{\lambda_i+1}$, and the classical characteristic polynomials of partition lattices [10, Chapter 3] recover λ (Corollary 10.17). The classification is of lattice types, not of all triple systems with the same parameters.

Cyclic content and assembly freedom are linked by an exact budget. For a connected system of positive cycle rank, the lattice rank $N = k - 1$, the cost of distributing positive rank among several atoms, and the total core-order excess above the fixed-rank minima sum to $s - 2 - \lceil 2\sqrt{\beta} \rceil$ (Proposition 10.18). Thus rank splitting and excess core order reduce the available decomposition rank. Zero cost means that one minimum-order cyclic atom carries all positive rank and the remaining atoms are single triples; it does not imply a unique isomorphism type for the cyclic core.

Assembly geometry also controls the number of decompositions at each prescribed piece count. If $d_j(F)$ counts decompositions into exactly $k - j$ nonempty original-edge pieces, then $\binom{N}{j} \leq d_j(F) \leq S(N + 1, N + 1 - j)$ for $0 \leq j \leq N$, where $S(a, b)$ is a second-kind Stirling number [11]. For $N \geq 2$, equality at any interior j characterizes binary joints at the lower bound and a single common joint at the upper bound (Corollary 10.19). Both extremal profiles can be attained with the atom list and component count fixed. The example following that corollary has identical atoms and global parameters but four or five decomposition choices. These counting consequences quantify the freedom left by the canonical structure; they are not assertions of new general Stirling-number identities.

The triple-system classification and substantial structural extensions have accepted Lean implementations. The distinction is between the mathematical results proved in the manuscript and the exact interfaces already exported by Lean, not between a numerical test and a proof. Full profile realization, lattice spectra, maximal-chain and height endpoints, and the new counting statements still have unfinished formal interfaces. The finite results above have manuscript proofs; they are not all claimed as accepted Lean exports. Appendix A records the implementation routes, validation records and remaining obligations without interrupting the mathematical arguments.

Context and proof strategy. The private-vertex expansion J^+ also appears in finite extremal hypergraph theory. Kostochka, Mubayi and Verstraëte study its Turán extremal function [12]. Here the same operation supplies the canonical cyclic pieces, but the host condition is uncountable chromatic number rather than edge density.

The graph analogue was proved by Erdős and Hajnal: the obligatory finite graphs are exactly the bipartite graphs [13, Corollary 5.6 and Theorem 7.4]. The classical nonlinearity obstruction for uniform hypergraphs is due to Erdős, Hajnal, and Rothschild [14, Theorem 2, p. 532]. Obligatory triple systems were studied further by Erdős et al. [5], Komjáth [15], Hajnal and Komjáth [16], and Komjáth [17]; the expansion theorem of Reiher [4, Theorem 1.2] supplies the positive atoms used below.

In the finite setting, arbitrarily large chromatic number is compatible with arbitrarily large girth: Erdős proved this for graphs [18, pp. 35–37], and Erdős and Lovász established a uniform-hypergraph counterpart [19,

Theorem 1']. Later constructions control sparsity quantitatively: Kostochka and Rödl obtain high-girth uniform hypergraphs with controlled maximum degree [20, Theorem 3], while Alon, Kostochka, Reiniger, West and Zhu give an explicit construction [21, Theorem 3.2]. Uncountable chromatic number is essential here. For finite hosts, Wang et al. [22, Proposition 1.3(ii)] record that the weak chromatic numbers of F -free uniform hypergraphs are bounded exactly when F contains no Berge cycle. This does not classify obligatory systems: the expansion C_4^+ contains a Berge cycle and can be avoided by finite triple systems of arbitrarily large chromatic number, yet it occurs in every uncountably chromatic triple system by Theorem A. Finite high-chromatic avoidance and uncountable obligatoriness are therefore different questions.

The proof separates into positive and negative directions. For the positive direction, a probabilistic rainbow lemma provides the local injectivity needed to force complete bipartite expansions. A rooted-abundance lemma then establishes closure under one-point amalgamation, including at singular uncountable cardinals. After the bridges of the Levi graph are deleted, each remaining active component is the expansion of a finite bipartite graph. The quotient is a forest, whose running-intersection property determines a valid amalgamation order.

For the negative direction, we use an independently developed one-apex sequence lift; Li [3, Sections 3–4] gives a parallel formulation. The lift transfers uncountable chromatic number from a graph to a triple system while controlling every finite linear trace. The trace theorem is proved by decomposing the trace into expansion fibres whose support-incidence graph is a forest, so distinct fibres can meet only at cut points. Consequently, neither a bridge-free hyperedge-node nor a Berge cycle can be assembled across fibres. Each of the three failures of the intrinsic criterion has a corresponding host: an Erdős–Rado linear host excludes nonlinearity, the lift of K_{ω_1} excludes a missing bridge, and a classical Erdős–Hajnal graph of uncountable chromatic number whose odd girth exceeds a prescribed finite bound excludes an odd Berge cycle.

All embeddings are injective and non-induced, and the argument is carried out in ZFC. For the triple-system classification proof, the imported black-box inputs are the de Bruijn–Erdős compactness theorem [23, Theorem 1, pp. 371–373], the Erdős–Hajnal high-odd-girth theorem [13, Theorem 7.4, p. 76], and the pair case of the Erdős–Rado partition theorem [24, Theorem 4(i), formula (95), p. 471]. The later block-formulation corollary also uses Komjáth’s block reduction [15]. For general background on cardinal partition relations, see Erdős et al. [25]; the specific imported statements are those listed above. The extension in Section 11 additionally uses the all-uniformity expansion theorem of Reiher and nonlinearity obstruction of Erdős–Hajnal–Rothschild, in the precise forms cited there. The chromatic spectra in Section 12 also use Erdős–Galvin–Hajnal’s simultaneous-colour construction; the finite-family counterexample uses Hajnal–Komjáth’s linearly obligatory loose seven-cycle. The exact statements are cited where they enter the proofs.

1. PRELIMINARIES

A *triple system* is a simple 3-uniform hypergraph $H = (V(H), E(H))$, so $E(H) \subseteq [V(H)]^3$. A colouring $c : V(H) \rightarrow \lambda$ is *proper* when no hyperedge is monochromatic, and $\chi(H)$ is the least cardinal λ admitting such a colouring. An embedding of a triple system F in H is an injective map $f : V(F) \rightarrow V(H)$ such that $f[e] \in E(H)$ for every $e \in E(F)$; embeddings are not required to be induced. A finite triple system F is *obligatory* if every triple system H with $\chi(H) > \aleph_0$ contains an embedding of F . A point is *isolated* if it belongs to no hyperedge, and F° denotes the result of deleting all isolated points.

All ordinary graphs in this paper are simple. A triple system is *linear* when any two distinct hyperedges meet in at most one point. Its *Levi graph* $I(F)$ is the bipartite graph with classes $V(F)$ and $E(F)$, where a point-node p is adjacent to a hyperedge-node e exactly when $p \in e$. A *bridge* is an edge of a graph whose deletion increases the number of connected components. A *Berge cycle of length* $\ell \geq 2$ consists of distinct points $p_0, \dots, p_{\ell-1}$ and distinct hyperedges $e_0, \dots, e_{\ell-1}$ such that $p_i \in e_i \cap e_{i+1}$, with indices modulo ℓ . Equivalently, it is a simple cycle of length 2ℓ in $I(F)$. For graph terminology and the incidence-graph interpretation of hypergraph connectivity, see Diestel [9, Chapters 1 and 3] and Bahmanian and Šajna [26].

For a finite simple graph J , its *private-vertex expansion* J^+ has vertex set

$$V(J) \dot{\cup} \{p_a : a \in E(J)\}$$

and hyperedges $\{u, v, p_{\{u,v\}}\}$ for $\{u, v\} \in E(J)$, where the points p_a are new and pairwise distinct. A *one-point amalgamation* of vertex-disjoint triple systems F_0, F_1 is obtained by choosing $x_i \in V(F_i)$, identifying x_0 with x_1 , and making no other identifications or new hyperedges. Finite disjoint unions and finite edgeless systems include the empty cases. Finite reductions.

Lemma 1.1 (Isolated-vertex reduction). *Let F be a finite triple system and let F° be obtained by deleting all isolated vertices. Then*

$$F \text{ is obligatory} \iff F^\circ \text{ is obligatory,}$$

and

$$F \in \mathcal{B} \iff F^\circ \in \mathcal{B}.$$

Proof. If F° is obligatory and H has uncountable chromatic number, then H has infinitely many vertices. After embedding the finite system F° , map the isolated vertices of F injectively to unused host vertices. Containment is non-induced, so additional host edges are irrelevant. The converse follows because F° is a subhypergraph of F .

For membership in $\mathcal{B}$, the implication $F^\circ \in \mathcal{B} \Rightarrow F \in \mathcal{B}$ follows by taking the disjoint union with a finite edgeless system.

For the converse, use structural induction on a construction of F in $\mathcal{B}$. If $F = J^+$, delete the isolated graph vertices of J to obtain J_0 ; then $(J^+)^{\circ} = J_0^+ \in \mathcal{B}$. If F is edgeless, its reduction is the empty edgeless system. Reduction commutes with disjoint union.

Suppose finally that F is a one-point amalgamation of F_0 and F_1 , identifying $x_i \in V(F_i)$. By induction, $F_0^{\circ}, F_1^{\circ} \in \mathcal{B}$. If both selected vertices are nonisolated in their factors, then F° is the one-point amalgamation of the two reductions at those vertices. If exactly one selected vertex is nonisolated, the isolated side disappears at the identified point and F° is the disjoint union of the two reductions. The same description applies when both selected vertices are isolated, because the identified point is then deleted. Hence $F^{\circ} \in \mathcal{B}$ in every case. $\square$

We may therefore suppress isolated vertices in the substantive argument and restore them at the end.

Lemma 1.2 (Finite deletion). *If $\chi(H) > \aleph_0$ and $S \subseteq V(H)$ is finite, then*

$$\chi(H - S) > \aleph_0.$$

Proof. Otherwise a countable colouring of $H - S$, together with finitely many fresh colours for S , would colour H countably. $\square$

Colouring and closure tools.

Lemma 1.3 (Two elementary colouring facts). *We use the following two facts repeatedly.*

1. *If $E(H) = E_1 \cup \dots \cup E_r$ and every $(V(H), E_i)$ is countably chromatic, then H is countably chromatic.*
2. *If a graph G admits an orientation in which every vertex has outdegree at most $d < \omega$, then $\chi(G) \leq 2d + 1$.*

Proof. For the first assertion, take the product of the r countable colourings.

For the second, every finite subgraph has at most $d|V|$ edges and therefore a vertex of degree at most $2d$. It is $(2d + 1)$ -colourable by greedy deletion. The de Bruijn–Erdős compactness theorem [23, Theorem 1] then gives a $(2d + 1)$ -colouring of all of G . $\square$

Lemma 1.4 (Closure-chain lemma). *Let κ be an uncountable cardinal, let $1 \leq r < \omega$, and let*

$$\Phi : [\kappa]^r \longrightarrow [\kappa]^{<\omega}$$

have uniformly finite values. There is an increasing continuous chain

$$\emptyset = M_0 \subseteq M_1 \subseteq \dots \subseteq M_i \subseteq \dots \quad (i < \text{cf } \kappa)$$

such that

$$\bigcup_{i < \text{cf } \kappa} M_i = \kappa, \quad |M_i| < \kappa,$$

and every M_i is Φ -closed:

$$a \in [M_i]^r \implies \Phi(a) \subseteq M_i.$$

Proof. Choose sets $A_i \subseteq \kappa$, each of cardinality below κ , whose union is κ . At a successor stage close $M_i \cup A_i$ under Φ , iterating the operation ω times. Since Φ is finitary and finite-valued, this does not increase an infinite cardinal below κ . At a limit stage take the union. A union of an increasing chain of sets closed under a finitary operation remains closed; and because the stage is below $\text{cf } \kappa$, its cardinality remains below κ . $\square$

Write

$$I_i = M_{i+1} \setminus M_i.$$

Then $(I_i)_{i < \text{cf } \kappa}$ partitions κ , and $M_i = \bigcup_{j < i} I_j$.

2. A BIPARTITE SUBGRAPH LEMMA

The following finite complete-bipartite lemma is a consequence of Erdős and Hajnal [13, Corollary 5.6, p. 72]. We give a short closure-chain proof because the construction will also be used later.

Lemma 2.1 (Uncountable chromatic number forces complete bipartite graphs). *For every positive integer n , every graph of uncountable chromatic number contains $K_{n,n}$.*

Proof. Suppose otherwise, and choose a $K_{n,n}$ -free graph G of uncountable chromatic number whose vertex cardinality κ is minimal.

For $A \in [\kappa]^n$, let

$$N(A) = \{v \in \kappa : v \text{ is adjacent to every member of } A\}.$$

Because G is $K_{n,n}$ -free, $|N(A)| < n$. Apply Lemma 1.4 to $A \mapsto N(A)$, and let M_i, I_i be the resulting chain and layers.

By minimality of κ , every induced graph $G[I_i]$ is countably chromatic. Let $G^\times$ consist of the edges joining distinct layers, and orient each such edge from the later endpoint toward the earlier endpoint. If $v \in I_i$ had n earlier neighbours, those neighbours would form an n -set $A \subseteq M_i$, and closure would give

$$v \in N(A) \subseteq M_i,$$

contrary to $v \in I_i$. Thus the orientation of $G^\times$ has outdegree at most $n - 1$. By Lemma 1.3, $G^\times$ is finitely colourable.

Colour each $G[I_i]$ with colours in ω , reusing the same palette on all layers, and combine this with a finite proper colouring of $G^\times$. The resulting countable product colouring is proper on all of G , a contradiction. $\square$

3. BIPARTITE EXPANSIONS

Rainbow bipartite submatrices.

Lemma 3.1 (Rainbow bipartite submatrices). *For all positive integers n, t , there exists $q = q(n, t)$ with the following property. Suppose the edges of $K_{q,q}$ are coloured, with an arbitrary colour set, so that at every vertex each individual colour occurs on fewer than t incident edges. Then there are n -element subsets X' and Y' of the two vertex classes such that all n^2 edges between X' and Y' have distinct colours.*

Proof. Choose independent uniformly random n -subsets X' and Y' . For every unordered pair P of distinct same-coloured edges, let I_P be the indicator that both edges of P lie in the selected $K_{n,n}$, and put

$$Z = \sum_P I_P.$$

Thus $Z = 0$ exactly when all selected edges have distinct colours.

Let m_γ be the number of edges of colour γ . The local bound gives

$$m_\gamma \leq (t-1)q.$$

At a fixed vertex and for a fixed colour γ , fewer than t incident edges have colour γ . Summing $\binom{d_{v,\gamma}}{2} \leq (d_{v,\gamma})(t-1)/2$ over the $2q$ vertices shows that the number of same-coloured pairs sharing a vertex is at most $(t-1)q^2$. The total number of same-coloured pairs is at most

$$\sum_\gamma \binom{m_\gamma}{2} \leq \frac{1}{2} (\max_\gamma m_\gamma) \sum_\gamma m_\gamma \leq \frac{1}{2} (t-1)q^3.$$

If the two edges share a vertex in the left class, their exact selection probability is

$$\frac{n}{q} \frac{\binom{n}{2}}{\binom{q}{2}};$$

the same formula holds when the shared vertex is on the right. If the two edges are disjoint, the exact probability is

$$\left(\frac{\binom{n}{2}}{\binom{q}{2}} \right)^2.$$

For $q \geq 2n$, these are at most $2n^3/q^3$ and $4n^4/q^4$, respectively. By linearity of expectation,

$$\mathbb{E}[Z] \leq \frac{2(t-1)n^3}{q} + \frac{2(t-1)n^4}{q}.$$

This is below 1 for sufficiently large q . Since Z is integer-valued and nonnegative, it cannot satisfy $Z \geq 1$ for every choice when $\mathbb{E}[Z] < 1$. Hence some choice has $Z = 0$, and the selected $K_{n,n}$ is rainbow. $\square$

The complete bipartite expansion atom.

Proposition 3.2 (The complete bipartite expansion atom). *For every positive integer n , the triple system $K_{n,n}^+$ is obligatory.*

Proof. Suppose not. Choose a $K_{n,n}^+$ -free triple system

$$H = (\kappa, E)$$

with $\chi(H) > \aleph_0$ and with κ minimal.

For a pair xy , write

$$d_H(xy) = |\{z : \{x, y, z\} \in E\}|.$$

Set $t = 3n^2 + 1$ and define a graph R on κ by

$$xy \in E(R) \iff d_H(xy) \geq t.$$

Claim 3.2.1 (the graph R is countably chromatic).

Otherwise Lemma 2.1 gives a $K_{n,n}$ in R , with core vertex set C , $|C| = 2n$. For each of its n^2 graph edges xy , choose a third vertex p_{xy} with

$$\{x, y, p_{xy}\} \in E,$$

avoiding C and all previously chosen private vertices. At every stage fewer than $2n + n^2 < t$ vertices are forbidden, while xy has at least t possible third vertices. Thus the p_{xy} can be chosen distinct and outside C , producing a copy of $K_{n,n}^+$, a contradiction. $\square$

Let

$$E_1 = \{e \in E : \text{some pair contained in } e \text{ is an edge of } R\}, \quad E_2 = E \setminus E_1.$$

A proper colouring of R is a proper colouring of (κ, E_1) , since every E_1 -edge contains an R -edge. Hence $\chi(\kappa, E_1) \leq \aleph_0$, and therefore

$$\chi(\kappa, E_2) > \aleph_0. \tag{3.1}$$

For a pair xy , define

$$\Phi(xy) = \begin{cases} \{z : \{x, y, z\} \in E\}, & d_H(xy) < t, \\ \emptyset, & d_H(xy) \geq t. \end{cases}$$

Apply Lemma 1.4 to Φ , obtaining M_i, I_i .

Every $e \in E_2$ has at least two vertices in its highest layer. Indeed, suppose the highest layer met e only in $z \in I_i$. Then the other pair $xy = e \setminus \{z\}$ lies in M_i . Since $e \in E_2$, $xy \notin R$, so $d_H(xy) < t$. Closure gives $z \in \Phi(xy) \subseteq M_i$, a contradiction.

Thus every E_2 -edge is either contained in one layer, or has exactly two vertices in its highest layer I_i and its third vertex in M_i . Let E_i^* be the E_2 -edges contained in I_i . Since $|I_i| < \kappa$ and H is $K_{n,n}^+$ -free, minimality of κ gives $\chi(I_i, E_i^*) \leq \aleph_0$. Consequently

$$E^* = \bigcup_i E_i^*$$

is countably chromatic. By (3.1), the remaining crossing-edge system

$$E^{**} = E_2 \setminus E^*$$

has uncountable chromatic number.

For each i , define a graph G_i on I_i by declaring $xy \in E(G_i)$ when there is a $\beta \in M_i$ such that $\{x, y, \beta\} \in E^{**}$. If every G_i were countably chromatic, colour each I_i properly in G_i , using the same countable palette on every layer. Every edge of E^{**} would then have differently coloured top-layer vertices, giving a countable colouring of (κ, E^{**}) , a contradiction. Hence some G_i is uncountably chromatic.

Choose $q = q(n, t)$ from Lemma 3.1. Lemma 2.1 gives a $K_{q,q}$ in G_i , with sides X, Y . For every $xy \in X \times Y$, choose

$$\beta_{xy} \in M_i$$

such that $\{x, y, \beta_{xy}\} \in E^{**}$, and regard β_{xy} as the colour of the edge xy .

This edge-colouring is locally $(t-1)$ -bounded. For example, if for fixed x and fixed β there were t different vertices y with $\beta_{xy} = \beta$, then the pair $\{x, \beta\}$ would have codegree at least t . It would belong to R , forcing the triples $\{x, y, \beta\}$ into E_1 , contrary to their membership in $E^{**} \subseteq E_2$. The same argument applies at vertices of Y .

Lemma 3.1 gives n -sets $X' \subseteq X$, $Y' \subseteq Y$ for which the β_{xy} , $x \in X'$, $y \in Y'$, are all distinct. They lie in M_i , whereas $X' \cup Y' \subseteq I_i$, so none is a core vertex. The triples

$$\{x, y, \beta_{xy}\}, \quad x \in X', \quad y \in Y',$$

form $K_{n,n}^+$, the final contradiction. $\square$

Corollary 3.3 (Every finite bipartite expansion is obligatory). *If J is any finite bipartite graph, then J^+ is obligatory.*

Proof. Embed the two vertex classes of J in the two classes of some $K_{n,n}$. Then J^+ is a subhypergraph of $K_{n,n}^+$. Obligatoriness is downward closed under non-induced containment: a copy of the larger finite system contains a copy of the smaller one. Apply Proposition 3.2. $\square$

Corollary 3.3 also follows from Reiher [4, Theorem 1.2]. The argument above gives a direct proof in the notation used here.

4. CLOSURE PROPERTIES

Closure under disjoint unions and one-point amalgamations is known; see Komjáth [15, pp. 233–238] and the summary in Reiher [4, p. 1]. We include complete proofs because the subsequent argument uses the rooted-abundance construction explicitly.

Disjoint-union closure.

Lemma 4.1 (Disjoint-union closure). *The class of finite obligatory triple systems is closed under finite disjoint unions.*

Proof. Let F_1, F_2 be obligatory and let H be uncountably chromatic. First find a copy of F_1 . Delete its finite vertex set. By Lemma 1.2 the remaining triple system is still uncountably chromatic, so it contains F_2 . The two copies are disjoint. Iteration proves the finite case. $\square$

Rooted abundance and one-point amalgamation.

Definition 4.2 (Rooted abundance). Let (F, r) be a finite rooted triple system with $|V(F)| > 1$. For a triple system H and $m \geq 1$, let $R_m(F, r; H)$ be the set of vertices v for which there are m rooted copies of F , all sending r to v , whose off-root vertex sets are pairwise disjoint.

Lemma 4.3 (Rooted-abundance lemma). *If (F, r) is finite, $|V(F)| > 1$, and F is obligatory, and if $\chi(H) > \aleph_0$, then*

$$\chi(H[R_m(F, r; H)]) > \aleph_0.$$

Proof. Put

$$B = V(H) \setminus R_m(F, r; H).$$

For each $v \in B$, consider the family of off-root vertex sets of all copies of F rooted at v . Because $v \notin R_m(F, r; H)$, this family has matching number at most $m - 1$. Choose a maximal pairwise disjoint subfamily. It contains at most $m - 1$ members; let S_v be their union. Then

$$|S_v| \leq (m - 1)(|V(F)| - 1) =: D.$$

By maximality, S_v meets the off-root part of every rooted copy at v . Otherwise that off-root set would be disjoint from every member of the chosen subfamily and could be added to it.

Form a graph D_B on B by joining v to each member of $S_v \cap B$. If an edge is generated in both directions, assign it to one of its generators, and orient it away from that generator. Every vertex has outdegree at most D , so Lemma 1.3 gives a finite proper colouring of D_B .

Let A be one colour class. Then

$$S_v \cap A = \emptyset \quad (v \in A).$$

Consequently $H[A]$ contains no copy of F . Indeed, if φ were such a copy and $v = \varphi(r)$, its off-root part would have to meet S_v . All those off-root vertices lie in A , contradicting $S_v \cap A = \emptyset$. Since F is obligatory, every F -free triple system is countably chromatic. Thus each of the finitely many colour classes of D_B induces a countably chromatic subsystem. Tagging a vertex by its finite D_B -colour and its countable colour inside that class gives a countable proper colouring of $H[B]$.

If $H[R_m(F, r; H)]$ were countably chromatic as well, tagging the two sides would give a countable colouring of H . This contradicts $\chi(H) > \aleph_0$. $\square$

Proposition 4.4 (One-point amalgamation closure). *The class of finite obligatory triple systems is closed under one-point amalgamations.*

Proof. Let F_1, F_2 be obligatory, with selected vertices r_1, r_2 , and let F be their one-point amalgamation. If one factor consists only of its selected vertex, the assertion is immediate. Otherwise set

$$m = |V(F_2)|.$$

By Lemma 4.3, the set

$$R = R_m(F_1, r_1; H)$$

induces an uncountably chromatic subsystem of every uncountably chromatic H . Hence $H[R]$ contains a copy of F_2 ; write φ for its embedding and let $v = \varphi(r_2)$.

At v there are m rooted copies of F_1 with pairwise disjoint off-root sets. The set $\varphi(V(F_2) \setminus \{r_2\})$ has $m - 1$ vertices, so it meets at most $m - 1$ of those off-root sets. One rooted F_1 -copy therefore avoids the F_2 -copy outside v . Their union is the required one-point amalgamation. $\square$

Corollary 3.3, Lemma 4.1, Proposition 4.4, and the trivial edgeless case prove

$$(2) \implies (1). \tag{4.1}$$

5. THE INTRINSIC DECOMPOSITION

We state the selected-incidence decomposition and its running-intersection argument in the explicit form used by the Lean development. For a distinct structural formulation, compare Li [3, Sections 4–5].

We prove $(2) \iff (3)$. Isolated vertices may be ignored by Lemma 1.1. Forward preservation.

Proposition 5.1 (The intrinsic conditions are preserved by the generators and operations). *Every generator of $\mathcal{B}$ satisfies the intrinsic conditions, and the conditions are preserved under finite disjoint unions and one-point amalgamations.*

Proof. For a bipartite graph J , the expansion J^+ is linear. Every hyperedge-node is joined to its private point by a bridge. A Berge cycle cannot use a private point, so the Berge cycles of J^+ are precisely the ordinary graph cycles of J , with the same lengths; these are even. Edgeless systems satisfy the conditions vacuously, and disjoint union plainly preserves them.

Consider a one-point amalgamation. Hyperedges belonging to different factors intersect only in the identified point, so linearity is preserved. Its Levi graph is the vertex-sum of the two Levi graphs at a point-node. Every simple cycle in a vertex-sum lies wholly in one factor: a cycle crossing into the other factor would have to pass through the cut vertex twice. Consequently bridges in the factors remain bridges, and every Berge cycle lies in one factor and retains its parity. This uses the standard cut-vertex structure of graph blocks [9, Chapter 3]; the corresponding hypergraph separation terminology is discussed in Bahmanian and Šajna [26]. $\square$

Thus

$$(2) \implies (3). \tag{5.1}$$

Bridge-block reconstruction.

Proposition 5.2 (Bridge-block decomposition). *Suppose F is finite, has no isolated vertices, and satisfies the three intrinsic conditions. Then $F \in \mathcal{B}$.*

Proof. If $E(F) = \emptyset$, then the assumption that F has no isolated vertices gives $V(F) = \emptyset$, and F is an edgeless generator. Hence assume $E(F) \neq \emptyset$. Put $L = I(F)$, let $B(L)$ be its set of bridges, and let $\mathcal{C}$ be the set of connected components of $L - B(L)$.

Consider a hyperedge-node e . Since e is incident with a bridge and has degree three, it has at most two nonbridge incidences. It cannot have exactly one: a nonbridge incidence lies on a cycle, and that cycle must leave e through a second nonbridge incidence. Therefore every hyperedge-node has either no nonbridge incidences or exactly two.

The nontrivial components. Let $C \in \mathcal{C}$ contain a nonbridge incidence. Every hyperedge-node $e \in C$ has exactly two point-neighbours in C . Contract each such e to an ordinary graph edge between those two point-neighbours. This produces a finite graph J_C .

The graph J_C is simple: parallel edges would correspond to two hyperedges of F sharing two points, contrary to linearity. Every cycle in J_C gives a Berge cycle of the same length in F . Hence J_C has no odd cycle and is bipartite.

Each hyperedge-node $e \in C$ has a third point-neighbour p_e across a bridge. These third points satisfy:

1. $p_e \notin C$; otherwise a path in C from p_e to e , together with the incidence ep_e , would place that incidence on a cycle.
2. If $e \neq f$ lie in C , then $p_e \neq p_f$; otherwise a path in C from e to f , together with ep_e and p_ff , would place those bridge incidences on a cycle.

Define a map from J_C^+ to the subsystem P_C formed by these hyperedges as follows. Send every graph vertex of J_C to the corresponding point-node of C , send the private vertex indexed by the contracted edge associated with e to p_e , and send that expanded edge to the hyperedge e of F . The two observations above show that the vertex map is injective: the private points are pairwise distinct and lie outside C . Every e has exactly the two contracted endpoints and p_e as its three vertices, so this map is incidence-preserving and surjective on the vertices and hyperedges of P_C . Hence it is an isomorphism $J_C^+ \cong P_C$. Components consisting of one hyperedge-node. If a hyperedge-node e has no nonbridge incidence, it is an isolated component after the bridges are deleted. Its single triple is the expansion of a one-edge bipartite graph. Thus every hyperedge of F belongs to a piece P_C isomorphic to J^+ for a finite bipartite graph J . The hyperedge sets of these pieces partition $E(F)$. Their vertex sets cover $V(F)$: every point is incident with a hyperedge, and it either lies in that hyperedge-node's bridge-deleted component or is the point endpoint of a bridge leaving it.

Quotient forest. Contract every component of $L - B(L)$. The original bridges become edges of a quotient multigraph T . Two distinct bridges cannot join the same pair of bridge-deleted components: paths inside those two components, together with the two bridges, would form a cycle containing both, contradicting that they are bridges. Thus T is a simple graph.

To see that T is a forest, suppose that it contained a cycle. Within each contracted component on this cycle, choose a path joining the endpoints of the two incident quotient edges. Concatenating these paths with the corresponding bridges gives a closed walk in L that traverses each of those bridges exactly once. This is impossible: deleting a bridge separates its endpoints, so every closed walk crosses the resulting cut an even number of times. Hence T is a forest.

A vertex $C \in V(T) = \mathcal{C}$ is called *active* if the bridge-deleted component C contains a hyperedge-node. For an active C , the piece P_C consists of the hyperedges represented in C , all point-nodes of C , and the point endpoints of bridge incidences leaving hyperedge-nodes in C . An inactive component contains no hyperedge-node and hence no incidence of $L - B(L)$, so it is a singleton point-node.

Every hyperedge now belongs to a bipartite expansion piece. It remains to order the pieces so that each new piece meets the assembled system in at most one point. The quotient forest records the attachment geometry needed for this running-intersection property.

Claim (5.2.1. Running intersection). *Every component of T contains an active vertex, because F has no isolated vertices. Choose an active root in each component, and order the active vertices by nondecreasing depth, with arbitrary order among equal-depth vertices. When an active component C is added, its piece P_C meets the union of the earlier pieces in at most one point.*

Proof of the claim. For a point $p \in V(F)$, let $X(p) \in \mathcal{C}$ be the component of $L - B(L)$ containing the point-node p . The active pieces containing p are precisely $P_{X(p)}$, when $X(p)$ is active, together with the pieces P_D at active neighbours D of $X(p)$ whose corresponding original bridge has point endpoint p . Thus the vertices of T indexing pieces that contain p are contained in the closed star centred at $X(p)$.

Suppose that $p \in P_C$ also lies in an earlier piece. If $X(p) = C$, every other piece containing p is indexed by a neighbour of C . A tree has only one neighbour of C at smaller depth, namely its parent, so only that piece can be earlier. If $X(p) \neq C$, then C is a neighbour of $X(p)$. The vertex $X(p)$ cannot be a child of C , because then both $X(p)$ and every other vertex in its closed star would be deeper than C . Hence $X(p)$ is the parent of C . In either case, p is the point endpoint of the unique parent edge of C . The same reasoning applies to every point shared with an earlier piece, so all such points coincide with this parent point p_C , and

$$P_C \cap \bigcup_{D \text{ earlier}} P_D \subseteq \{p_C\}.$$

For a root active component the intersection is empty. $\square$

Starting with the root piece in each component, add the active pieces in this order. If the new intersection is empty, use a disjoint union. If it is the singleton $\{p_C\}$, use a one-point amalgamation at that point. Several earlier pieces may already contain the same inactive attachment point, but the new step still identifies only that one existing vertex.

The identity maps from the pieces into F assemble inductively to an incidence isomorphism onto the union of the pieces already added. By the running-intersection claim, the next piece and the preceding union are either disjoint or meet exactly at the point used in the amalgamation, so no unintended identification is introduced. The piece hyperedge sets are disjoint and partition $E(F)$, and their vertex sets cover $V(F)$. After all active pieces have been added, the resulting subsystem is F . Different components of T are combined by disjoint union, proving $F \in \mathcal{B}$. $\square$

This proves

$$(3) \implies (2). \tag{5.2}$$

Combining (5.1) and (5.2),

$$(2) \iff (3). \tag{5.3}$$

Theorem 5.3 (Canonical atom normal form). *Let F be a finite triple system without isolated points satisfying the three intrinsic conditions in Theorem A. Its hyperedges admit a canonical partition into atoms with the following properties.*

- (1) Every atom is either one triple or J^+ for a finite 2-connected simple bipartite graph J .
- (2) Two distinct atoms meet in at most one point. The bipartite incidence graph between atoms and points belonging to at least two atoms is a forest.
- (3) The atom partition, the nontrivial cores J up to graph isomorphism, and the atom–point forest are determined by F .

Conversely, every forest assembly of these atoms by disjoint unions and one-point amalgamations satisfies the intrinsic conditions.

Proof. Let $L = I(F)$. A *cyclic block* is a maximal 2-connected subgraph of L containing a cycle. Since every hyperedge-node has degree three and is incident with a bridge, its degree inside a cyclic block is two. Suppressing the hyperedge-nodes in such a block C gives a finite graph J_C on the point-nodes of C . It is simple by linearity and 2-connected by standard block theory [8, 9]. Its cycles correspond to Berge cycles of the same lengths, so J_C is bipartite.

For a hyperedge-node $e \in C$, the third incidence is a bridge. Its point endpoint lies outside C , since otherwise that incidence would lie on a cycle. These third points are pairwise distinct as e varies in C ; otherwise a path inside C between two hyperedge-nodes, completed through the common third point, would put both bridge incidences on a cycle. The hyperedges represented in C therefore form exactly J_C^+ .

A hyperedge-node outside every cyclic block has no nonbridge incidence and contributes one single-triple atom. Thus the atoms partition the hyperedges. Two atoms cannot share two points, since paths inside the two atoms would produce a Levi cycle crossing two cyclic blocks. Similarly, a cycle in the atom–shared-point incidence graph expands to a closed Levi walk and hence to a Levi cycle crossing atom boundaries. The incidence graph is therefore a forest. Cyclic blocks and the block–cut forest are canonical, proving the forward statement. The converse follows because an atomic expansion is linear, has a private bridge at each hyperedge-node, and has only even Berge cycles, while a forest of one-point sums preserves these properties. $\square$

The graph primitives in this normal form are classical algorithmic objects: bridges and bridge-connected components can be found by Tarjan’s algorithm [27], while biconnected components admit a depth-first-search construction [28]. These algorithms are distinct from the structural identification above; no verified running-time bound for the Lean development is asserted.

Call a connected triple system with at least one hyperedge *one-point indecomposable* if it cannot be written as a one-point amalgamation of two systems that both contain a hyperedge.

Corollary 5.4 (Minimal generators and one-point indecomposables). *Up to isomorphism, the class $\mathcal{B}$ is generated by finite edgeless systems, one triple, and J^+ for finite 2-connected simple bipartite graphs J , under finite disjoint unions and one-point amalgamations. A connected reduced obligatory system with at*

least one hyperedge is one-point indecomposable if and only if it is one triple or J^+ for a finite 2-connected simple bipartite graph J .

Proof. The theorem decomposes every reduced member of $\mathcal{B}$ into the displayed atoms; isolated points are restored by an edgeless factor. If the atom forest has at least two atom-nodes, a leaf atom separates from the rest at its unique shared point. Conversely, one triple is indecomposable, and if J is 2-connected then deleting any point of J^+ cannot separate its hyperedge-nodes into two nonempty classes: a private point is a leaf, while a core point leaves J connected. Hence J^+ is indecomposable. $\square$

6. THE SEQUENCE LIFT

We use a one-apex construction specialised to ω_1 , presented through base fibres and their support-incidence graph to match the formalisation. Li [3, Sections 3–4 and Theorem 4.6] gives an alternative one-apex and bridge-trace formulation.

The one-apex construction. Let $G = (X, A)$ be an arbitrary graph, where $A = E(G)$ is also regarded as an alphabet. Let

$$T = A^{<\omega_1}$$

be the tree of all sequences of graph edges of countable ordinal length. For $s \in T$ and $x \in X$, write $x_s = (s, x)$.

Definition 6.1 (The one-apex lift). Define the triple system $\mathcal{L}(G)$ by

$$V(\mathcal{L}(G)) = T \times X.$$

For $s \in T$, an edge $a = \{x, y\} \in A$, a sequence $t \in T$ extending $s \frown a$, and $z \in X$, put

$$\{x_s, y_s, z_t\} \in E(\mathcal{L}(G)). \quad (6.1)$$

The two vertices x_s, y_s form the base pair and z_t is the apex. Since t properly extends s , these are three distinct vertices. In the resulting triple, the sequence node occurring twice uniquely identifies s ; its two graph labels identify the edge $a = \{x, y\}$, and the remaining vertex identifies t, z . Thus each hyperedge arises from unique data, up to interchanging x and y , and the system is simple and 3-uniform.

Chromatic and finite-trace properties.

Lemma 6.2 (Chromatic lower bound). *If $\chi(G) > \aleph_0$, then*

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

Proof. Suppose $c : T \times X \rightarrow \omega$ is proper. For every $s \in T$, the restriction $x \mapsto c(x_s)$ is a countable colouring of G . Hence there is an edge

$$a_s = \{x_s^0, x_s^1\} \in E(G)$$

whose two vertices at node s have a common colour k_s .

Construct $s_\alpha \in A^\alpha$ for $\alpha < \omega_1$ by

$$s_0 = \emptyset, \quad s_{\alpha+1} = s_\alpha \widehat{a}_{s_\alpha},$$

and at limit ordinals take unions.

If $\alpha < \beta$, then s_β extends $s_\alpha \widehat{a}_{s_\alpha}$. If $k_{s_\alpha} = k_{s_\beta}$, take either endpoint z of the monochromatic edge at s_β . By (6.1),

$$\{(s_\alpha, x_{s_\alpha}^0), (s_\alpha, x_{s_\alpha}^1), (s_\beta, z)\}$$

is a monochromatic hyperedge. Therefore the colours k_{s_α} , $\alpha < \omega_1$, are pairwise distinct, impossible in ω . $\square$

The following trace theorem uses the fibre decomposition described above. Li gives a parallel bridge-trace formulation [3, Theorem 4.6].

Theorem 6.3 (Finite linear trace decomposition). *Every finite, not necessarily induced, linear subhypergraph K of $\mathcal{L}(G)$ is obtainable, by finite disjoint unions and one-point amalgamations, from systems J^+ where J is a finite subgraph of G .*

Proof. We use three local objects in the assembly. A *base node* is a sequence s at which a lift hyperedge begins. The two vertices with sequence coordinate s are its *core vertices*; the third vertex, whose coordinate properly extends s , is its *apex*. The *base fibre* K_s consists of all hyperedges of K based at s , together with their incident vertices. A base node is *active* when its fibre is nonempty. We first identify each fibre with an expansion piece and then show that the fibres form a forest of one-point overlaps.

First set aside every isolated vertex of K . Each is a one-vertex edgeless factor, hence isomorphic to J^+ for a one-vertex edgeless subgraph J of G , and may be restored by disjoint union at the end. We may therefore assume that every vertex of K lies in a hyperedge.

Every hyperedge of $\mathcal{L}(G)$ has a unique base node: exactly two of its vertices have sequence coordinate s , while the apex has a strictly longer sequence coordinate. We group the hyperedges of K by this coordinate and use the notation K_s from the dictionary.

Claim 6.3.1 (each base fibre is one expansion piece). For every active base node s , there is a finite subgraph J_s of G such that $K_s \cong J_s^+$.

Proof of Claim 6.3.1. For a fixed graph edge $a = \{x, y\} \in E(G)$, at most one hyperedge of K_s can have base pair $\{x_s, y_s\}$, since two such hyperedges would share two vertices and violate linearity. Let J_s be the finite subgraph of G whose edges are precisely the graph edges used by hyperedges of K_s , with vertex set consisting of their endpoints.

Define a map $J_s^+ \rightarrow K_s$ by sending each graph vertex $x \in V(J_s)$ to the core vertex x_s , and by sending the private vertex of $a = \{x, y\}$ to the apex of the unique hyperedge of K_s with base pair $\{x_s, y_s\}$. Distinct graph edges have distinct apices: if z_t were the apex for a and a' , the coordinate of t immediately after s would have to equal both a and a' . No apex is a core vertex, because an apex has sequence coordinate strictly extending s , whereas every core vertex has coordinate s . The map is therefore injective, and by construction it is bijective on hyperedges and preserves each incidence. Hence $K_s \cong J_s^+$. $\square$

Claim 6.3.2 (two fibres have at most one common point). For distinct active base nodes s and u ,

$$|V(K_s) \cap V(K_u)| \leq 1.$$

Proof of Claim 6.3.2. If a point z_t lies in both K_s and K_u , then both s and u are prefixes of t , so they are comparable. Suppose $s \subsetneq u$. A common point cannot be a core point of K_s ; it is an apex of K_s . The graph edge used by its hyperedge in K_s is the coordinate of u immediately following s . Hence all points in $K_s \cap K_u$ would be apices of the same base edge of K_s . There is at most one such apex. Therefore

$$|V(K_s) \cap V(K_u)| \leq 1. \tag{6.2}$$

$\square$

Let Q be the bipartite incidence graph of the base fibres. Its left class is the set of active base nodes. Its right class consists of points that lie in at least two fibres. Join a base node s to a point p exactly when $p \in V(K_s)$.

Claim 6.3.3 (the overlap graph is a forest). The graph Q is acyclic.

Proof of Claim 6.3.3. Suppose that Q has a cycle, and choose on it a base node s of minimum sequence length. Its two neighbouring base nodes on the cycle are proper descendants of s . The two intervening shared points are distinct, so they are distinct apices in K_s , arising from two distinct graph edges $a \neq a'$. Thus the two neighbouring base nodes lie in different immediate branches below s , one beginning with a and the other with a' .

Starting from one neighbour of s , every base node on the remaining part of the cycle lies in the same immediate branch below s . Indeed, consecutive base nodes share a point and are therefore comparable. For two comparable proper descendants of s , the first sequence coordinate after s is the same, so a length-two step through a shared point cannot move from one immediate branch to another. Induction along the remaining path therefore keeps every base node in the first branch, contradicting that its final node lies in the other branch. Hence Q is acyclic. $\square$

Root each component of Q at a base node, and order its base nodes by nondecreasing even distance from the root. For a nonroot base node s , let p_s be the point-node adjacent to s on the path to the root.

Claim (6.3.4. Base-fibre running intersection). *Every earlier factor that meets K_s meets it at p_s .*

Proof of Claim 6.3.4. Suppose K_u is earlier and contains a point $p \in K_s$. Then Q contains the length-two path $s - p - u$. Let the distance of s from the root be $2d$. The point-node p has distance either $2d - 1$ or $2d + 1$. In the second case u has distance $2d + 2$, so it cannot be earlier. Thus p has distance $2d - 1$, and the uniqueness of the path to the root in the tree Q gives $p = p_s$. This also covers an earlier sibling u , which shares the same parent point p_s . $\square$

Claim 6.3.5 (the ordered fibres assemble to K). Let U_s be the union of the factors preceding K_s . Equation (6.2) and Claim 6.3.4 give

$$V(K_s) \cap U_s \subseteq \{p_s\}.$$

Starting with the root factor, add the remaining factors in this order by disjoint unions or one-point amalgamations as appropriate. At each step the natural maps into K identify exactly the displayed common point and nothing else, so they assemble to an isomorphism onto the union of the factors already added. Different components of Q , and the isolated factors removed at the start, are combined by disjoint union. This is the running-intersection step: Claim 6.3.1 supplies the expansion pieces, while Claims 6.3.2–6.3.4 certify that each new piece is attached along at most one previous point. $\square$

Corollary 6.4 (Restrictions on finite linear traces). *Every finite linear $K \subseteq \mathcal{L}(G)$ satisfies:*

1. *every hyperedge-node of $I(K)$ is incident with a bridge;*
2. *every Berge cycle of K is contained in a single factor J_s^+ ;*
3. *a Berge cycle of length ℓ in K yields an ordinary cycle of length ℓ in G .*

Proof. In every J_s^+ the incidence with the private vertex is a bridge. Bridgehood is preserved by one-point amalgamation. A simple Levi cycle cannot cross a one-point cut, so every Berge cycle lies in one factor. Berge cycles in J_s^+ are exactly ordinary graph cycles in J_s . $\square$

7. A CLASSICAL HIGH-ODD-GIRTH INPUT

For the odd-cycle obstruction we use the following classical graph theorem directly, without passing through a shift-graph surrogate.

Theorem 7.1 (Erdős–Hajnal). *For every positive integer m there is a graph G such that*

$$\chi(G) > \aleph_0$$

and G contains no odd cycle of length at most m .

Source. Theorem 7.4 of Erdős and Hajnal [13, p. 76], as quoted in Erdős et al. [5, Theorem C, p. 428], gives for every positive integer i an uncountably chromatic graph containing no cycle C_{2j+1} for $0 < j < i$. Choose i so that $2i - 1 \geq m$. $\square$

8. AVOIDANCE HOSTS

The intrinsic criterion can fail in three ways: nonlinearity, a missing bridge, or an odd Berge cycle. Li [3, Sections 2–4] organises the negative direction in a parallel way. We prove the first two avoidance constructions in our notation and apply the Erdős–Hajnal theorem from Section 7 to the third.

We prove

$$\neg(3) \implies \neg(1).$$

Nonlinearity obstruction.

Proposition 8.1 (Avoidance of every nonlinear finite triple system). *There is an uncountably chromatic linear triple system. Consequently, every finite nonlinear triple system is non-obligatory.*

Proof. We use the Erdős–Rado relation [24, Theorem 4(i), formula (95), p. 471]

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)_{\aleph_0}^2. \quad (8.1)$$

Let $\kappa = (2^{\aleph_0})^+$. Define a triple system T_κ whose vertices are the pairs in $[\kappa]^2$, and whose hyperedges are

$$\{\{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}\}, \quad \alpha < \beta < \gamma < \kappa.$$

This triple system is linear: two distinct triangles of a complete graph cannot share two graph edges, because two graph edges of a triangle determine the third edge and the three underlying vertices.

It is uncountably chromatic. A countable vertex-colouring of T_κ is a countable edge-colouring of K_κ . By (8.1) there is a monochromatic subset of cardinality $\aleph_1$, and in particular a monochromatic graph triangle. This is a monochromatic hyperedge of T_κ .

If a finite triple system F is nonlinear, it has two distinct edges sharing at least two vertices. No injective embedding of those edges can exist in the linear host T_κ . Thus T_κ is F -free. $\square$

Missing-bridge obstruction.

Proposition 8.2 (Avoidance when the bridge condition fails). *Let F be finite and linear, and suppose some hyperedge-node of $I(F^\circ)$ is incident with no bridge. Then F is non-obligatory.*

Proof. This is the K_{ω_1} specialisation of the sequence lift defined above. Li gives an alternative missing-bridge obstruction [3, Section 4]. By Lemma 1.1 it suffices to avoid F° , so assume that F has no isolated vertices. Take $G = K_{\omega_1}$. Then $\chi(G) > \aleph_0$, and Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, retain exactly the host hyperedges corresponding to the edges of F . Because the embedding is injective and F is linear, these selected host hyperedges form a finite linear subhypergraph isomorphic to F . Corollary 6.4 says that every hyperedge-node of such a trace is incident with a bridge, contradicting the chosen hyperedge of F . Therefore $\mathcal{L}(K_{\omega_1})$ is an uncountably chromatic F -free host. $\square$

Odd-cycle obstruction.

Proposition 8.3 (Avoidance of an odd Berge cycle). *Let F be finite and linear, and suppose F° has an odd Berge cycle. Then F is non-obligatory.*

Proof. Suppress isolated vertices and put $m = |E(F)|$. By Theorem 7.1 choose a graph G with $\chi(G) > \aleph_0$ and no odd cycle of length at most m . Lemma 6.2 gives $\chi(\mathcal{L}(G)) > \aleph_0$.

Suppose that F embeds in $\mathcal{L}(G)$ and retain exactly the host hyperedges corresponding to the edges of F . The selected trace is finite and linear. Corollary 6.4 localises every Berge cycle in one expansion fibre and sends a Berge cycle of length ℓ to an ordinary cycle of the same length in G . The given odd Berge cycle has $\ell \leq m$, contradicting the choice of G . Thus $\mathcal{L}(G)$ is an uncountably chromatic F -free host. $\square$

The three propositions cover every failure of the intrinsic conditions and therefore prove

$$\neg(3) \implies \neg(1). \tag{8.2}$$

9. PROOF OF THE CLASSIFICATION

Sections 3 and 4 establish $(2) \implies (1)$, Section 5 establishes $(2) \iff (3)$, and Section 8 proves the contrapositive $\neg(3) \implies \neg(1)$. Together these implications prove all equivalences in Theorem A. In particular,

$$F \text{ is obligatory} \iff F \in \mathcal{B},$$

$$F^\circ \text{ is linear,}$$

$$F \in \mathcal{B} \iff \begin{array}{l} \text{every hyperedge-node of } I(F^\circ) \text{ is incident with a bridge,} \\ \text{every Berge cycle of } F^\circ \text{ has even length.} \end{array}$$

All embeddings are non-induced, so additional host hyperedges on the image vertices are irrelevant. The proof uses only ZFC, graph-colouring compactness [23], the classical Erdős–Hajnal high-odd-girth theorem [13], and the standard Erdős–Rado relation (8.1) [24]. It assumes neither CH nor GCH and uses no forcing axiom or large-cardinal hypothesis.

10. FINITE PARAMETER CONSEQUENCES

In this section, a triple system is *connected* when its Levi graph is connected. We first exclude isolated vertices. For an integer $r \geq 1$, set

$$q(r) = \lceil 2\sqrt{r} \rceil.$$

Proposition 10.1 (Edge-deletion form of the bridge condition). *Let F be a finite triple system and let $e = \{x, y, z\} \in E(F)$. The hyperedge-node e has no incident bridge in $I(F)$ if and only if x, y, z lie in one connected component after the hyperedge e is deleted and all points are retained. Consequently the bridge condition in Theorem A is equivalent to the following hypergraph-native condition:*

*for every $e \in E(F^\circ)$, deleting e
separates its three points into at least two components.*

Proof. Fix $x \in e$. The Levi incidence xe is not a bridge precisely when there is an alternative path from e to x after that incidence is deleted. A simple such path leaves e through y or z and thereafter lies entirely in the Levi graph of $F - e$. Thus xe is non-bridging exactly when x is connected in $F - e$ to at least one of the other two points of e .

If all three incidences at e are non-bridging, no one of x, y, z can form a singleton component among those three points. A partition of three points with this property has one block, so all three are connected in $F - e$. The converse follows by reversing the same paths. $\square$

Lemma 10.2 (Bipartite shadow). *Let F be an obligatory finite triple system with no isolated vertices and at least one hyperedge. There is a finite bipartite graph J , without isolated vertices, such that*

$$|E(J)| = |E(F)|, \quad |V(J)| = |V(F)| - |E(F)|,$$

and J has the same number of connected components as F . Conversely, for every finite bipartite graph J without isolated vertices, the expansion J^+ is obligatory and has these parameter relations.

Proof. By Theorem A and Proposition 5.2, each connected component H of F is obtained by one-point amalgamations from connected expansion pieces

$$J_1^+, \dots, J_k^+,$$

where each J_i is a connected bipartite graph. Disconnected graph pieces may simply be split into their connected components. Write $m_i = |E(J_i)|$ and $s_i = |V(J_i)|$. Since H is connected, its k pieces are joined by exactly $k - 1$ one-point identifications. Hence

$$|E(H)| = \sum_{i=1}^k m_i, \quad |V(H)| = \sum_{i=1}^k (m_i + s_i) - (k - 1). \quad (10.1)$$

Take one-point sums of the ordinary graphs $J_1, \dots, J_k$ along any tree on the k indices. Before each identification, interchange the two bipartition classes of

the new factor if necessary. The resulting graph J_H is connected and bipartite, and (10.1) gives

$$|E(J_H)| = |E(H)|, \quad |V(J_H)| = |V(H)| - |E(H)|.$$

Taking the disjoint union of these shadows over the components of F proves the forward assertion. The converse is the expansion-generator case of Theorem A, with the displayed counts immediate from the definition of J^+ . $\square$

Theorem 10.3 (Exact order–size–component spectrum). *Let $m \geq 1$, $1 \leq c \leq m$, and n be integers. There exists an obligatory triple system F with no isolated vertices, exactly m hyperedges, exactly n vertices, and exactly c connected components if and only if*

$$m + 2(c - 1) + \left\lceil 2\sqrt{m - c + 1} \right\rceil \leq n \leq 2m + c. \quad (10.2)$$

Every integer in this interval occurs.

Proof. We begin with the corresponding elementary graph fact. A connected simple bipartite graph with $r \geq 1$ edges and s vertices exists if and only if

$$q(r) \leq s \leq r + 1. \quad (10.3)$$

Indeed, connectedness gives $r \geq s - 1$. If the bipartition has sizes a, b , then

$$r \leq ab \leq \left\lfloor \frac{s^2}{4} \right\rfloor,$$

which is equivalent to $s \geq q(r)$. Conversely, $K_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}$ contains a spanning tree, and starting with that tree and adding arbitrary unused edges realises every edge count between $s - 1$ and $\lfloor s^2/4 \rfloor$.

Apply Lemma 10.2. Let the c connected components of the shadow have positive edge counts $m_1, \dots, m_c$, where $\sum_i m_i = m$. If their orders are s_i , then (10.3) gives

$$\sum_{i=1}^c q(m_i) \leq \sum_{i=1}^c s_i \leq m + c. \quad (10.4)$$

For positive integers a, b ,

$$q(a) + q(b) \geq q(a + b - 1) + 2. \quad (10.5)$$

To see this, note that

$$\sqrt{a} + \sqrt{b} \geq \sqrt{a + b - 1} + 1,$$

because, after squaring, the assertion reduces to $(a - 1)(b - 1) \geq 0$; taking ceilings gives (10.5). Repeatedly applying (10.5) to the left side of (10.4) yields

$$\sum_{i=1}^c q(m_i) \geq q(m - c + 1) + 2(c - 1).$$

The shadow has $n - m$ vertices, so these inequalities give (10.2).

For sharpness put $M = m - c + 1$. Take $c - 1$ components equal to K_2 . By (10.3), a final connected bipartite component with M edges can have every order from $q(M)$ through $M + 1$. Their disjoint union therefore realises every shadow order from $q(M) + 2(c - 1)$ through $m + c$. Expanding it adds exactly m private vertices and proves attainability of every integer in (10.2). $\square$

Corollary 10.4 (Connected and unrestricted spectra). *For $m \geq 1$:*

- (1) *a connected obligatory triple system without isolated vertices has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \leq n \leq 2m + 1;$$

- (2) *without prescribing the number of components, a reduced obligatory triple system has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \leq n \leq 3m;$$

- (3) *if isolated vertices are permitted, an obligatory triple system with m hyperedges and n vertices exists if and only if*

$$n \geq m + \lceil 2\sqrt{m} \rceil.$$

Proof. The first assertion is Theorem 10.3 with $c = 1$. For the second, the connected interval covers through $2m + 1$; for $n > 2m + 1$, take $c = n - 2m$ and use the upper endpoint $n = 2m + c$ of (10.2). The largest possible value is obtained from m disjoint triples. The final assertion follows by adjoining arbitrary isolated vertices. $\square$

Corollary 10.5 (Size spectrum at fixed order). *Let $c \geq 1$ and $n \geq 3c$. A reduced obligatory triple system with exactly n vertices, c connected components, and m hyperedges exists if and only if*

$$\left\lceil \frac{n - c}{2} \right\rceil \leq m \leq n - 2c + 4 - \left\lceil 2\sqrt{n - 3c + 4} \right\rceil. \quad (10.6)$$

Consequently, a reduced obligatory triple system with exactly c connected components and n vertices exists if and only if

$$n = 3c, \quad n = 3c + 2, \quad \text{or} \quad n \geq 3c + 4. \quad (10.7)$$

Proof. The upper-order inequality in (10.2) is equivalent to $m \geq \lceil (n - c)/2 \rceil$. For the other inequality put

$$M = m - c + 1, \quad N = n - 3c + 3.$$

It becomes

$$M + \lceil 2\sqrt{M} \rceil \leq N. \quad (10.8)$$

For integers $N \geq 3$ and $M \geq 1$, the largest M satisfying (10.8) is

$$U = N + 2 - \lceil 2\sqrt{N + 1} \rceil. \quad (10.9)$$

Indeed, write $k = \lceil 2\sqrt{N+1} \rceil$. Since $k^2 \geq 4(N+1)$,

$$4U = 4N + 8 - 4k \leq (k-2)^2,$$

so $\lceil 2\sqrt{U} \rceil \leq k-2$ and $U + \lceil 2\sqrt{U} \rceil \leq N$. On the other hand, $(k-1)^2 < 4(N+1)$ gives

$$(k-3)^2 < 4(N+3-k) = 4(U+1),$$

so $\lceil 2\sqrt{U+1} \rceil \geq k-2$ and $(U+1) + \lceil 2\sqrt{U+1} \rceil \geq N+1$. The left side of (10.8) is strictly increasing in M , proving (10.9). Translating back to m gives (10.6).

For (10.7), Corollary 10.4 shows that a connected reduced obligatory system has order 3, order 5, or any order at least 7. The first two statements come from $m = 1, 2$. For $m \geq 3$, the connected intervals have no gaps because $\lceil 2\sqrt{m+1} \rceil \leq m+1$ for $m+1 \geq 4$. Thus every component has order

$$3 + r, \quad r \in \{0, 2\} \cup [4, \infty).$$

The sum of c such increments is again 0, 2, or any integer at least 4: use one nonzero increment and take all remaining increments to be zero. The increments 1 and 3 are impossible. This proves (10.7). $\square$

Corollary 10.6 (Exact Levi cycle-rank spectrum). *Let F be reduced and obligatory, with m hyperedges, n vertices, and c connected components. The cyclomatic number of its Levi graph is*

$$\beta(I(F)) = |E(I(F))| - |V(I(F))| + c = 2m - n + c. \quad (10.10)$$

For fixed m, c , every integer in the interval

$$0 \leq \beta(I(F)) \leq m - c + 2 - \lceil 2\sqrt{m - c + 1} \rceil \quad (10.11)$$

occurs, and no other value occurs. In particular,

$$|V(F)| = 2|E(F)| + c \iff I(F) \text{ is a forest.}$$

Proof. The Levi graph has $n+m$ vertices, $3m$ incidence edges, and c components, which gives (10.10). Substitution of the two endpoints in (10.2) gives (10.11), and every intermediate value occurs because every intermediate order does. A finite graph has cyclomatic number zero exactly when it is a forest. $\square$

Corollary 10.7 (Rigidity at balanced lower endpoints). *Let F be connected, reduced, and obligatory.*

(1) *If $|E(F)| = t^2$ and $|V(F)| = t^2 + 2t$, then $F \cong K_{t,t}^+$.*

(2) *If $|E(F)| = t(t+1)$ and $|V(F)| = t(t+1) + 2t + 1$, then $F \cong K_{t,t+1}^+$.*

Here $t \geq 1$, and the complete bipartite graphs are unique up to interchanging the two classes.

Proof. In either case the shadow from Lemma 10.2 attains $|E(J)| = \lfloor |V(J)|^2/4 \rfloor$. Equality forces every possible cross edge to be present and the two bipartition classes to be balanced, so the shadow is $K_{t,t}$ or $K_{t,t+1}$, respectively.

For $t \geq 2$ these complete bipartite graphs have no cut vertex. On the other hand, the shadow built from two or more connected expansion pieces

is a nontrivial one-point sum and has a cut vertex. Hence the bridge-block reconstruction has only one piece, and F is the expansion of the displayed complete bipartite graph. The cases $t = 1$ are immediate. $\square$

Canonical atoms and structural spectra. For a connected reduced obligatory system F , put

$$s = |V(F)| - |E(F)|, \quad \beta = 2|E(F)| - |V(F)| + 1.$$

Thus s is the order of a connected bipartite shadow and β is its cycle rank. Let $a(F)$ denote the number of canonical atoms from Theorem 5.3.

Lemma 10.8 (Minimum order at fixed cyclic rank). *Let J be a finite 2-connected simple bipartite graph of cycle rank $r = |E(J)| - |V(J)| + 1 \geq 1$. Then*

$$|V(J)| \geq 2 + q(r). \quad (10.12)$$

For $r \geq 2$, every order $v \geq 2 + q(r)$ occurs. For $r = 1$, the possible orders are exactly the even integers $v \geq 4$.

Proof. Writing $v = |V(J)|$, the bipartite extremal bound gives

$$r \leq \left\lfloor \frac{v^2}{4} \right\rfloor - v + 1 = \left\lfloor \frac{(v-2)^2}{4} \right\rfloor,$$

which is equivalent to (10.12). If v is even, start with the spanning cycle C_v in the balanced complete bipartite graph and add edges until the desired rank is reached. If v is odd, start with C_{v-1} and add the remaining vertex with two neighbours in the opposite bipartition class; this is 2-connected and has rank two. Add arbitrary missing cross-edges. These constructions realise every allowed $r \geq 2$. A 2-connected graph of rank one is a cycle, which is bipartite exactly when its order is even. $\square$

Theorem 10.9 (Exact canonical atom-count spectrum). *Fix nonnegative integers s, β satisfying either $\beta = 0$ and $s \geq 2$, or $\beta \geq 1$ and $s \geq 2 + q(\beta)$. Let F range over connected reduced obligatory systems with at least one hyperedge, shadow order s and cycle rank β . The possible values $k = a(F)$ are exactly*

$$\begin{aligned} \beta = 0 : & \quad k = s - 1, \\ \beta = 1 : & \quad 1 \leq k \leq s - 3, \quad k \equiv s + 1 \pmod{2}, \\ \beta \geq 2 : & \quad 1 \leq k \leq s - 1 - q(\beta). \end{aligned} \quad (10.13)$$

For $\beta \geq 1$, the maximum is

$$a_{\max}(s, \beta) = s - 1 - q(\beta). \quad (10.14)$$

Every maximiser has one minimum-order cyclic atom carrying the full rank β , and every other atom is one triple. The parity obstruction in the rank-one line is the only gap in any atom-count spectrum.

Proof. Let the atom cores be J_i , with $v_i = |V(J_i)|$, $e_i = |E(J_i)|$, and $r_i = e_i - v_i + 1$. Since a tree of k atoms makes $k - 1$ one-point identifications,

$$s = 1 + \sum_i (v_i - 1), \quad \beta = \sum_i r_i. \quad (10.15)$$

If $\beta = 0$, every atom is one triple and (10.15) gives $k = s - 1$. If $\beta = 1$, there is one cyclic atom, whose core is an even cycle of order $v \geq 4$; the remaining atoms are single triples, so $s = k + v - 1$. This is exactly the middle line of (10.13).

Suppose $\beta \geq 2$. For positive integers a, b , the inequality $\sqrt{a} + \sqrt{b} \geq \sqrt{a+b-1} + 1$ follows after squaring from $(a-1)(b-1) \geq 0$. Hence $q(a) + q(b) \geq q(a+b-1) + 2 \geq q(a+b) + 1$, where the second inequality uses that q increases by at most one between consecutive integers. Together with Lemma 10.8, repeated application gives

$$s \geq 1 + k + q(\beta),$$

so $k \leq s - 1 - q(\beta)$. Conversely, for every integer in the displayed interval choose one cyclic core of rank β and order $s - k + 1$, supplied by the lemma, and attach $k - 1$ single-triple atoms. Equality in (10.14) forces one positive-rank atom and minimum order, proving the rigidity statement. $\square$

The rigidity in the maximum concerns the distribution of cyclic rank: all positive rank lies in one minimum-order atom. It does not assert that the numerical parameters determine the graph-isomorphism type of that atom's core.

For $s \geq 3$, define

$$\alpha(s) = \begin{cases} s, & s \text{ even,} \\ s + 1, & s \text{ odd,} \end{cases} \quad \delta(s) = \left\lfloor \frac{(s-1)^2}{4} \right\rfloor + 1.$$

Theorem 10.10 (Indecomposability phase diagram). *Apart from the single triple $(|E(F)|, |V(F)|) = (1, 3)$, a connected reduced one-point-indecomposable obligatory system with shadow order s and m hyperedges exists if and only if*

$$s \geq 4, \quad \alpha(s) \leq m \leq \left\lfloor \frac{s^2}{4} \right\rfloor. \quad (10.16)$$

A connected reduced one-point-decomposable obligatory system exists if and only if

$$s \geq 3, \quad s - 1 \leq m \leq \delta(s). \quad (10.17)$$

Consequently, for $s \geq 3$, the feasible connected parameters split into three exact zones:

$$\begin{aligned} s - 1 \leq m < \alpha(s) &\implies \text{every system is decomposable,} \\ \alpha(s) \leq m \leq \delta(s) &\implies \text{both behaviours occur,} \\ \delta(s) < m \leq \lfloor s^2/4 \rfloor &\implies \text{every system is indecomposable.} \end{aligned} \quad (10.18)$$

For $m \geq 2$, the last zone is equivalently

$$s \leq \left\lceil 2\sqrt{m-1} \right\rceil. \quad (10.19)$$

Proof. By Corollary 5.4, the nontrivial indecomposable systems are J^+ with J finite, simple, bipartite and 2-connected. If $|V(J)| = s$ and $|E(J)| = m$, minimum degree two gives $m \geq s$; equality makes J a cycle and therefore forces s even. The upper bound is the usual $m \leq \lfloor s^2/4 \rfloor$. Even cycles, and for odd s a cycle on $s-1$ vertices plus one new vertex with two neighbours, give the lower endpoints; adding cross-edges fills the intervals.

A decomposable system admits a connected bipartite shadow with a cut vertex. If its bipartition sizes are $a+b=s$ and the cut vertex lies in the a -side, then deleting it gives components with sizes (a_i, b_i) and

$$m \leq b + \sum_i a_i b_i \leq b + (a-1)(b-1) = ab - a + 1.$$

The symmetric estimate gives $m \leq ab - \min(a, b) + 1 \leq \delta(s)$. The bound is sharp: take a connected bipartite graph on $s-1$ vertices with $m-1$ edges and adjoin one leaf. This also fills every value in (10.17). The three zones and the elementary inversion (10.19) follow. $\square$

Corollary 10.11 (Rigidity on the structural boundaries). *Let F be connected, reduced and obligatory, with shadow order s and $m = |E(F)|$ hyperedges.*

- (1) *If F is indecomposable, s is even and $m = s$, then $F \cong C_s^+$.*
- (2) *If F is indecomposable, s is odd and $m = s+1$, then $F \cong \Theta(2a, 2b, 2c)^+$ for positive $a \leq b \leq c$ with $a+b+c = (s+1)/2$.*
- (3) *If F is indecomposable and $m = \lfloor s^2/4 \rfloor$, then $F \cong K_{\lfloor s/2 \rfloor, \lfloor s/2 \rfloor}^+$.*
- (4) *If F is decomposable and $m = \delta(s)$, then F is a one-point amalgamation of $K_{a,b}^+$ and one triple, where $a+b = s-1$ and $|a-b| \leq 1$.*

Here $\Theta(p, q, r)$ is the theta graph with three internally disjoint paths of lengths p, q, r and common endpoints.

Proof. The first and third assertions are the equality cases of minimum degree and the bipartite extremal bound. In the second case $\sum_v (\deg v - 2) = 2$. A single degree-four vertex would be a cut vertex, so there are exactly two degree-three vertices and all others have degree two; the three resulting paths have the same parity. Their total length is $s+1$, which is even, so all three paths are even. For the fourth assertion when $s \geq 5$, choose the shadow by assembling the canonical atom cores. Its cyclic blocks are precisely the nontrivial canonical cores. Equality in the cut-vertex estimate forces a balanced complete bipartite graph on $s-1$ vertices plus one leaf, and hence the displayed complete-bipartite atom and one singleton atom. The tree cases $s = 3, 4$ are immediate. $\square$

Proposition 10.12 (Componentwise atom-count spectrum). *Let F be reduced and obligatory with at least one hyperedge and $c \geq 1$ connected components,*

shadow order $s = |V(F)| - |E(F)|$, cycle rank $\beta = 2|E(F)| - |V(F)| + c$, and k canonical atoms. For every feasible (s, β, c) , the possible values of k are exactly

$$\begin{aligned} \beta = 0 : & \quad k = s - c, \\ \beta = 1 : & \quad c \leq k \leq s - c - 2, \quad k \equiv s - c \pmod{2}, \\ \beta \geq 2 : & \quad c \leq k \leq s - c - q(\beta). \end{aligned} \tag{10.20}$$

The parameters themselves are feasible exactly when $s \geq 2c$ for $\beta = 0$ and $s \geq 2c + q(\beta)$ for $\beta \geq 1$.

Proof. If the atom cores are J_i , a forest with k atom-nodes and c components makes $k - c$ identifications. Hence

$$s = c + \sum_i (|V(J_i)| - 1), \quad \beta = \sum_i (|E(J_i)| - |V(J_i)| + 1).$$

The connected proof applies componentwise. Necessity gives the lower atom floor $k \geq c$ and the three upper formulas. For sufficiency, take $c - 1$ single-triple components and realise the corresponding connected atom count in the remaining component. The feasibility thresholds follow by concentrating all positive rank in one minimum-order cyclic atom. $\square$

Remark 10.13 (Blocks and the earlier reduction). Komjáth proved that a finite triple system is obligatory if and only if all of its 2-connected components are obligatory [15]. General hypergraph block decomposition and its relation to the incidence graph are classical; see Whitney [8], Diestel [9, Chapter 3], and, for the hypergraph formulation, Bahmanian and Šajna [26]. For the reduced 3-uniform systems considered here, the canonical atoms in Theorem 5.3 are precisely these maximal nonseparable edge-supported blocks. Thus the new content of the canonical atom theorem is not the existence of a block decomposition, but the exact identification of the block types that can occur in an obligatory triple system, together with the quantitative spectra derived from them.

Corollary 10.14 (Block formulation of the classification). *Let F be a finite triple system. After isolated points are deleted, F is obligatory if and only if every nonempty block of F° is either one triple or J^+ for a finite 2-connected simple bipartite graph J .*

Proof. If $E(F^\circ) = \emptyset$, then F° is empty and the assertion follows from Lemma 1.1. Otherwise suppose first that F is obligatory. Lemma 1.1 gives that F° is obligatory, and Komjáth's reduction gives obligatoriness of each of its blocks. A block is one-point indecomposable, so Corollary 5.4 gives exactly the displayed list.

Conversely, each displayed block is obligatory by the positive expansion result (the one-triple case included). Komjáth's reduction therefore makes F° obligatory, and Lemma 1.1 restores the isolated points. $\square$

Lemma 10.15 (Local product for one-point decompositions). *Let F be reduced and obligatory with at least one hyperedge. Let $\mathcal{A}(F)$ be its canonical atoms, let*

$S(F)$ be the points belonging to at least two atoms, and put

$$\mu(p) = |\{A \in \mathcal{A}(F) : p \in V(A)\}|.$$

Let $\mathcal{D}(F)$ be the set of partitions of $E(F)$ into nonempty connected supported pieces such that two pieces meet in at most one point and the piece-shared-point incidence graph is a forest, ordered by refinement. Then $\mathcal{D}(F)$ is a lattice and

$$\mathcal{D}(F) \cong \prod_{p \in S(F)} \Pi_{\mu(p)}, \quad (\dagger)$$

where Π_r denotes the partition lattice of an r -element set. If F has k canonical atoms and c connected components, then

$$\text{rk } \mathcal{D}(F) = k - c = \sum_{p \in S(F)} (\mu(p) - 1). \quad (\ddagger)$$

The empty product in $(\dagger)$ is the one-element lattice.

Proof. A supported one-point forest decomposition cannot split a block: if two hyperedges of one block lay in different pieces, the unique path between those pieces in the piece-shared-point forest would contain a point whose deletion separates the two hyperedges, contradicting nonseparability of the block. Thus every element of $\mathcal{D}(F)$ is a coarsening of the canonical atom decomposition.

For a shared point p , restrict such a coarsening to the set of the $\mu(p)$ atoms through p ; this gives an element of $\Pi_{\mu(p)}$. Conversely, choose independently a partition of the atoms through every shared point and take the equivalence relation on all atoms generated by these local identifications. To track its classes without contracting overlapping subtrees, replace the star at each shared point p by the following two-level tree. Retain a central node p , insert one node (p, B) for each block B of the prescribed local partition, join p to (p, B) , and join (p, B) to each atom in B . Replacing a star of a forest by a tree with the same attachment nodes preserves acyclicity, so the resulting auxiliary graph is a forest.

Delete the edges from the central nodes p to their nodes (p, B) . The components containing atoms are vertex-disjoint and correspond exactly to the generated equivalence classes; their atom unions are connected supported pieces. Two different blocks at one p cannot lie in the same component, since a path between them, together with their two edges to p , would give a cycle in the auxiliary graph. Contract these components in the full auxiliary graph, with the deleted edges restored, while retaining the central nodes p . There are therefore no repeated incidences between a piece and p . Delete each degree-one central node, which is now internal to one piece. The resulting graph is precisely the piece-shared-point incidence graph and is still a forest. In particular, two pieces cannot meet at two shared points. Restriction recovers the prescribed local partitions. Conversely, a connected supported piece is recovered from its local identifications, since its atom nodes are connected through the shared points it contains. Thus the constructions are inverse and preserve refinement, proving $(\ddagger)$. For the related theory of connected set partitions and bond lattices,

see Simon et al. [29] and Stanley [30, Lecture 2]. The factors here are full partition lattices on the atoms through each shared point, not the bond lattice of the incidence forest itself. Binary joints contribute two-element-chain factors, whereas three atoms through one point already give the diamond $\Pi_3 \cong M_3$, the familiar obstruction to distributivity [31, Chapter II, Theorem 101].

Finally the canonical incidence forest has $k + |S(F)|$ vertices, c components and $\sum_p \mu(p)$ edges. Hence

$$\sum_p (\mu(p) - 1) = k - c,$$

which is also the rank of the product lattice by the standard rank formula for partition lattices and their products [10, Chapter 3]. $\square$

Lemma 10.16 (Capacity-safe realization of every attachment profile). *Fix pairwise disjoint canonical atoms $A_1, \dots, A_k$ and an integer $1 \leq c \leq k$. For every integer partition*

$$\lambda = (\lambda_1, \dots, \lambda_t) \vdash k - c$$

(with the empty partition allowed when $k = c$), there is a forest assembly of the A_i into exactly c connected components whose shared-point excess profile is λ ; that is, its nontrivial shared points have multiplicities $\lambda_i + 1$. Every atom is incident with at most two distinct shared points. Consequently, for a fixed atom list and fixed c , all these assemblies have the same order, size, shadow order and cycle rank.

Proof. Put $N = k - c$. If $N = 0$, take the $k = c$ atoms as separate components. Assume $N > 0$. Use $N + 1$ atoms in one nontrivial component and leave the remaining $c - 1$ atoms as singleton components.

If $t = 1$, identify one chosen vertex from each of the $N + 1 = \lambda_1 + 1$ active atoms to one common shared point. If $t \geq 2$, arrange shared points $p_1, \dots, p_t$ in a chain. Use $t - 1$ atoms as connectors, the i th connector meeting p_i and p_{i+1} . The two endpoint shared points already have one incident connector and the internal shared points have two. Supply the remaining incidences with leaf atoms so that $\deg(p_i) = \lambda_i + 1$. The number of required leaf atoms is

$$\lambda_1 + \lambda_t + \sum_{i=2}^{t-1} (\lambda_i - 1) = N - t + 2,$$

so together with the $t - 1$ connectors this uses exactly $N + 1$ atoms. Every connector atom uses two distinct vertices and every leaf atom one; canonical atoms have at least three vertices, so the identifications can be made with distinct ports inside each atom. The atom–shared-point incidence graph is a tree and has the required multiplicities.

Disjoint union and one-point amalgamation preserve obligatoriness by Lemma 4.1 and Proposition 4.4. Each shared point separates the incident atom branches in this forest assembly. A nonseparable supported block therefore cannot contain edges from two different atoms: the intervening

shared point would separate those edges. Each supplied atom is itself one-point indecomposable, so its image is a maximal nonseparable supported block. By Theorem 5.3, these images are precisely the canonical atoms of the assembled system. Thus the constructed profile is the canonical attachment profile, not merely an auxiliary incidence pattern.

Any forest assembly of the fixed k atoms into c components makes exactly $k - c$ vertex identifications. Thus

$$|E(F)| = \sum_i |E(A_i)|, \quad |V(F)| = \sum_i |V(A_i)| - (k - c),$$

so the stated global parameters do not depend on the chosen profile. $\square$

Corollary 10.17 (Exact one-point factorization-lattice spectrum). *Fix a feasible triple (s, β, c) for reduced obligatory systems with at least one hyperedge, where*

$$s = |V(F)| - |E(F)|, \quad \beta = 2|E(F)| - |V(F)| + c, \quad q(r) = \lceil 2\sqrt{r} \rceil.$$

As F ranges over all such systems with these fixed parameters, put k for its number of canonical atoms and $N = k - c$. The possible values of N are exactly

$$\begin{aligned} \beta = 0 : & \quad N = s - 2c, \\ \beta = 1 : & \quad 0 \leq N \leq s - 2c - 2, \quad N \equiv s - 2c \pmod{2}, \\ \beta \geq 2 : & \quad 0 \leq N \leq s - 2c - q(\beta). \end{aligned} \quad (\star)$$

For every allowed N , the possible lattice isomorphism types of $\mathcal{D}(F)$ are precisely

$$\prod_{i=1}^t \Pi_{\lambda_i+1}, \quad \lambda = (\lambda_1, \dots, \lambda_t) \vdash N, \quad (\star\star)$$

where for $N = 0$ the empty partition gives the one-element lattice. Every lattice in $(\star\star)$ occurs, and different partitions λ give nonisomorphic lattices.

Proof. The rank spectrum $(\star)$ is Proposition 10.12 after subtracting c from the atom count k . Fix an allowed N and choose one system with $k = N + c$ atoms and the prescribed global parameters. By Lemma 10.16, its fixed atom list can be reassembled with the same c and the same global parameters to realize every partition $\lambda \vdash N$. Lemma 10.15 then gives exactly the product in $(\star\star)$. Conversely every system has an excess profile partitioning N , so there are no other types.

Finally, the standard characteristic polynomial of the partition lattice [10, Chapter 3] is

$$\chi_{\Pi_r}(x) = \prod_{j=1}^{r-1} (x - j),$$

and characteristic polynomials multiply under direct products, by the product rule for Möbius functions [10, 32]. For the equivalent braid-arrangement

viewpoint, see Stanley [30, Lectures 1–2]. Hence the multiplicity of the root j in the characteristic polynomial of the lattice in $(\star\star)$ is

$$|\{i : \lambda_i \geq j\}|.$$

These multiplicities recover the partition λ , so distinct partitions give nonisomorphic lattices. $\square$

Uniqueness of indecomposable direct factors also has a classical incidence-poset formulation; compare Schmitt [33, Lemma 6.1]. The argument here recovers the attachment profile directly from characteristic-polynomial root multiplicities.

The numerical and lattice spectra record different parts of the same decomposition. The shadow order, cyclic rank and component count restrict the possible excess $N = k - c$. For each allowed N , the attachment profile $\lambda \vdash N$ determines the product of partition lattices, and every such profile is realizable. These assertions classify the available factorization-lattice types; they do not identify all systems with the same numerical parameters.

Cyclic costs and assembly freedom. The atom-count bound and the local product measure complementary constraints. The first records how much cyclic structure costs; the second records the decomposition choices left after the atoms have been fixed.

Proposition 10.18 (Exact cyclic-cost budget). *Let F be connected, reduced and obligatory, with positive Levi cycle rank β . Put $s = |V(F)| - |E(F)|$, let k be its canonical atom count, and set $N = k - 1$. For the positive-rank atoms, write r_i for the core rank, v_i for the core order, and h for their number. Define*

$$C = \sum_{r_i > 0} q(r_i) - q(\beta), \quad L = \sum_{r_i > 0} (v_i - 2 - q(r_i)).$$

Then

$$N + C + L = s - 2 - q(\beta), \quad C \geq h - 1, \quad L \geq 0.$$

In particular,

$$N + (h - 1) + L \leq s - 2 - q(\beta).$$

The cost $C + L$ is zero exactly when the atom count is maximal: there is one minimum-order cyclic atom carrying rank β , and all other atoms are single triples.

Proof. Zero-rank atoms have core order two. Equation (10.15) therefore gives

$$s = 1 + k + \sum_{r_i > 0} q(r_i) + L, \quad \sum_{r_i > 0} r_i = \beta.$$

Substituting the definitions of N and C gives the identity. Lemma 10.8 gives $L \geq 0$. Since $\beta > 0$, there is at least one positive-rank atom. The strict aggregation inequality in the proof of Theorem 10.9 gives $C \geq h - 1$. Thus zero cost forces $h = 1$ and zero core-order slack; the converse follows from the same identity. $\square$

Since $N = \text{rk } \mathcal{D}(F)$, the budget also describes the rank available to the factorization lattice. Splitting cyclic rank among atoms and increasing their core orders reduce this rank. It is a numerical constraint, not a graph-edit-distance stability theorem or a uniqueness statement about the cores.

We use the usual second-kind Stirling numbers: they count partitions into nonempty blocks, with $S(0, 0) = 1$ [11, Section 5.1].

Corollary 10.19 (Sharp counts at a prescribed piece number). *Let F be reduced and obligatory with at least one hyperedge, k canonical atoms and c connected components. Put $N = k - c$ and let $\lambda = (\lambda_1, \dots, \lambda_t) \vdash N$ be its attachment excess profile. For $0 \leq j \leq N$, define*

$$d_j(F) = |\{D \in \mathcal{D}(F) : |D| = k - j\}|,$$

where $|D|$ counts the actual nonempty original-edge pieces. Write $S(a, b)$ for the number of partitions of an a -element set into b blocks. Then

$$\sum_{j=0}^N d_j(F) z^j = \prod_{i=1}^t \left(\sum_{r=0}^{\lambda_i} S(\lambda_i + 1, \lambda_i + 1 - r) z^r \right),$$

and

$$\binom{N}{j} \leq d_j(F) \leq S(N + 1, N + 1 - j) \quad (0 \leq j \leq N).$$

If $N \geq 2$, then for every $1 \leq j \leq N - 1$ the lower equality holds exactly when all shared points are binary, that is, $\lambda = (1, \dots, 1)$; the upper equality holds exactly when $\lambda = (N)$. Each bound is attained at every j by its corresponding extremal profile, even with the atom list and component count fixed. For $N = 0$ the generating polynomial is 1, and for $N = 1$ the two extremal profiles coincide.

Proof. Let π_i be the local partition at the i th shared point. In the auxiliary forest from Lemma 10.15, delete the central point-nodes and their incident edges. The remaining forest has $k + \sum_i |\pi_i|$ vertices and $\sum_i (\lambda_i + 1)$ edges. Every component contains an atom and corresponds to one actual piece. Hence

$$|D| = k - \sum_i (\lambda_i + 1 - |\pi_i|).$$

Independent local choices give the generating polynomial. This is the standard partition-lattice enumeration [10, Example 3.10.4]; connected-partition polynomials and their articulation factorization are discussed in Simon et al. [29, Sections 3–5].

For the lower bound, give the i th local set one anchor and λ_i ordinary elements. A j -element subset of the total N ordinary elements defines local partitions by joining its elements to their respective anchors and leaving all others singleton. This is an injection into the choices counted by $d_j(F)$.

For the upper bound, identify all local anchors to one anchor on a set of $N + 1$ elements. Fuse the anchor blocks of the local partitions and retain every other block. Restriction to each anchored local set recovers its partition, so the

map is injective. A tuple of total excess j gives $N + 1 - j$ blocks, proving the upper bound. This construction also shows that merging two positive profile parts never decreases any coefficient.

For strictness in the lower bound, suppose some $\lambda_i \geq 2$. Form an anchor-free pair in that group and choose $j - 1$ of the remaining $N - 2$ ordinary elements to join their local anchors. This tuple has excess j but is outside the lower image. For strictness in the upper bound, suppose there are at least two profile groups. Choose $j + 1$ ordinary elements meeting two groups, put them in one anchor-free block and leave all other elements singleton. This partition has $N + 1 - j$ blocks but is outside the upper image. Both choices are possible for $1 \leq j \leq N - 1$. The two extremal profiles attain the respective bounds. Finally, Lemma 10.16 realizes both profiles using the same atoms and component count, with all global parameters preserved. $\square$

For example, take C_4^+ and two single triples. A chain assembly with two binary joints has decomposition polynomial $1 + 2z + z^2$; an assembly with both triples attached at one point of C_4^+ has polynomial $1 + 3z + z^2$. Both systems have twelve vertices, six hyperedges, three canonical atoms and cycle rank one, and both maximize the atom count for these parameters. Their four and five decomposition choices differ only because of attachment geometry. Thus extremal cyclic concentration does not eliminate assembly freedom. The counting inequalities above are elementary partition-lattice consequences; their role here is to quantify that freedom, not to assert a new general enumeration theorem. The individual factors are reversed rows of second-kind Stirling numbers, whose generating-function and concavity properties are classical [34]. The bounds above instead compare different attachment profiles at fixed total excess. Pitman's probabilistic representations and coefficient bounds give further context for these classical Stirling rows [35]. Our question is which products of the rows are realized by attachment profiles, not a new general identity for Stirling numbers.

11. THE CLASSIFICATION IN EVERY FINITE UNIFORMITY

The same structural criterion extends to every finite uniformity, with the number of required bridge incidences increasing with the edge size. The positive expansions are supplied by Reiher's theorem for all uniformities [4, Theorem 1.2]. For the negative direction, we iterate the one-apex construction from Section 6 and Li [3, Sections 3–4], allowing a hypergraph of arbitrary finite uniformity at its base. The decisive point is that a finite linear trace projects to finite *linear* base systems at every stage of the iteration.

Fix an integer $r \geq 3$. All r -uniform hypergraphs below are simple, and embeddings are injective and non-induced. A proper vertex colouring has no monochromatic edge. For a finite graph J , let $J^{[r]}$ be its r -uniform private expansion: add $r - 2$ new vertices to each graph edge, with all these added vertices distinct. Thus $J^{[3]} = J^+$. Let $\mathcal{B}_r$ be the class generated, up to isomorphism, by

$J^{[r]}$ for finite bipartite graphs J and by finite edgeless systems, under finite disjoint unions and one-point amalgamations. The Levi graph, Berge cycles and isolated reduction have the same definitions as before, with each hyperedge-node now of degree r .

Theorem 11.1 (Classification in finite uniformity). *For every integer $r \geq 3$ and every finite r -uniform hypergraph F , the following are equivalent:*

- (1) F occurs in every r -uniform hypergraph of uncountable chromatic number;
- (2) $F \in \mathcal{B}_r$;
- (3) F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with at least $r - 2$ bridges, and every Berge cycle of F° has even length.

For $r = 3$ this is Theorem A. At $r = 2$, the bridge requirement becomes vacuous and the cycle condition is the classical bipartite characterization [13]. We first prove the trace statement needed for the higher-uniformity negative direction.

A lift with a hypergraph base. If D is a k -uniform hypergraph, write D^+ for its $(k + 1)$ -uniform *one-private-vertex inflation*: each edge receives one new vertex, used by no other edge. On graphs this agrees with the expansion notation used earlier, and $(J^{[k]})^+ \cong J^{[k+1]}$.

Let $H = (X, A)$ be a k -uniform hypergraph, where $k \geq 2$ and $A = E(H)$ is regarded as an alphabet. Put $T = A^{<\omega_1}$ and write $\ell(s) = \text{dom}(s)$ for the ordinal length of $s \in T$. Define $\mathcal{L}(H)$ on $T \times X$ by including exactly the edges

$$\{(s, x) : x \in a\} \cup \{(t, z)\}, \quad a \in A, \quad s \frown a \preceq t, \quad z \in X, \quad (1)$$

where $\preceq$ denotes the prefix order. The k vertices at s are the base and the vertex at t is the apex. Since t properly extends s , these are $k + 1$ distinct vertices. The node occurring $k \geq 2$ times uniquely identifies the base, so each edge has unique defining data.

Lemma 11.2 (Chromatic lower bound for the rank-raising lift). *If $\chi(H) > \aleph_0$, then $\chi(\mathcal{L}(H)) > \aleph_0$.*

Proof. Suppose that $c : T \times X \rightarrow \omega$ is proper. At each $s \in T$, its restriction to $\{s\} \times X$ has a monochromatic H -edge a_s , of colour q_s . Recursively set $s_0 = \emptyset$ and $s_{\alpha+1} = s_\alpha \frown a_{s_\alpha}$, taking unions at countable limit ordinals. For $\alpha < \beta < \omega_1$, the node s_β extends $s_\alpha \frown a_{s_\alpha}$. If $q_{s_\alpha} = q_{s_\beta}$, any vertex of the monochromatic base edge at s_β can serve as an apex for the base edge at s_α , producing a monochromatic edge of $\mathcal{L}(H)$. Thus the colours q_{s_α} are pairwise distinct for $\alpha < \omega_1$, a contradiction. $\square$

Lemma 11.3 (Finite linear traces over a hypergraph base). *Let $k \geq 2$ and let H be a k -uniform hypergraph. Every finite linear subhypergraph $K \subseteq \mathcal{L}(H)$ is obtained by disjoint unions and one-point amalgamations from systems D^+ , where D is a finite linear subhypergraph of H , and finite edgeless systems. Each such D has at most $|E(K)|$ edges.*

Proof. Set aside the isolated vertices of K . For each node s supporting at least one selected edge, let K_s consist of the edges of K based at s and all their incident vertices.

The fibre pieces. No two distinct edges of K_s can use the same base $a \in E(H)$: they would share $k \geq 2$ vertices, contrary to linearity. Let $D_s \subseteq H$ consist of the bases used at s , with their incident vertices. Two of these bases meet in at most one vertex, because their corresponding K -edges do. Thus D_s is linear. Moreover, distinct bases have distinct apices. If a point (t, z) were an apex for both a and a' , then the coordinate of t immediately following s would equal both a and a' . An apex cannot be a core vertex of K_s , since its node properly extends s . Sending each base vertex x to (s, x) and its private inflation vertex to the corresponding apex therefore gives $D_s^+ \cong K_s$. The number of edges of D_s is exactly the number of edges of K_s .

Intersections of fibres. If (t, z) belongs to both K_s and K_u , then s, u are prefixes of t and hence are comparable. Suppose $s \subsetneq u$. A common point is necessarily an apex of K_s , and the base of its edge is the coordinate $u(\ell(s))$. All common points would consequently be apices for this one base. There is at most one selected edge with that base and one apex of that edge. Hence distinct fibres meet in at most one point.

The support forest. Let Q be the bipartite graph between active base nodes and points belonging to at least two fibres, with adjacency given by membership. Suppose Q contains a cycle and choose on it a base node s of minimum length. The two adjacent base nodes along the cycle are proper descendants of s . Their two shared points with K_s are distinct apices, so they come from distinct bases a, a' . The adjacent nodes therefore lie in different immediate branches below s .

On the remaining path of the cycle, consecutive base nodes are comparable. Every node on that path has length at least $\ell(s)$ and is different from s . A node with these properties that is comparable with a proper descendant of s is again a proper descendant of s : it either extends that descendant or is its prefix of length greater than $\ell(s)$. Comparable proper descendants have the same coordinate immediately after s . Induction along the path therefore keeps all its nodes in a single branch below s , contradicting the two different branches at its endpoints. Thus Q is a forest.

Assembly. Root each component of Q at a base node and order its base nodes by nondecreasing distance from the root. For a nonroot node s , let p_s be its parent point. If an earlier fibre K_u meets K_s at p , then $s - p - u$ is a path in Q . The point p cannot be a child of s , since every other base node adjacent to that child has greater depth. Therefore $p = p_s$. Each new fibre meets the preceding union in at most this one point, so the ordered fibres assemble by one-point amalgamations. Different components and the removed isolated vertices are combined by disjoint union. The natural maps into K identify exactly these common points; their union is an isomorphism onto K . $\square$

The restriction to linear traces is essential here. It is not imposed on the base host H , or on $\mathcal{L}(H)$ itself.

Lemma 11.4 (Inflation and assembly). *One-private-vertex inflation commutes, up to isomorphism, with finite disjoint unions and one-point amalgamations. In particular, if D is assembled from systems $J_i^{[k]}$, then D^+ is assembled from the corresponding systems $J_i^{[k+1]}$ in the same way.*

Proof. Index each new private vertex by its original edge. The edge sets of one-point factors are disjoint because their edges have size at least two. After the factors are inflated, identify their selected old vertices and make no identification of new private vertices. This gives exactly the inflation of their amalgamation. The argument applies to any selected old vertex, including one that was private in an earlier expansion. Disjoint unions are immediate, and inflation leaves edgeless systems unchanged. $\square$

For a graph G , define the iterated hosts

$$H_2(G) = G, \quad H_j(G) = \mathcal{L}(H_{j-1}(G)) \quad (j \geq 3).$$

Proposition 11.5 (Iterated finite traces). *For every $r \geq 2$, each finite linear subhypergraph $K \subseteq H_r(G)$ is assembled by disjoint unions and one-point amalgamations from $J^{[r]}$, with J a finite subgraph of G , and finite edgeless systems. Every graph J used in this construction has at most $|E(K)|$ edges. If $\chi(G) > \aleph_0$, then $\chi(H_r(G)) > \aleph_0$.*

Proof. For $r = 2$, take the finite graph K itself, with the convention $J^{[2]} = J$. At the next stage, Lemma 11.3 expresses K as a forest assembly of D_s^+ , where each D_s is a finite linear trace in $H_{r-1}(G)$. Apply induction to D_s and then Lemma 11.4. This changes every constituent $J^{[r-1]}$ into $J^{[r]}$ and preserves the assembly. Since $|E(D_s)| = |E(K_s)| \leq |E(K)|$, the asserted bound on constituent edge counts also follows by induction. The chromatic assertion is repeated application of Lemma 11.2. $\square$

Reconstruction and obligatory systems.

Lemma 11.6 (Intrinsic reconstruction). *A finite reduced r -uniform hypergraph belongs to $\mathcal{B}_r$ if and only if it is linear, every hyperedge-node has at least $r - 2$ incident Levi bridges, and every Berge cycle is even.*

Proof. For $J^{[r]}$ with J bipartite, the $r - 2$ private incidences at each hyperedge are bridges. Its Berge cycles are exactly the graph cycles of J . One-point amalgamation preserves linearity and bridges, and every Levi cycle of a one-point sum is contained in one factor. These observations prove preservation of all three conditions.

Conversely, let $L = I(F)$ and delete all bridges. At each hyperedge-node there are at most two surviving incidences. There cannot be exactly one, since any nonbridge incidence lies on a cycle, which uses a second nonbridge incidence at that node. Thus the surviving degree is zero or two. In a nontrivial remaining

component C , suppress the hyperedge-nodes to obtain a graph J_C on its point-nodes. Linearity excludes parallel edges, and its cycles correspond to Berge cycles in F , so J_C is simple and bipartite.

Every hyperedge-node in C has $r-2$ further point-neighbours, across bridges. Each lies outside C , since a path inside C would otherwise give an alternative route around that bridge. No such neighbour can belong to two different hyperedges represented in C : an internal path between those hyperedge-nodes and their two incidences to the common point would again put bridges on a cycle. The hyperedges represented in C , with all their vertices, therefore form exactly $J_C^{[r]}$. A hyperedge-node of surviving degree zero gives a single r -edge, which is $K_2^{[r]}$.

Contract the bridge-deleted components. The bridges give a simple forest T : a loop, parallel pair or quotient cycle would provide an alternative route around an original bridge. A component is active if it contains a hyperedge-node; the inactive components are singleton point-nodes. The pieces just constructed partition the original hyperedge set and cover all its vertices.

For a point p , let $X(p)$ be its bridge-deleted component. The pieces containing p are the piece at $X(p)$ when this component is active, together with active neighbours whose connecting bridge has point endpoint p . Root each tree at an active component and order active components by depth. If the piece at C shares p with an earlier piece, then either $X(p) = C$, in which case the earlier neighbour is its parent, or $X(p)$ is the parent of C . It cannot be a child, whose other neighbouring active components all have greater depth. In both cases p is the point endpoint of the unique parent edge of C . Consequently each piece meets the preceding union in at most one point. The running-intersection assembly from Proposition 5.2 now applies without change. If F has no edges, reducedness makes it empty, which is an edgeless generator. $\square$

Proof of Theorem 11.1. Isolated vertices can be deleted and restored exactly as in Lemma 1.1. The same structural induction proves that this reduction preserves membership in $\mathcal{B}_r$. We may therefore assume F is reduced; the empty case is immediate. Lemma 11.6 establishes $(2) \iff (3)$.

For $(2) \implies (1)$, Reiher's theorem makes every $J^{[r]}$ with J finite bipartite obligatory [4, Theorem 1.2]. The disjoint-union proof of Lemma 4.1 applies in any finite uniformity. So do the rooted-abundance proof and the one-point amalgamation proof in Lemma 4.3 and Proposition 4.4: the bound $(m-1)(|V(F)|-1)$ on a hitting set for off-root copies does not depend on uniformity, and the auxiliary bounded-outdegree graph is unchanged. Thus obligatoriness is preserved by the operations defining $\mathcal{B}_r$.

For $(1) \implies (2)$, a nonlinear finite r -uniform hypergraph is not obligatory by the Erdős–Hajnal–Rothschild nonlinearity obstruction [14, Theorem 2]; its all-uniformity scope is also recorded in Reiher [4, Section 1]. It remains to treat linear $F \notin \mathcal{B}_r$. Put $m = |E(F)| \geq 1$. By Theorem 7.1 choose a graph G of uncountable chromatic number with no odd cycle of length at most m .

Proposition 11.5 makes $H_r(G)$ uncountably chromatic. If F embedded in this host, retain exactly the host hyperedges corresponding to its image. They form a finite linear trace, which the proposition assembles from graph expansions $J^{[r]}$ with $J \subseteq G$ and $|E(J)| \leq m$. Each such J is bipartite: an odd cycle in it would have length at most $|E(J)| \leq m$, contrary to the choice of G . This would give $F \in \mathcal{B}_r$, a contradiction. Hence $H_r(G)$ is an uncountably chromatic F -free host. $\square$

The avoidance host depends on the finite edge bound m . No claim is made that one uncountably chromatic host has only obligatory finite subgraphs; the trace assertion used above is restricted to linear subhypergraphs of the required finite size.

Corollary 11.7 (Canonical atoms in finite uniformity). *A finite reduced obligatory r -uniform hypergraph has a canonical partition of its hyperedges into single r -edges and systems $J^{[r]}$ with J finite, simple, bipartite and 2-connected. Distinct atoms meet in at most one point, and their incidence graph with shared points is a forest. These atoms are precisely its one-point indecomposable edge-supported blocks.*

Proof. Every hyperedge-node has at most two nonbridge incidences. In each cyclic block of the Levi graph it therefore has degree two, and it cannot belong to two different cyclic blocks. Suppressing the hyperedge-nodes gives a simple 2-connected bipartite graph J . Its other $r - 2$ incidences per edge are bridges, with endpoints outside the block and pairwise distinct between its edges, by the same alternative-path argument as in Lemma 11.6. Thus the edges of this block form $J^{[r]}$. A hyperedge-node in no cyclic block has only bridge incidences and gives a single-edge atom.

For completeness, the atom forest can be read directly from the block-cut forest of the Levi graph. Assign each bridge block to the atom containing its hyperedge endpoint. A cyclic-block atom consists of that block, its hyperedge-node cut vertices and the bridge blocks assigned to them; a single-edge atom consists of its hyperedge-node and its incident bridge blocks. Each such set is connected in the block-cut forest, and the sets for different atoms are disjoint. Contract these connected sets. The remaining nodes are point-nodes; discard those incident with only one atom. The resulting graph is exactly the atom-shared-point incidence graph and is still a forest. In particular two atoms cannot share two points. This construction also proves canonicity. Finally, a component containing several atoms has a leaf atom that separates at its unique shared point. Within an expansion of a 2-connected graph, deleting a point-node from the Levi graph cannot separate its hyperedge-nodes into two nonempty classes: a private point is a leaf, and deleting a core point leaves the core connected. This gives the claimed indecomposable blocks. $\square$

Remark 11.8 (The increased bridge requirement is strict). Consider the 4-uniform hypergraph with five edges

$$\begin{aligned} e &= \{a_1, a_2, a_3, p\}, & f &= \{b_1, b_2, b_3, q\}, \\ g_i &= \{a_i, b_i, c_i, d_i\} & (i &= 1, 2, 3), \end{aligned}$$

where all displayed vertices are distinct. It is linear. After its point leaves are removed, its Levi graph consists of three internally disjoint paths $e-a_i-g_i-b_i-f$, each of length four. Every simple cycle uses two of these paths and hence has Levi length eight, or Berge length four. Each edge has a bridge incidence, but e and f have exactly one, to p and q respectively. The requirement of two bridge incidences in Theorem 11.1 therefore excludes this system. In particular, retaining the triple-system requirement of only one bridge would give a false higher-uniformity characterization.

Finite structure in higher uniformity. The classification in Theorem 11.1 also transports the finite structure of Section 10. Fix an integer $r \geq 3$. As in the triple case, all systems in this subsection are finite, reduced, obligatory and have at least one edge. Their canonical atoms are single r -edges and the expansions $J^{[r]}$ of finite 2-connected simple bipartite graphs (Corollary 11.7). Put

$$n = |V(F)|, \quad m = |E(F)|, \quad s = n - (r-2)m, \quad \beta = (r-1)m - n + c,$$

where c is the number of connected components. Thus β is the cycle rank of the Levi graph, and s is the order of a bipartite shadow.

Corollary 11.9 (Finite spectra at every uniformity). *Let $m \geq 1$, $1 \leq c \leq m$, and n be integers. A reduced obligatory r -uniform hypergraph with n vertices, m edges and c components exists if and only if*

$$(r-2)m + 2(c-1) + \left\lceil 2\sqrt{m-c+1} \right\rceil \leq n \leq (r-1)m + c.$$

Every integer in this interval occurs.

With the definitions of s, β, c above, the canonical atom-count spectra and their feasibility conditions in Theorem 10.9 and Proposition 10.12 are unchanged, retaining the connectedness hypothesis of the former. For connected systems, the indecomposability phase diagram and its boundary descriptions in Theorem 10.10 and Corollary 10.11 are likewise unchanged when J^+ is replaced by $J^{[r]}$ and a single triple by a single r -edge; the singleton exception is now $(m, n) = (1, r)$.

Let $\mathcal{D}_r(F)$ denote the partitions of the original edge set into nonempty connected supported pieces that pairwise meet in at most one point and whose piece-shared-point incidence graph is a forest, ordered by refinement. If k is the number of canonical atoms and $\mu(p)$ is their multiplicity at a shared point, then

$$\mathcal{D}_r(F) \cong \prod_p \Pi_{\mu(p)}, \quad \text{rk } \mathcal{D}_r(F) = k - c.$$

The exact lattice spectrum of Corollary 10.17, the cyclic budget of Proposition 10.18, and the piece-count bounds and equality cases of Corollary 10.19

hold with the same formulas and their stated hypotheses. Every integer partition $\lambda \vdash k - c$ is still realizable as an attachment excess profile with the actual atom list and component count fixed.

Proof. For a singleton atom take its core to be K_2 . Write v_i, e_i for the orders and sizes of all canonical cores. The atom-point forest makes exactly $k - c$ vertex identifications, so

$$m = \sum_i e_i, \quad n = \sum_i (v_i + (r - 2)e_i) - (k - c).$$

Consequently

$$s = c + \sum_i (v_i - 1), \quad \beta = \sum_i (e_i - v_i + 1).$$

Separately within each of the c original components, take one-point sums of the ordinary cores, with bipartitions interchanged as needed. Their disjoint union is a simple bipartite shadow with s vertices, m edges and c components. Conversely the expansion $J^{[r]}$ of any such shadow has exactly these parameters. The ordinary bipartite-graph calculation in the proof of Theorem 10.3 therefore gives the displayed interval and every integer realization.

The two additive identities are precisely those used for the atom-count spectra and the cyclic budget. Their bounds, the core constructions at each allowed rank and order, and the one-point assemblies that attain them remain available. One-point indecomposability still means having one canonical atom. The same minimum-degree and cut-vertex arguments on the ordinary cores thus give the phase diagram and boundary cases.

For the lattice statements, a supported forest decomposition cannot split a canonical atom. The local star replacement and disjoint component contractions in the proof of Lemma 10.15 apply to the canonical atom-point forest without reference to edge size. They give the displayed product and its rank. To realize a prescribed excess profile, use the chain-and-leaves construction of Lemma 10.16. Each connector uses two distinct vertices, whereas every supplied atom has at least $r \geq 3$ vertices. The resulting shared points separate atom branches, so the supplied atoms remain the actual canonical atoms of the assembly. This construction preserves the displayed additive counts. The profile recovery and the counting injections consequently have exactly the same domains, ranks and actual piece numbers as before, proving all remaining assertions. $\square$

12. EXACT CHROMATIC SPECTRA AND SIMULTANEOUS AVOIDANCE

We now prescribe the chromatic number of the avoiding hosts. The sequence lift treats linear forbidden systems, while a labelled reservoir construction treats nonlinear ones. Together they give the full uncountable avoidance spectrum and explain the distinction between individual and simultaneous avoidance. For a family $\mathcal{F}$ of finite hypergraphs, a host is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$.

Exact chromatic number of the sequence lift. For an infinite cardinal κ , let $\mathcal{L}_\kappa(H)$ denote the construction in (1), with $T = E(H)^{<\kappa}$ in place of $E(H)^{<\omega_1}$. Thus an edge has a k -edge as its base at s and one apex at a node $t \succeq s \frown a$. The rank-raising lift admits an exact calibration, including at singular heights.

Lemma 12.1 (Sharp chromatic cap). *Let κ be an infinite cardinal and let H be a k -uniform hypergraph with nonempty vertex set, where $k \geq 2$ is finite. Then*

$$\chi(\mathcal{L}_\kappa(H)) = \min\{\chi(H), \kappa\}.$$

Proof. A proper colouring of H induces a proper colouring of the lift by giving (s, x) the colour of x : the base of every lifted edge is already nonmonochromatic. Independently, colouring (s, x) by its ordinal length $\ell(s) < \kappa$ is proper, because an apex has strictly greater length than its base. These give both upper bounds. If H is edgeless, both sides of the claimed equality are one.

For the lower bound, suppose that $\theta < \chi(H)$ and $\theta < \kappa$, and consider a colouring $c : E(H)^{<\kappa} \times V(H) \rightarrow \theta$. At every node s , its restriction to $\{s\} \times V(H)$ has a monochromatic H -edge a_s , of some colour b_s . Starting with $s_0 = \emptyset$, recursively put $s_{\alpha+1} = s_\alpha \frown a_{s_\alpha}$ and take unions at limit ordinals. The node s_α has length $\alpha < \kappa$, so every stage needed here lies in the tree. Since $\theta < \kappa$, there are $\alpha < \beta < \kappa$ with $b_{s_\alpha} = b_{s_\beta}$. The base a_{s_α} at s_α , together with any vertex of the selected monochromatic base at s_β as apex, is a monochromatic edge of $\mathcal{L}_\kappa(H)$. Thus no such θ colours suffice. The recursion uses ordinal lengths and unions and does not assume that κ is regular. $\square$

Write

$$H_{2,\kappa}(G) = G, \quad H_{j,\kappa}(G) = \mathcal{L}_\kappa(H_{j-1,\kappa}(G)) \quad (j \geq 3).$$

Lemma 11.3 and Proposition 11.5 remain valid for these hosts. Their finite-trace proofs use only the prefix order, the unique base and apex of a selected edge, and finitely many active nodes; the tree height plays no further role. In particular, every finite linear trace $K \subseteq H_{r,\kappa}(G)$ is assembled from $J^{[r]}$ with $J \subseteq G$ finite and $|E(J)| \leq |E(K)|$, together with finite edgeless systems. The lemma also gives

$$\chi(G) = \kappa \implies \chi(H_{r,\kappa}(G)) = \kappa.$$

At a successor $\kappa = \mu^+$, the order of these iterated hosts has a bound independent of r . Put $\rho = 2^\mu$. For any finite-uniform H with $|V(H)| \leq \rho$ we have $|E(H)| \leq \rho$, and hence

$$|E(H)^{<\mu^+}| \leq \mu^+ \cdot \rho^\mu = \rho, \quad |V(\mathcal{L}_{\mu^+}(H))| \leq \rho. \quad (2)$$

Here every ordinal below μ^+ has cardinality at most μ , and $(2^\mu)^\mu = 2^\mu$. Thus a graph of order μ^+ starts a finite iteration all of whose hosts have order at most 2^μ .

Simultaneous avoidance of finite linear families. We use the cardinal form of the classical graph input: for every uncountable cardinal κ and positive integer m , there is a graph G with

$$|V(G)| = \chi(G) = \kappa$$

and no odd cycle of length at most m [13, Theorem 7.4, p. 76]; this exact statement is also recorded in Erdős et al. [5, Theorem C, p. 428]. For an infinite cardinal κ , write $2^{<\kappa} = \sup_{\nu < \kappa} 2^\nu$, where the supremum runs over cardinals ν .

Theorem 12.2 (Simultaneous linear-family avoidance). *Let $r \geq 3$ be finite, and let $\mathcal{F}$ be a finite family of finite linear r -uniform hypergraphs. The following are equivalent:*

- (1) *No member of $\mathcal{F}$ belongs to $\mathcal{B}_r$.*
- (2) *For every uncountable cardinal κ , there is an $\mathcal{F}$ -free r -uniform hypergraph H with $\chi(H) = \kappa$ and $|V(H)| \leq 2^{<\kappa}$.*
- (3) *There is an $\mathcal{F}$ -free r -uniform hypergraph of uncountable chromatic number.*

The hosts in this assertion are not required to be linear.

Proof. The second assertion implies the third, and the third implies the first by Theorem 11.1. Suppose the first assertion holds. For each $F \in \mathcal{F}$, choose a connected component C_F that does not belong to $\mathcal{B}_r$. Such a component exists because membership in $\mathcal{B}_r$ is componentwise, either by its intrinsic criterion or by disjoint-union closure. Each C_F is linear and has at least one edge, since isolated vertices belong to $\mathcal{B}_r$. Put

$$m = \max(\{1\} \cup \{|E(C_F)| : F \in \mathcal{F}\}).$$

First let $\kappa = \mu^+$, where μ is infinite, and take an exact- κ graph G of order κ with no odd cycle of length at most m . The host $H = H_{r,\kappa}(G)$ has chromatic number κ and, by (2), order at most 2^μ . If some C_F embedded in H , its image would be a finite linear trace assembled from $J^{[r]}$ with $J \subseteq G$ and $|E(J)| \leq |E(C_F)| \leq m$. Each such J is bipartite: a nonbipartite J would contain an odd cycle of length at most $|E(J)| \leq m$. The assembly would therefore put C_F in $\mathcal{B}_r$, a contradiction. Thus H avoids every C_F and hence every F . Since $2^{<\mu^+} = 2^\mu$, this proves the required bound at successors.

If κ is an uncountable limit cardinal, choose uncountable successors $\kappa_j = \mu_j^+$, for $j < \text{cf}(\kappa)$, cofinal in κ . For each j use the preceding construction with the same components C_F , and take the disjoint union of the resulting hosts H_j . Every C_F is connected, so an embedding of it into this union would lie in one summand. The union therefore remains C_F -free for every F , and is consequently $\mathcal{F}$ -free. Its chromatic number is $\sup_j \kappa_j = \kappa$, and its order is at most

$$\text{cf}(\kappa) \cdot \sup_j 2^{\mu_j} \leq 2^{<\kappa}.$$

The last inequality uses $2^{<\kappa} \geq \sup_{\nu < \kappa} \nu^+ = \kappa$ at a limit cardinal. Passing first to connected forbidden components is what makes this disjoint-union step valid even when the original members of $\mathcal{F}$ are disconnected. $\square$

Exact calibration with bounded intersections. We use the simultaneous-colour property of Erdős–Galvin–Hajnal [5, Definition 6.2, p. 448]. An edge labelling $\ell : E(S) \rightarrow I$ has property $P(S, I, \kappa)$ if every colouring $c : V(S) \rightarrow \theta$ with $0 < \theta < \kappa$ has one colour $a < \theta$ such that every label in I occurs on an edge contained in $c^{-1}\{a\}$. For nonempty I , this implies $\chi(S) \geq \kappa$. The following lemma extends Li’s reservoir calibration for triple systems [3, Section 6]: it adds several old vertices to each base edge while retaining an explicit proper colouring.

Lemma 12.3 (Reservoir calibration). *Let μ be infinite, put $\kappa = \mu^+$, and fix a label set I of cardinality 2^μ . Suppose $\sigma \geq 2^\mu$ and $\text{cf}(\sigma) > \mu$. Let $k \geq 2$ and $1 \leq d \leq i$ be finite integers. Suppose that S is a simple k -uniform hypergraph on at most σ vertices, distinct edges of S meet in at most i vertices, and $P(S, I, \kappa)$ holds. Then there is a simple $(k + d)$ -uniform hypergraph L such that distinct edges meet in at most i vertices and*

$$|V(L)| \leq 2^\sigma, \quad \chi(L) = \kappa.$$

Proof. Put $R = \sigma^+$ and $\Lambda = 2^\sigma$. Then $R \leq \Lambda$ and $\Lambda^\sigma = \Lambda$. Fix $q : R \rightarrow \kappa$ whose fibres are cofinal in R ; a partition of R into κ sets of cardinality R supplies such a map. Take pairwise disjoint levels $(V_\alpha)_{\alpha < R}$ of cardinality Λ . Write $V_{<\alpha} = \bigcup_{\beta < \alpha} V_\beta$, and let $\text{rk}(x) = \alpha$ for $x \in V_\alpha$.

An admissible reservoir at level α is a family $\mathbf{X} = (X_p)_{p \in I}$ of pairwise disjoint subsets of $V_{<\alpha}$, each empty or of cardinality σ , such that $q(\text{rk}(x)) \neq q(\alpha)$ whenever $x \in \bigcup_p X_p$. There are at most Λ such families at each level: each coordinate has at most $\Lambda^\sigma = \Lambda$ possibilities, and $|I| \leq \sigma$. For every admissible $\mathbf{X}$, install σ labelled copies of S in V_α , with all installed copies at that level mutually vertex-disjoint. This requires at most $\Lambda \cdot \sigma \cdot \sigma = \Lambda$ vertices.

In a copy B installed for $\mathbf{X}$, call $e \in E(B)$ active when $X_{\ell(e)} \neq \emptyset$. Choose d -element sets $A_B(e) \subseteq X_{\ell(e)}$ for the active edges, pairwise disjoint within B . These choices are possible because $|E(S)| \leq \sigma$, d is finite, and every nonempty reservoir has cardinality σ . Put $V(L) = \bigcup_{\alpha < R} V_\alpha$ and take as edges precisely the sets

$$e \cup A_B(e) \quad (e \text{ active in an installed copy } B).$$

All choices are fixed before considering a vertex colouring.

The highest-level vertices of an edge recover its base edge and its installed copy, so L is simple and $(k + d)$ -uniform. For two edges in the same copy, the old vertex sets are disjoint, so their intersection has size at most i . Edges from different copies at one level have disjoint bases and can intersect only in their old vertex sets, of size d . For edges based at different levels, the intersection is contained in the d old vertices of the higher edge. Thus the intersection bound i is preserved. The map $x \mapsto q(\text{rk}(x))$ is a proper κ -colouring by reservoir admissibility, and $|V(L)| = R\Lambda = \Lambda$.

To prove the lower bound, consider any $c : V(L) \rightarrow \theta$ with $0 < \theta < \kappa$, and write $C_a = c^{-1}\{a\}$. Call a colour large when $|C_a| \geq \sigma$, and let D be the union of the other colour classes. There are at most μ colours, so $\text{cf}(\sigma) > \mu$ gives

$|D| < \sigma$. At least one colour is large. For $\xi < \kappa$, let $Q_\xi = \bigcup \{V_\beta : q(\beta) = \xi\}$. A fixed large colour a has at most one value ξ for which $|C_a \setminus Q_\xi| < \sigma$: two such values would express C_a as a union of two sets of cardinality below σ . Since there are at most μ large colours, choose $\xi < \mu^+$ exceptional for none of them.

Assign distinct labels $p_a \in I$ to the large colours and choose $X_{p_a} \in [C_a \setminus Q_\xi]^\sigma$, putting all other $X_p = \emptyset$. Their union has size σ , so its ranks are bounded in $R = \sigma^+$. Choose $\alpha \in q^{-1}\{\xi\}$ above those ranks. This is an admissible reservoir at level α . Fewer than σ of its σ disjoint installed copies meet D , since the copies meeting D inject into D . Choose an installed copy B avoiding D .

By $P(S, I, \kappa)$ applied to $c|_B$, some colour a contains a base edge of every label. It is large because B avoids D . In particular, there is a monochromatic base edge e of label p_a . It is active, and $A_B(e) \subseteq X_{p_a} \subseteq C_a$. Thus $e \cup A_B(e)$ is a monochromatic edge of L . No colouring with fewer than κ colours is proper, proving $\chi(L) = \kappa$. $\square$

Define $\beth_0(\mu) = \mu$ and $\beth_{j+1}(\mu) = 2^{\beth_j(\mu)}$ for finite j .

Theorem 12.4 (Exact bounded-intersection calibration). *Let μ be an infinite cardinal, $r \geq 3$ finite, and $1 \leq i \leq r - 2$. Put $t = \lceil (r - 1)/i \rceil$. There is a simple r -uniform hypergraph L whose distinct edges intersect in at most i vertices and such that*

$$\chi(L) = \mu^+, \quad |V(L)| \leq \beth_t(\mu).$$

In particular, for $i = 1$ there is a linear r -uniform hypergraph of chromatic number exactly μ^+ on at most $\beth_{r-1}(\mu)$ vertices.

Proof. Write $k = r - i$, $j = \min\{i, k - 1\}$ and $m = t - 1$. Then $k \geq 2$, $m \geq 1$, $1 \leq j < k$, and $k \leq mj + 1$. Indeed, if $i \leq k - 1$, then $j = i$ and $m = \lceil (k - 1)/i \rceil$; otherwise $j = k - 1$ and $m = 1$.

Erdős–Galvin–Hajnal’s Corollary 12.3 [5, p. 485] gives a (k, j) -system S of size $\sigma = \beth_m(\mu)$ with $P^*(S, \sigma, \mu^+, j)$. Taking $V(S) = \bigcup E(S)$ gives $|V(S)| \leq \sigma$. Here distinct edges of a (k, j) -system intersect in at most j vertices, and their stronger property P^* implies P . Coarsen the σ labels by a surjection onto a set I of cardinality $2^\mu \leq \sigma$; the property P is preserved. Since $m \geq 1$, König’s theorem gives

$$\text{cf}(\sigma) = \text{cf}(2^{\beth_{m-1}(\mu)}) > \beth_{m-1}(\mu) \geq \mu.$$

Apply Lemma 12.3 with $d = i$. The output has rank $k + i = r$, intersection bound i , chromatic number exactly μ^+ , and order at most $2^\sigma = \beth_t(\mu)$. $\square$

Corollary 12.5 (All uncountable chromatic cardinals). *For every finite $r \geq 3$, every $1 \leq i \leq r - 2$, and every uncountable cardinal κ , there is a simple r -uniform hypergraph of chromatic number exactly κ whose distinct edges intersect in at most i vertices. If κ is a limit cardinal and $t = \lceil (r - 1)/i \rceil$, the host can be chosen with order at most $\sup_{\aleph_0 \leq \mu < \kappa} \beth_t(\mu)$. In particular, its order is exactly κ when κ is an uncountable strong limit cardinal.*

Proof. The successor case is Theorem 12.4. For a limit cardinal κ , take a cofinal family of uncountable successor cardinals $\kappa_\eta < \kappa$ and form a disjoint union of the corresponding hosts. The intersection bound is preserved, and the chromatic number of the union is $\sup_\eta \kappa_\eta = \kappa$. Put $M = \sup_{\aleph_0 \leq \mu < \kappa} \beth_t(\mu)$. Since $M \geq \kappa$, the union has order at most $\text{cf}(\kappa) \cdot M = M$. At a strong limit κ , every finite iterate $\beth_t(\mu)$ with $\mu < \kappa$ is below κ , so $M = \kappa$; the reverse order bound follows from $\chi(H) \leq |V(H)|$. This argument includes singular cardinals. $\square$

Remark 12.6 (Classical size benchmark). Conditional optimality under GCH already follows from Erdős–Galvin–Hajnal. Their Corollary 2.2 [5, p. 433] implies that an (r, i) -system on at most μ^{+m} vertices is μ -colourable when $r \geq mi + 2$. Taking $m = t - 1$ gives the ZFC lower bound $|V(H)| \geq \mu^{+t}$ whenever $\chi(H) > \mu$. Under GCH, their Corollary 12.3, applied directly with rank r and parameter t , supplies such a host on μ^{+t} vertices. Applying Corollary 2.2 again with base μ^+ and parameter $t - 1$ gives chromatic number at most μ^+ , hence exactly μ^+ . Thus the exact GCH threshold is a classical consequence. The construction above combines their simultaneous-colour bases with Li’s reservoir method to retain exact chromatic calibration in ZFC. At the remaining endpoint $i = r - 1$, the complete r -uniform hypergraph on μ^+ vertices already has order and chromatic number μ^+ .

The uncountable avoidance spectrum. For a finite r -uniform hypergraph F , define its uncountable avoidance spectrum to be the class

$$\text{Spec}_r^{\text{unc}}(F) = \{\kappa > \aleph_0 : \text{some } F\text{-free } r\text{-uniform } H \text{ has } \chi(H) = \kappa\},$$

where κ ranges over cardinals. For a nonlinear F , put

$$\tau(F) = \max\{|e \cap f| : e, f \in E(F), e \neq f\}.$$

Thus $2 \leq \tau(F) \leq r - 1$.

Theorem 12.7 (Exact uncountable avoidance spectra). *For every finite $r \geq 3$ and finite r -uniform hypergraph F ,*

$$\text{Spec}_r^{\text{unc}}(F) = \begin{cases} \emptyset, & F \in \mathcal{B}_r, \\ \{\kappa : \kappa > \aleph_0 \text{ is a cardinal}\}, & F \notin \mathcal{B}_r. \end{cases}$$

If $F \notin \mathcal{B}_r$ and $\kappa = \mu^+$ with μ infinite, an F -free host of chromatic number κ can be chosen with

$$|V(H)| \leq \begin{cases} 2^\mu, & F \text{ is linear,} \\ \beth_{\lceil (r-1)/(\tau(F)-1) \rceil}(\mu), & F \text{ is nonlinear.} \end{cases}$$

For linear $F \notin \mathcal{B}_r$, the stronger bound $|V(H)| \leq 2^{<\kappa}$ holds at every uncountable κ .

Proof. If $F \in \mathcal{B}_r$, the classification theorem makes F obligatory, so the spectrum is empty. If F is linear and does not belong to $\mathcal{B}_r$, apply Theorem 12.2 to the singleton family $\{F\}$.

Suppose instead that F is nonlinear, and set $i = \tau(F) - 1$. Then $1 \leq i \leq r - 2$. For $\kappa = \mu^+$, Theorem 12.4 supplies an r -uniform host of chromatic number κ , of order at most $\beth_{\lceil (r-1)/i \rceil}(\mu)$, in which distinct edges intersect in at most i points. This host is F -free: an embedding of F would preserve the $\tau(F) = i + 1$ shared vertices of a pair of its edges. This proves both the successor assertion and its stated order bound.

For an uncountable limit κ , choose cofinal uncountable successors and take a disjoint union of the corresponding hosts with the same intersection bound i . The union still has pairwise edge intersections at most i and therefore omits F ; its chromatic number is the supremum of the successor chromatic numbers, namely κ . This completes all cases, including singular cardinals. $\square$

The triple-system spectrum is established in Li [3, Theorem 7.1]. Here the classification in every uniformity and the bounded-intersection calibration extend the dichotomy to all finite $r \geq 3$. The quantitative bounds are witness bounds; no minimum order is asserted for an individual forbidden hypergraph.

A finite-family obstruction in every uniformity. The assumption that every forbidden member is linear in the simultaneous avoidance result cannot be dropped. The following construction extends the triple-system example used by Li [3, Section 7] in connection with Erdős Problem #1177.

Proposition 12.8 (Individually avoidable, jointly obligatory). *For every finite $r \geq 3$ there is a finite family $\mathcal{F}_r$ of finite simple r -uniform hypergraphs such that no member of $\mathcal{F}_r$ is obligatory, but every $\mathcal{F}_r$ -free r -uniform hypergraph has countable chromatic number.*

Proof. For $2 \leq t \leq r - 1$, let $N_{r,t}$ consist of two r -edges whose intersection has size t , with no isolated vertices, and put

$$\mathcal{N}_r = \{N_{r,t} : 2 \leq t \leq r - 1\}.$$

A simple r -uniform host avoids $\mathcal{N}_r$ precisely when it is linear. Every member of $\mathcal{N}_r$ is non-obligatory, by the classical nonlinear obstruction [14].

Let $C_7^{[3]}$ be the private triple expansion of the 7-cycle. Define $\mathcal{E}_r$ to contain one representative of each isomorphism type of a reduced linear r -uniform hypergraph E with exactly seven edges $e_1, \dots, e_7$ such that there are triples $a_j \subseteq e_j$ for which the triple system $\{a_1, \dots, a_7\}$ is isomorphic to $C_7^{[3]}$. This is a finite family: every such E has at most $7r$ vertices. It is nonempty, since $C_7^{[r]}$ is a member. Every $E \in \mathcal{E}_r$ retains the seven distinct cycle vertices and seven distinct edges of the displayed loose triple cycle, so it contains a Berge cycle of odd length seven. Hence no member of $\mathcal{E}_r$ belongs to $\mathcal{B}_r$, and the classification theorem shows that each is non-obligatory.

Set $\mathcal{F}_r = \mathcal{N}_r \cup \mathcal{E}_r$, and suppose that an r -uniform host H has uncountable chromatic number. If H is nonlinear, it contains a member of $\mathcal{N}_r$. If H is linear, choose a triple $a_e \subseteq e$ from every edge $e \in E(H)$, and let H_3 be the resulting triple system on $V(H)$. The chosen triples are distinct and form a linear triple

system, because the original edges meet pairwise in at most one point. Every proper colouring of H_3 is a proper colouring of H , so

$$\chi(H_3) \geq \chi(H) > \aleph_0.$$

The theorem of Hajnal–Komjáth that $C_7^{[3]}$ is linearly obligatory [16] supplies a copy of that triple system in H_3 . Restore the seven original r -edges from which its triples were selected, and take their union as vertex set. The resulting reduced seven-edge subhypergraph of H is a member of $\mathcal{E}_r$ up to isomorphism. Thus every uncountably chromatic host contains a member of $\mathcal{F}_r$, as required. $\square$

For $r = 3$, the family $\mathcal{E}_3$ consists only of $C_7^{[3]}$, and $\mathcal{N}_3$ consists only of two triples meeting in two points. For larger r , the definition of $\mathcal{E}_r$ allows all identifications among the added points that preserve linearity. This finite completion family is needed because choosing triples from a linear r -uniform host does not force the restored edges to have globally private added vertices.

13. OBLIGATORY ATOMS AND SIDORENKO EXPANSIONS

The canonical description also connects obligatoriness with a finite counting question. A bipartite private expansion can be forced by uncountable chromatic number while having fewer copies than the random density benchmark in a finite host. We establish this for a single canonical atom in every uniformity, using the specific graph and rank-layer construction of OpenAI [6].

The conclusions about that graph in this section are conditional on the rank-layer limit in the pinned version of the preprint. We give the transfer proofs, including an additional proper-subgraph limit extracted from its coordinate expansion. The preprint’s finite-field estimate is an external input. The classification and chromatic spectra in the preceding sections do not use it.

The distinction is also relevant to a published conjecture of Nie–Spiro [7, Conjecture 5.1], which asks whether every bipartite graph has a Sidorenko private expansion in some uniformity. The same fixed graph below has no such expansion. The transfer requires constructing an actual symmetric hypergraph kernel; a graph density counterexample alone does not provide one. We begin with a universal marginal-realization theorem and quantitative density bounds that are independent of the external rank-layer input. They describe both the available realizations and the size of possible density errors as the uniformity grows.

The smallest examples explain the quantities involved. A star expansion measures the moments of the one-vertex degree function; a cycle expansion measures correlations that remain when that function is constant. When atoms are joined, their rooted densities must be paired at the attachment vertex. We therefore pass from pair marginals for single atoms to higher marginals for general linear assemblies. The finite-alphabet obstruction, the density estimates and the attachment examples all have independent proofs below. Only the final explicit negative example uses the cited rank-layer limit.

Universal pair marginals and matching symmetrization. All probability spaces below are finite and all their atoms have positive mass. For a symmetric nonnegative r -variable kernel U and a finite simple r -uniform hypergraph F , put

$$t(F, U) = \mathbb{E} \prod_{e \in E(F)} U((X_v)_{v \in e}),$$

where the X_v are independent samples. Symmetry makes the ordering within an edge immaterial. Write $p(U) = \mathbb{E}U$. For a finite simple r -uniform host G of order $n > 0$, use

$$t(F, G) = \frac{\text{Hom}(F, G)}{n^{|V(F)|}}, \quad p_r(G) = \frac{r! |E(G)|}{n^r}.$$

Thus all edge-preserving maps are counted, including noninjective maps. We call F *Sidorenko* if $t(F, G) \geq p_r(G)^{|E(F)|}$ for every such host. This is the ordinary hypergraph homomorphism-density convention of Conlon et al. [36] and Nie and Spiro [7]. For $n \geq r$, the normalization $p_r(G)$ differs from $|E(G)|/\binom{n}{r}$ by the factor $(n)_r/n^r$, which tends to one as n tends to infinity.

For a simple graph J , let $J^{[r]}$ be its private r -uniform expansion, with $r - 2$ distinct new vertices on each graph edge. The convention at rank two is $J^{[2]} = J$. The two-variable marginal of U is

$$M_U(x, y) = \mathbb{E}_{Z_3, \dots, Z_r} U(x, y, Z_3, \dots, Z_r).$$

The private vertices occur in exactly one edge, so Fubini's theorem gives

$$t(J^{[r]}, U) = t(J, M_U), \quad p(U) = \mathbb{E}M_U. \quad (3)$$

An arbitrary graph kernel need not be such a marginal. We first characterize which diluted pair marginals can always be realized; the symmetric construction needed for expansions follows as a special case.

For $r \geq 2$, write $\mathfrak{M}_r$ for the convex hull in $\mathbb{R}^{\binom{r}{2}}$ of the incidence vectors of all matchings of K_r , including the empty matching. Edmonds' matching-polytope theorem [37, Section 2, Theorem (P)] gives its description

$$\lambda_{ij} \geq 0, \quad \sum_{j \neq i} \lambda_{ij} \leq 1, \quad \sum_{\{i, j\} \subseteq S} \lambda_{ij} \leq \frac{|S| - 1}{2} \quad (|S| \text{ odd}). \quad (4)$$

The last family may be restricted to $|S| \geq 3$. The following identifies this polytope as the exact region of coefficients that realize every collection of finite pair kernels. In this general statement the spaces may differ, and the joint density need not be symmetric. We write $\lambda_{ji} = \lambda_{ij}$ when an unordered pair is indexed in the opposite order.

Theorem 13.1 (Universal pair-marginal profiles). *Let $r \geq 2$ and $\lambda \in \mathbb{R}^{\binom{r}{2}}$. The following are equivalent.*

- (1) *For every collection of finite probability spaces (Ω_i, μ_i) and nonnegative kernels $V_{ij} : \Omega_i \times \Omega_j \rightarrow \mathbb{R}$ satisfying*

$$\mathbb{E}_{x_j} V_{ij}(x_i, x_j) = 1, \quad \mathbb{E}_{x_i} V_{ij}(x_i, x_j) = 1,$$

there is a nonnegative density U of mean one on $\prod_{i=1}^r \Omega_i$ whose (i, j) -marginal is

$$M_{ij} = 1 + \lambda_{ij}(V_{ij} - 1) \quad (1 \leq i < j \leq r).$$

(2) The preceding assertion holds when all spaces are the same finite probability space and all V_{ij} are the same symmetric nonnegative kernel V with constant row integral one.

(3) $\lambda \in \mathfrak{M}_r$.

Proof. The first assertion implies the second. For the third implying the first, write $\lambda = \sum_M w_M \mathbf{1}_M$, where $w_M \geq 0$ and $\sum_M w_M = 1$, and set

$$U(x_1, \dots, x_r) = \sum_M w_M \prod_{\{i,j\} \in M} V_{ij}(x_i, x_j).$$

Each product has mean one because its factors use disjoint variables. Its (i, j) -marginal is V_{ij} if $\{i, j\} \in M$ and is one otherwise: in the latter case every factor touching i or j has its other variable integrated out. This proves the required formula.

It remains to prove the inequalities (4) from the second assertion. For any even N , take the uniform probability space $[N]$, a fixed-point-free involution f of $[N]$, and

$$V_N(x, y) = N \mathbf{1}_{\{y=f(x)\}}.$$

This is a symmetric nonnegative kernel with row integral one. Nonnegativity of each prescribed pair marginal, both on and off the graph of f , gives $1 + \lambda_{ij}(N - 1) \geq 0$ and $1 - \lambda_{ij} \geq 0$. Letting even N tend to infinity yields $0 \leq \lambda_{ij} \leq 1$.

Give $(Z_1, \dots, Z_r)$ the joint probability law with density U relative to the uniform product measure. Fix i and put $A_j = \{Z_j = f(Z_i)\}$ for $j \neq i$. The specified pair marginals give

$$\mathbb{P}(A_j) = \lambda_{ij} + \frac{1 - \lambda_{ij}}{N}.$$

For distinct $j, k \neq i$, the event $A_j \cap A_k$ implies $Z_j = Z_k$. Since f has no fixed point, the (j, k) -marginal therefore gives

$$\mathbb{P}(A_j \cap A_k) \leq \mathbb{P}(Z_j = Z_k) = \frac{1 - \lambda_{jk}}{N} \leq \frac{1}{N}.$$

For any finite family of events, $\sum_j \mathbf{1}_{A_j} \leq 1 + \sum_{j < k} \mathbf{1}_{A_j \cap A_k}$ pointwise. Consequently

$$\sum_{j \neq i} \left(\lambda_{ij} + \frac{1 - \lambda_{ij}}{N} \right) \leq 1 + \frac{\binom{r-1}{2}}{N}.$$

Letting N tend to infinity proves the degree inequalities.

For the odd-set inequalities use the uniform space $\{-1, 1\}$ and $V(a, b) = 1 - ab$. The prescribed pair marginals imply $\mathbb{E}Z_i Z_j = -\lambda_{ij}$. For every odd

subset S , the integer $\sum_{i \in S} Z_i$ is odd, and thus

$$1 \leq \mathbb{E} \left(\sum_{i \in S} Z_i \right)^2 = |S| - 2 \sum_{\{i,j\} \subseteq S} \lambda_{ij}.$$

These are precisely the remaining inequalities. Edmonds' theorem now gives $\lambda \in \mathfrak{M}_r$. $\square$

Lemma 13.2 (Matching symmetrization). *Let V be a finite symmetric nonnegative kernel with $\mathbb{E}_y V(x, y) = 1$ for every x . For an integer $r \geq 2$, let $\mathcal{M}_r$ be the set of matchings of size $\lfloor r/2 \rfloor$ in the complete graph on $[r]$, and define*

$$U_r(x_1, \dots, x_r) = \frac{1}{|\mathcal{M}_r|} \sum_{M \in \mathcal{M}_r} \prod_{\{a,b\} \in M} V(x_a, x_b).$$

Then U_r is symmetric and nonnegative, has mean one, and

$$M_{U_r} = 1 + \lambda_r(V - 1), \quad \lambda_r = \frac{2\lfloor r/2 \rfloor}{r(r-1)} = \begin{cases} (r-1)^{-1}, & r \text{ even}, \\ r^{-1}, & r \text{ odd}. \end{cases}$$

Proof. Each matching product has mean one, because its factors have disjoint variable sets. In its two-variable marginal, a matching containing $\{1, 2\}$ contributes $V(x_1, x_2)$. Every other matching contributes one: any partner of vertex 1 or 2 is integrated out using the constant row integral, and the remaining disjoint pairs have mean one. The proportion of matchings containing a given pair is their common number of edges divided by $\binom{r}{2}$, namely λ_r . Permutation invariance of the set of matchings proves symmetry. $\square$

The coefficient in Lemma 13.2 is optimal in the following universal sense: the assertion that every finite symmetric nonnegative kernel V with row integral one admits a symmetric nonnegative mean-one r -kernel with pair marginal $1 + \lambda(V - 1)$ holds precisely for $0 \leq \lambda \leq \lambda_r$. Indeed, Theorem 13.1 gives $(r-1)\lambda \leq 1$ when r is even and $\binom{r}{2}\lambda \leq (r-1)/2$ when r is odd; its nonnegativity condition excludes negative coefficients. The lemma attains the endpoint, and mixing its kernel with the constant one attains the remaining interval. The scalar parity obstruction is the familiar minimum-correlation bound for exchangeable signs [38]. Products averaged over perfect or near-perfect matchings also occur in the quantum marginal construction of Allerstorfer et al. [39, Lemma 4.4]. Theorem 13.1 addresses the full array of coefficients for arbitrary finite classical pair kernels.

Constructing and testing a marginal profile. Theorem 13.1 has a direct probabilistic interpretation. First choose a matching M with probabilities w_M . For each $\{i, j\} \in M$, draw (X_i, X_j) with joint density V_{ij} relative to $\mu_i \otimes \mu_j$, independently for the different matched pairs; draw every unmatched coordinate from its original law. The two constant marginal integrals of V_{ij} ensure that each coordinate retains its original distribution. A specified pair is drawn from its prescribed joint law with probability λ_{ij} , and otherwise has the independent

product law. This gives exactly $1 + \lambda_{ij}(V_{ij} - 1)$ as its density. Thus a matching decomposition is an explicit sampling rule, and the same decomposition works for every collection of the pair kernels. These mean-one densities need not be bounded by one; bounded kernels are considered separately in the density estimates below.

Three and four coordinates. For $r = 3$, the matching polytope is simply

$$\lambda_{12}, \lambda_{13}, \lambda_{23} \geq 0, \quad \lambda_{12} + \lambda_{13} + \lambda_{23} \leq 1.$$

Writing $s = \lambda_{12} + \lambda_{13} + \lambda_{23}$, the construction is the explicit density

$$U = 1 - s + \lambda_{12}V_{12} + \lambda_{13}V_{13} + \lambda_{23}V_{23}.$$

In particular, three copies of the same symmetric kernel with equal coefficient $1/3$ are realized by $(V(x_1, x_2) + V(x_1, x_3) + V(x_2, x_3))/3$.

For $r = 4$, the only constraints besides nonnegativity are the four degree inequalities and the four triangle inequalities. They also give a closed formula for the construction. Partition the six edges into the three pairs of opposite edges

$$\mathcal{P} = \{\{12, 34\}, \{13, 24\}, \{14, 23\}\}.$$

For $\{e, f\} \in \mathcal{P}$, put $a_{ef} = \min\{\lambda_e, \lambda_f\}$, and put $w_0 = 1 - \sum_{\{e, f\} \in \mathcal{P}} \max\{\lambda_e, \lambda_f\}$. Then

$$U = w_0 + \sum_{\{e, f\} \in \mathcal{P}} [a_{ef}V_eV_f + (\lambda_e - a_{ef})V_e + (\lambda_f - a_{ef})V_f]. \quad (5)$$

Here V_{ij} is evaluated at (x_i, x_j) . To see that $w_0 \geq 0$, choosing one edge from each opposite pair gives one of the four three-edge stars or four triangles of K_4 . The sum of the three maxima is therefore the largest of their eight coefficient sums, each at most one. The coefficients in (5) are nonnegative and sum to one; they give weights on the empty matching, the singleton matchings and the three perfect matchings. Each edge has total inclusion probability λ_e , which verifies all six marginals. At the common coefficient $1/3$, this reduces to

$$U = \frac{1}{3} (V_{12}V_{34} + V_{13}V_{24} + V_{14}V_{23}).$$

The triangle constraints cannot be omitted. For instance, the profile $\lambda_{12} = \lambda_{13} = \lambda_{23} = 1/2$, with all other coefficients zero, satisfies every degree inequality but has no universal realization. Already the common two-point kernel $V(a, b) = 1 - ab$, for uniform $a, b \in \{-1, 1\}$, would require

$$\mathbb{E}(Z_1 + Z_2 + Z_3)^2 = 3 - 2(\lambda_{12} + \lambda_{13} + \lambda_{23}) = 0,$$

although the sum of three signs is never zero. More generally, on five coordinates the common coefficient $1/4$ satisfies all degree and triangle inequalities but violates the five-element odd-set inequality. The admissible common endpoint there is $1/5$. These examples show separately why the local degree conditions and the larger odd-set conditions occur.

A quantitative test on one finite alphabet. The necessity proof can be made finite and stable under small errors in the proposed marginals. Let $\text{dist}_\infty(\lambda, \mathfrak{M}_r)$

denote the distance in the maximum-coordinate norm, and let total variation between probability laws mean the supremum of their differences on events.

Proposition 13.3 (Finite-alphabet obstruction). *Let $r \geq 2$, $L = \binom{r}{2}$, and $\lambda \in [0, 1]^{\binom{r}{2}}$. For an even integer $N \geq 2$, fix a fixed-point-free involution f of $[N]$, and define the pair probability laws*

$$Q_{ij}(x, y) = \frac{1}{N^2} \left[1 + \lambda_{ij} \left(N \mathbf{1}_{\{y=f(x)\}} - 1 \right) \right].$$

Suppose a joint law on $[N]^r$ has each of its pair marginals within total variation ε of Q_{ij} . Then

$$\text{dist}_\infty(\lambda, \mathfrak{M}_r) \leq \frac{L}{N} + (L+1)\varepsilon. \quad (6)$$

Consequently, if this distance is $\delta > 0$, every even $N > L/\delta$ gives a single finite kernel $V_N(x, y) = N \mathbf{1}_{\{y=f(x)\}}$ for which the prescribed pair marginals cannot be realized exactly.

Proof. Sample $(Z_1, \dots, Z_r)$ from the joint law and let B be the event that two of its coordinates coincide. Put $R = \sum_{i < j} (1 - \lambda_{ij})$, so $0 \leq R \leq L$. The diagonal has Q_{ij} -probability $(1 - \lambda_{ij})/N$, because f has no fixed points. A union bound gives

$$\mathbb{P}(B) \leq R/N + L\varepsilon.$$

When B does not occur, the pairs satisfying $Z_j = f(Z_i)$ form a matching: two such pairs sharing a coordinate would force their other coordinates to coincide. On B , choose the empty matching instead. The inclusion-probability vector θ of this random matching belongs to $\mathfrak{M}_r$.

For $e = \{i, j\}$, the event $A_e = \{Z_j = f(Z_i)\}$ has Q_{ij} -probability $\lambda_e + (1 - \lambda_e)/N$. Since $\theta_e = \mathbb{P}(A_e \setminus B)$, it follows that

$$\frac{1 - \lambda_e - R}{N} - (L+1)\varepsilon \leq \theta_e - \lambda_e \leq \frac{1 - \lambda_e}{N} + \varepsilon.$$

Here $R \geq 1 - \lambda_e$. Both absolute deviations are therefore at most $L/N + (L+1)\varepsilon$, proving (6). The final assertion is the case $\varepsilon = 0$. $\square$

Thus a violation separated from the matching polytope by δ has a finite witness with alphabet size of order r^2/δ . For a fixed alphabet the same argument gives the explicit lower bound $(\delta - L/N)/(L+1)$ on the maximum pair-marginal error, whenever that expression is positive. This tests realizability of the whole profile, rather than one chosen inequality.

Uniform density estimates as uniformity increases. For bounded symmetric kernels, increasing the uniformity constrains the pair marginal. We use the first two terms of the classical Hoeffding orthogonal decomposition [40] and give the short projection argument in full. Nie-Spiro's exponent estimate [7, Theorem 1.13] already yields one-sided approximate Sidorenko bounds for expansions at fixed positive density. The results below give explicit two-sided estimates, uniform over bounded kernels, together with optimal rates and a

girth-sensitive refinement under regularity. All L^2 norms refer to the relevant product probability measure.

Lemma 13.4 (Two-coordinate projection bound). *Let $r \geq 2$ and let $U : \Omega^r \rightarrow [0, 1]$ be symmetric. Set $p = \mathbb{E}U$, $d(x) = \mathbb{E}[U \mid X_1 = x]$, and $M(x, y) = \mathbb{E}[U \mid X_1 = x, X_2 = y]$. With*

$$h = d - p, \quad k(x, y) = M(x, y) - p - h(x) - h(y),$$

one has

$$r\|h\|_2^2 + \binom{r}{2}\|k\|_2^2 \leq p(1-p), \quad \|M - p\|_2^2 \leq \frac{2p(1-p)}{r}. \quad (7)$$

If $d \equiv p$, the second bound improves to

$$\|M - p\|_2^2 \leq \frac{2p(1-p)}{r(r-1)}. \quad (8)$$

Proof. The mean of h is zero and both one-variable integrals of k vanish. Hence the random variables $h(X_i)$ and $k(X_i, X_j)$, over all singletons and pairs, are mutually orthogonal in L^2 . Their inner products with $U - p$ are respectively $\|h\|_2^2$ and $\|k\|_2^2$, by conditional expectation. Subtracting their sum from $U - p$ leaves an orthogonal remainder. Thus their squared norms sum to at most $\text{Var } U \leq p(1-p)$, proving the first bound. Furthermore

$$\|M - p\|_2^2 = 2\|h\|_2^2 + \|k\|_2^2 \leq \frac{2}{r} \left(r\|h\|_2^2 + \binom{r}{2}\|k\|_2^2 \right),$$

because $r \geq 2$. If $h = 0$, use the first bound directly. $\square$

Theorem 13.5 (Two-sided density approximation). *Let F be a finite simple graph with $m \geq 1$ edges and $r \geq 2$. Every symmetric $[0, 1]$ -valued r -kernel U of mean p satisfies*

$$|t(F^{[r]}, U) - p^m| \leq \frac{m(m-1)}{r} p(1-p). \quad (9)$$

The same inequality holds for every finite simple r -uniform host G , with $p = p_r(G)$ and homomorphism-density normalization. The order r^{-1} cannot be improved uniformly, even for $F = P_3$ and $p = 1/2$.

Proof. Write $D = M_U - p$ and $f(s) = t(F, p + sD)$ for $0 \leq s \leq 1$. The private-expansion identity gives $f(1) = t(F^{[r]}, U)$. We have $f(0) = p^m$ and $f'(0) = 0$, since $\mathbb{E}D = 0$. Each of the $m(m-1)$ terms in $f''(s)$ has two factors D and $m-2$ factors in $[0, 1]$. Cauchy-Schwarz on the entire product probability space bounds its absolute value by $\|D\|_2^2$, whether the two corresponding graph edges meet or are disjoint. Taylor's formula with integral remainder yields

$$|f(1) - f(0)| \leq \frac{1}{2}m(m-1)\|D\|_2^2.$$

Apply Lemma 13.4. The case $m = 1$ is exact and needs no differentiation. The adjacency kernel of a finite simple host, assigned zero on repeated coordinates, proves the finite-host assertion without an asymptotic error.

For sharpness let $r = 2n + 1$, take independent uniform signs, and set $U = \mathbf{1}_{\{X_1 + \dots + X_r > 0\}}$. Then $p = 1/2$ and

$$d(\pm 1) = \frac{1}{2} \pm \frac{1}{2} \frac{\binom{2n}{n}}{4^n}.$$

As P_3 has two edges,

$$t(P_3^{[r]}, U) - \frac{1}{4} = \text{Var}(d) = \frac{1}{4} \left(\frac{\binom{2n}{n}}{4^n} \right)^2 \sim \frac{1}{2\pi r},$$

where the last assertion follows from Stirling's formula. $\square$

Theorem 13.6 (Girth-sensitive approximation under regularity). *Let F be a finite simple graph with $m \geq 1$ edges, and let $U : \Omega^r \rightarrow [0, 1]$ be symmetric, where $r \geq 2$, with every one-coordinate marginal equal to its mean p . If F is a forest, then $t(F^{[r]}, U) = p^m$ exactly. Otherwise, if g is the girth of F , then*

$$\left| t(F^{[r]}, U) - p^m \right| \leq (2^m - 1) \left(\frac{2p(1-p)}{r(r-1)} \right)^{g/2}. \quad (10)$$

The same assertion holds for finite regular r -uniform hosts, with $p = p_r(G)$. For every fixed cyclic graph F , the order r^{-g} is optimal among such kernels, already at density $p = 1/2$.

Proof. Put $D = M_U - p$ and $q = (2p(1-p)/(r(r-1)))^{1/2}$. Both one-variable integrals of D are zero, $|D| \leq 1$, and Lemma 13.4 gives $\|D\|_2 \leq q \leq 1$. Expand

$$t(F, M_U) - p^m = \sum_{\emptyset \neq A \subseteq E(F)} p^{m-|A|} t(F_A, D), \quad F_A = (V(F), A).$$

A vertex of degree one in F_A makes its term vanish by integration. Every other nonempty F_A contains a cycle C of length $\ell \geq g$. After taking absolute values, discard the factors outside C to obtain

$$|t(F_A, D)| \leq t(C, |D|) \leq \|D\|_2^\ell \leq q^\ell.$$

For the middle inequality, the symmetric kernel $|D|$ defines a self-adjoint operator on the finite-dimensional space $L^2(\Omega)$. If its eigenvalues are η_i , its cycle integral is $\sum_i \eta_i^\ell \leq \sum_i |\eta_i|^\ell \leq (\sum_i \eta_i^2)^{\ell/2} = \|D\|_2^\ell$. There are at most $2^m - 1$ nonempty edge subsets and $p^{m-|A|} \leq 1$, proving (10). For a forest every nonempty term vanishes. The adjacency kernel of a finite regular host has constant one-coordinate marginal, so the finite-host statement follows without an additional limiting argument.

For sharpness take independent uniform signs, put $\theta_r = r^{-1/2}$ and $c_r = \cos \theta_r$, and define

$$U_r(x_1, \dots, x_r) = \frac{1}{2} + \frac{1}{4} \left[\cos \left(\theta_r \sum_{i=1}^r x_i \right) - c_r^r \right].$$

This kernel is symmetric and takes values in $[0, 1]$. Its cosine term has expectation c_r^r . Global sign reversal leaves U_r unchanged, so each one-coordinate marginal

is constant on $\{-1, 1\}$. The mean and these marginals are therefore exactly $1/2$. Its pair marginal is

$$M_r(a, b) = \frac{1}{2} - b_r ab, \quad b_r = \frac{1}{4} \sin^2 \theta_r c_r^{r-2} \sim \frac{e^{-1/2}}{4r}.$$

Expanding in the signs shows that only edge subsets with all degrees even survive. If $N_g(F)$ is the number of cycles of length g in F , the smallest nonempty such subsets are precisely these cycles. Consequently

$$t(F^{[r]}, U_r) - 2^{-m} = N_g(F)(-b_r)^g 2^{-(m-g)} + O_F(b_r^{g+1}).$$

Since $N_g(F) > 0$, the absolute difference is of order r^{-g} . In particular, for $F = C_g$ the identity is exactly $t(C_g^{[r]}, U_r) - 2^{-g} = (-b_r)^g$. $\square$

The estimates are uniform over the size of the underlying finite probability space. For the canonical atom $J^{[r]}$, the girth in Theorem 13.6 is the length of its shortest Berge cycle. Thus the same cyclic structure used in the classification controls a quantitative density estimate in regular finite hosts.

Remark 13.7 (Stars, cycle atoms, and what regularity removes). Let U be a symmetric $[0, 1]$ -valued r -kernel of mean p , write $d(x) = \mathbb{E}[U \mid X_1 = x]$, and let $M = M_U$ be its pair marginal. For the star $S_k = K_{1,k}$, where $k \geq 1$, integrating each leaf gives the exact identity

$$t(S_k^{[r]}, U) = \mathbb{E}d(X)^k \geq p^k.$$

In particular,

$$t(P_3^{[r]}, U) - p^2 = \text{Var}(d).$$

Thus the first nontrivial tree example measures the degree variance exactly. For the adjacency kernel of a finite host G on n vertices, $d(v) = (r-1)! \deg_G(v)/n^{r-1}$, so this is the variance of its normalized vertex degrees. For $r \geq 3$, the star expansion is an obligatory assembly of k single-edge atoms at one point; it has no cyclic atom. For a regular kernel its density is exactly the random benchmark, regardless of r .

The even-cycle lower bound is also classical without regularity: the even spectral moment of the self-adjoint operator T_M dominates $\|T_M\|_{\text{op}}^{2\ell}$, which in turn dominates $|\langle 1, T_M 1 \rangle|^{2\ell} = p^{2\ell}$. Now assume $d \equiv p$ and put $D = M - p$. Let T_D be the self-adjoint kernel operator on $L^2(\Omega)$ and let its eigenvalues be $\eta_1, \dots, \eta_N$, including multiplicities. Since $T_D 1 = 0$, the constant and centered parts act on orthogonal subspaces. For $\ell \geq 2$ this gives

$$t(C_{2\ell}^{[r]}, U) - p^{2\ell} = \text{tr}(T_D^{2\ell}) = \sum_j \eta_j^{2\ell} \geq 0. \quad (11)$$

For $r \geq 3$, the cycle expansion is one canonical atom, with 2ℓ edges, $2\ell(r-1)$ vertices, cyclic rank one, and shortest Berge cycle of length 2ℓ . Lemma 13.4 therefore gives

$$0 \leq t(C_{2\ell}^{[r]}, U) - p^{2\ell} \leq \left(\frac{2p(1-p)}{r(r-1)} \right)^\ell.$$

For $C_4^{[r]}$, the first cyclic atom, the upper bound has no $2^4 - 1$ prefactor. Equality with the random benchmark holds precisely when $M \equiv p$, because the even powers in (11) are nonnegative. This condition concerns the pair marginal; it does not require the full r -kernel to be constant.

From atoms to arbitrary linear patterns. Private graph expansions have two shared vertices on each cyclic edge. For a general finite linear hypergraph, the corresponding parameter is the number of vertices of an edge that also lie in another edge. The projection argument extends to these larger shared sets. This includes assemblies in which a private vertex of one atom is identified with a vertex of another atom; the resulting pattern need not itself be a private graph expansion.

The counting theorem of Kohayakawa et al. [41, Theorem 4] treats fixed uniformity under a quasirandomness assumption on the host. The estimates here hold for every bounded symmetric host kernel and track the uniformity through the number of shared vertices. They use the higher terms of the same Hoeffding decomposition [40].

Lemma 13.8 (Higher-coordinate projection bound). *Let $r \geq 2$, let $U : \Omega^r \rightarrow [0, 1]$ be symmetric, let $p = \mathbb{E}U$, and let U_b be its marginal on any b coordinates, where $0 \leq b \leq r$. Then*

$$\|U_b - p\|_2^2 \leq \frac{b}{r} p(1 - p).$$

If every one-coordinate marginal of U equals p , then

$$\|U_b - p\|_2^2 \leq \frac{b(b-1)}{r(r-1)} p(1 - p).$$

In particular the latter marginal is constant when $b \leq 1$.

Proof. For each coordinate set $A \subseteq [r]$, recursively define

$$u_A = \mathbb{E}[U \mid X_A] - \sum_{B \subsetneq A} u_B.$$

Here $u_\emptyset = p$. Each nonconstant u_A has zero integral in every coordinate of A , so these functions are mutually orthogonal. Symmetry makes $\|u_A\|_2^2 = v_j$ depend only on $j = |A|$. Consequently

$$\text{Var } U = \sum_{j=1}^r \binom{r}{j} v_j, \quad \|U_b - p\|_2^2 = \sum_{j=1}^b \binom{b}{j} v_j.$$

For $1 \leq j \leq b$ the binomial ratio is at most b/r ; for $2 \leq j \leq b$ it is at most $b(b-1)/(r(r-1))$. Apply $\text{Var } U \leq p(1-p)$, and in the regular case use $v_1 = 0$. $\square$

For a finite simple linear r -uniform hypergraph H , write

$$b_e(H) = |\{x \in e : \deg_H(x) \geq 2\}|, \quad b(H) = \max_{e \in E(H)} b_e(H).$$

If H has $m \geq 1$ edges, then $b(H) \leq \min\{r, m-1\}$: each other edge can share at most one vertex with a given edge. The Berge girth of H is half the length of a shortest cycle in its incidence graph. For linear H it is at least three whenever it is finite.

Theorem 13.9 (Uniform approximation for linear hypergraphs). *Let H be a finite simple linear r -uniform hypergraph with $m \geq 1$ edges, where $r \geq 2$. Put $b_e = b_e(H)$ and $b = b(H)$. For every symmetric $U : \Omega^r \rightarrow [0, 1]$ of mean p ,*

$$|t(H, U) - p^m| \leq \frac{p(1-p)}{r} \sum_{\{e,f\} \subseteq E(H)} \sqrt{b_e b_f} \quad (12)$$

$$\leq \binom{m}{2} \frac{b}{r} p(1-p). \quad (13)$$

Suppose in addition that every one-coordinate marginal equals p . If H has no Berge cycle, then $t(H, U) = p^m$. Otherwise, writing g for its Berge girth, one has

$$|t(H, U) - p^m| \leq (2^m - 1) \left(\frac{b(b-1)}{r(r-1)} p(1-p) \right)^{g/2}. \quad (14)$$

Both conclusions hold for finite simple host hypergraphs with the manuscript's homomorphism-density normalization; regularity of the kernel corresponds to regularity of the host.

Proof. Integrate every vertex that occurs in only one edge of H . The factor on an edge e becomes the marginal U_{b_e} on its shared vertices. Put $D_e = U_{b_e} - p$ and interpolate by

$$f(s) = \mathbb{E} \prod_{e \in E(H)} (p + s D_e), \quad 0 \leq s \leq 1.$$

Every interpolated factor lies in $[0, 1]$. The first derivative at zero vanishes, and the term of $f''(s)$ indexed by ordered distinct edges e, f has absolute value at most $\|D_e\|_2 \|D_f\|_2$, by Cauchy–Schwarz on the full product space. Taylor's integral remainder and Lemma 13.8 prove (12). The case $m = 1$ is exact.

Now assume regularity and expand the original product in $U - p$:

$$t(H, U) - p^m = \sum_{\emptyset \neq A \subseteq E(H)} p^{m-|A|} \mathbb{E} \prod_{e \in A} (U(X_e) - p).$$

For each A , integrate vertices lying in exactly one selected edge. The resulting factor on $e \in A$ is $D_{e,A} = U_{b_e(A)} - p$, where $b_e(A)$ counts its vertices shared with another selected edge. If $b_e(A) \leq 1$, this factor is zero. Otherwise the incidence graph on the selected edges and their shared vertices has all degrees at least two, so it contains a cycle. This also proves that every nonempty term vanishes when H has no Berge cycle.

Choose a shortest Berge cycle among the selected edges, of length $\ell \geq g$. Linearity and minimality imply that its edges meet each other exactly as a loose cycle: consecutive edges meet in their specified cycle vertex, and nonconsecutive

edges are disjoint. An extra incidence with an existing cycle vertex is a chord of the shortest incidence cycle. A shared vertex outside the cycle would give a two-incidence shortcut between two cycle edges; combining it with a shortest arc gives a shorter cycle when the edges are nonconsecutive. An additional intersection of consecutive edges contradicts linearity. For a triangle, linearity and the three distinct cycle vertices also exclude a vertex in all three edges.

Since $|D_{e,A}| \leq 1$, discard all factors not on this cycle after taking absolute values. Each remaining variable lies in at most two cycle factors. Integrating those that occur in only one factor leaves a cyclic product of two-variable nonnegative kernels $K_1, \dots, K_\ell$. Jensen's inequality and the projection bound give

$$\|K_i\|_2 \leq \|D_{e_i,A}\|_2 \leq q := \left(\frac{b(b-1)}{r(r-1)} p(1-p) \right)^{1/2} \leq \frac{1}{2}.$$

The cyclic integral is at most $\prod_i \|K_i\|_2$. One way to see this is to represent each kernel by its matrix on $L^2(\Omega)$: the integral is the trace of their product, bounded by the product of their Hilbert–Schmidt norms. Indeed, $|\operatorname{tr}(AB)| \leq \|A\|_{\text{HS}} \|B\|_{\text{HS}}$, while $\|AB\|_{\text{HS}} \leq \|A\|_{\text{HS}} \|B\|_{\text{op}} \leq \|A\|_{\text{HS}} \|B\|_{\text{HS}}$; iteration gives the assertion. Thus the term indexed by A is at most $q^\ell \leq q^g$ in absolute value. Sum over the at most $2^m - 1$ nonempty subsets to obtain (14).

The adjacency kernel of a finite simple host is symmetric and $[0, 1]$ -valued. Its one-coordinate marginal is constant exactly when all host vertices have the same degree. Applying the kernel statements gives the asserted finite-host versions directly. $\square$

Proposition 13.10 (Sharpness for every cyclic incidence pattern). *Fix a finite linear incidence pattern with m named edges and finitely many shared vertices, each belonging to at least two edges. Complete its edges to size r with distinct private vertices, obtaining H_r for all sufficiently large r . If the fixed pattern has Berge girth $g < \infty$, then the order r^{-g} in (14) is optimal for this pattern, already at density $p = 1/2$.*

Proof. Use the regular cosine kernels from the proof of Theorem 13.6. Write $\theta = r^{-1/2}$, $c = \cos \theta$, and

$$U_r(x_1, \dots, x_r) = \frac{1}{2} + \frac{1}{4} \left[\cos \left(\theta \sum_{i=1}^r x_i \right) - c^r \right]$$

on independent uniform signs. After integrating private coordinates, its marginal on any b coordinates is

$$(U_r)_b = \frac{1}{2} + \sum_{\substack{\emptyset \neq S \subseteq [b] \\ |S| \text{ even}}} a_{r,|S|} \prod_{i \in S} x_i, \quad a_{r,j} = \frac{1}{4} (-1)^{j/2} \sin^j \theta c^{r-j}.$$

For each fixed even $j \geq 2$, $a_{r,j} = O(r^{-j/2})$, and $a_{r,2} = -b_r$ with $b_r \sim e^{-1/2}/(4r)$.

Choose a term from this expansion at each hyperedge of H_r . Averaging the independent signs makes the term zero unless every shared vertex has

even selected degree. Each selected hyperedge also has even degree, by the displayed marginal formula. The surviving terms therefore correspond precisely to even-degree edge subsets of the fixed incidence graph. Every nonempty such subset contains a cycle and uses at least $2g$ incidences. Subsets using exactly $2g$ incidences are precisely the shortest incidence cycles. If N_g is their number, they contribute

$$N_g(-b_r)^g 2^{-(m-g)}.$$

Every other surviving subset uses at least $2g + 2$ incidences and contributes $O(r^{-g-1})$. The incidence pattern is fixed, so there are only finitely many terms, independently of r . Consequently

$$t(H_r, U_r) - 2^{-m} = N_g(-b_r)^g 2^{-(m-g)} + O_H(r^{-g-1}).$$

Since $N_g > 0$, its absolute value is of order r^{-g} . $\square$

Corollary 13.11 (Counting obligatory assemblies in regular hosts). *Fix $r \geq 3$, and let $H \in \mathcal{B}_r$ have $m \geq 1$ edges. For every symmetric regular $[0, 1]$ -valued r -kernel U of mean p , a Berge-acyclic H satisfies $t(H, U) = p^m$. Otherwise, with $b = b(H)$,*

$$|t(H, U) - p^m| \leq (2^m - 1) \left(\frac{b(b-1)p(1-p)}{r(r-1)} \right)^2.$$

In particular, with m fixed the error is $O_m(r^{-4})$, uniformly over all obligatory patterns with m edges and all such kernels. The same assertions hold in finite simple regular hosts.

Proof. The classification makes H linear and every Berge cycle even. Thus a finite Berge girth is at least four. Apply Theorem 13.9, using $q \leq 1/2$ from its proof and $b \leq m - 1$. $\square$

This conclusion concerns the full assembled pattern. Its exponent improves to the actual shortest Berge-cycle length, and Proposition 13.10 proves optimality for every fixed cyclic incidence pattern. In particular, the exponent four cannot be increased for the obligatory class: the single atom $C_4^{[r]}$ already attains this absolute error scale. These are two-sided error estimates; their sharpness examples do not assert a negative Sidorenko deficit for bipartite atoms.

Degree fluctuations and one-point assembly. The regular estimate has a useful stable form. Degree variation contributes its variance, while the remaining error retains the Berge-girth exponent. The explicit constant in the next theorem is chosen to keep this separation uniform over all finite spaces.

Theorem 13.12 (Degree variance and cyclic error). *Let $r \geq 2$ and let H be a finite simple linear r -uniform hypergraph with $m \geq 1$ edges. Put $b = b(H)$. Let $U : \Omega^r \rightarrow [0, 1]$ be symmetric, with mean p , and write*

$$d(x) = \mathbb{E}[U \mid X_1 = x], \quad h = d - p, \quad \delta^2 = \|h\|_2^2.$$

Define

$$q^2 = \frac{b(b-1)}{r(r-1)}[p(1-p) - r\delta^2], \quad C_{m,b} = [(b+1)(b+2)]^m.$$

The bracket is nonnegative. If H has no Berge cycle, then

$$|t(H, U) - p^m| \leq C_{m,b}\delta^2. \quad (15)$$

Otherwise, if g is its Berge girth, then

$$|t(H, U) - p^m| \leq C_{m,b}(\delta^2 + q^g). \quad (16)$$

No optimality is claimed for $C_{m,b}$. Since $b \leq m-1$, it is at most $[m(m+1)]^m$.

Proof. For a coordinate set S of size at most b , let U_S be the corresponding marginal and put

$$K_S = U_S - p - \sum_{x \in S} h(X_x).$$

The higher-coordinate orthogonal decomposition gives

$$\|K_S\|_2^2 \leq \frac{|S|(|S|-1)}{r(r-1)}[p(1-p) - r\delta^2] \leq q^2. \quad (17)$$

Indeed the first-coordinate projections contribute exactly $r\delta^2$ to $\text{Var } U$, and the binomial-ratio bound applies to all the remaining orders, which are at least two. This also proves nonnegativity of the bracket. We have $|h| \leq 1$, $|K_S| \leq b+1 =: B$, and $0 \leq q \leq 1/2$. For every $T \subseteq S$, conditional expectation gives

$$\mathbb{E}[K_S \mid X_T] = K_T. \quad (18)$$

In particular $K_T = 0$ when $|T| \leq 1$. This statement concerns the marginal on T ; it does not assert that integrating any one coordinate of K_S while fixing all the others gives zero.

Integrate the private vertices of H and let S_e be the shared vertices of an edge e . Expand each remaining factor as

$$U_{S_e} = p + \sum_{x \in S_e} h(X_x) + K_{S_e}.$$

There are at most $(b+2)^m$ expansion terms. The term selecting p on every edge is p^m . In each other term let a be the number of selected h factors and t the number of selected K factors; ignore the constant coefficient $p^{m-a-t} \leq 1$.

If $a \geq 2$, keep two selected h factors and bound all the others by one and every K factor by B . If the two kept factors use the same vertex, their absolute integral is δ^2 . If they use distinct vertices, independence gives an integral $(\mathbb{E}|h|)^2 \leq \delta^2$. This term is therefore at most $B^m\delta^2$ in absolute value.

Suppose next that $a = 0$ or $a = 1$, marking the vertex x of the unique h factor in the latter case. Among the selected K edges integrate every vertex that occurs in just one of these edges, except for the marked vertex. Formula (18) replaces each K_{S_e} by the corresponding smaller K kernel, still bounded by

B with L^2 norm at most q . If its retained support has size at most one, the entire term vanishes.

In a nonzero term the resulting incidence graph thus has every edge-node of degree at least two. Every point-node also has degree at least two, except possibly the single marked point. If the marked point occurs in no selected K edge, its factor integrates to zero. Otherwise this incidence graph cannot be a forest: a finite tree with at least one edge has at least two leaves, whereas at most one node here can have degree one. This proves that all $a \leq 1$ terms vanish when H is Berge-acyclic. It also shows that any remaining term contains a Berge cycle among the selected K edges.

Choose a shortest such cycle, of length $\ell \geq g$. As in the proof of Theorem 13.9, its edge factors share variables only consecutively around the cycle. If $a = 0$, discard all other factors at cost $B^{t-\ell}$. The cyclic integral is at most the product of the L^2 norms, giving

$$|\text{term}| \leq B^{t-\ell} q^\ell \leq B^m q^g.$$

If $a = 1$, Cauchy–Schwarz first gives

$$|\text{term}| \leq B^{t-\ell} \delta \left(\mathbb{E} \prod_{e \text{ on the cycle}} K_e^2 \right)^{1/2}.$$

The same cyclic integral bound applies to the squared kernels. For each factor, $\|K_e^2\|_2 \leq B \|K_e\|_2 \leq Bq$, so

$$|\text{term}| \leq B^{t-\ell/2} \delta q^{\ell/2} \leq \frac{1}{2} B^m (\delta^2 + q^g).$$

Here we used $q \leq 1$ and the inequality $2uv \leq u^2 + v^2$. Summing the at most $(b+2)^m$ terms proves both assertions. $\square$

For a fixed finite linear incidence pattern, b and m remain fixed as private vertices are added. The last theorem therefore gives the uniform estimate

$$|t(H_r, U_r) - p_r^m| = O_H(\text{Var}(d_r) + r^{-g})$$

when the pattern is cyclic, and $O_H(\text{Var}(d_r))$ when it is Berge-acyclic. Thus the regular r^{-g} scale persists whenever the degree variance is $O(r^{-g})$, while the general r^{-1} scale permits degree variance of order r^{-1} . The theorem requires only control of the L^2 degree fluctuation, not pointwise regularity.

For a finite host G of order n , the same theorem applies with

$$d(v) = \frac{(r-1)! \deg_G(v)}{n^{r-1}}, \quad \delta^2 = \frac{1}{n} \sum_{v \in V(G)} (d(v) - p_r(G))^2.$$

Thus its hypothesis and error can be read directly from the normalized degree sequence.

The one-point operation also has an exact density formula. For a rooted pattern (A, a) , let $a_U(x)$ be its homomorphism density with the image of a fixed

at x . If $A \vee B$ identifies the roots of two otherwise disjoint patterns, conditional independence gives

$$t(A \vee B, U) = \mathbb{E}[a_U(X)b_U(X)] = t(A, U)t(B, U) + \text{Cov}(a_U(X), b_U(X)). \quad (19)$$

For two copies of the same rooted pattern the covariance is a variance. A single-edge atom has rooted density d , so regularity removes this extra term when edge atoms are attached. The next example shows what can happen for cyclic atoms, even when the atom list and attachment excess profile are fixed.

Remark 13.13 (The attachment point can change the density). Canonical atoms and their attachment multiplicities do not determine homomorphism density, even in a regular kernel. Fix $r \geq 3$ and let $B = C_4^{[r]}$. Form H_c by identifying a core vertex of one copy of B with a core vertex of another copy, and form H_p by instead identifying two private vertices. Both systems have the same two canonical atoms, one joint of atom multiplicity two, and hence the same attachment excess profile (1). Both have 8 edges, $8r - 9$ vertices and cyclic rank 2. Nevertheless, a single bounded regular kernel distinguishes their densities.

Take the uniform probability space $\Omega = \{-1, 0, 1\}$, write $s = \mathbb{E}Z^2 = 2/3$, and choose $0 < p < 1$ and

$$0 < \varepsilon \leq \frac{\min\{p, 1-p\}}{\binom{r}{2}}.$$

The symmetric kernel

$$U_r(z_1, \dots, z_r) = p + \varepsilon \sum_{i < j} z_i z_j$$

takes values in $[0, 1]$, has mean p , and has every one-coordinate marginal equal to p , since $\mathbb{E}Z = 0$. Its two- and three-coordinate marginals are

$$M(x, y) = p + \varepsilon xy, \quad N(x, y, z) = p + \varepsilon(xy + xz + yz).$$

Let $a_c(z)$ and $a_p(z)$ denote the density of B when a specified core or private vertex, respectively, is fixed at z , while all other vertices are sampled independently. Integrating the private vertices and expanding the four-cycle gives

$$a_c(z) = p^4 + \varepsilon^4 s^3 z^2, \quad t(B, U_r) = p^4 + \varepsilon^4 s^4.$$

Indeed, every proper nonempty edge subset of the cycle has an unfixed degree-one vertex, whose coordinate has mean zero.

For a fixed private vertex z on the edge with core endpoints x, y , let $K_3(x, y)$ be the integral of the three M -factors on the complementary path. Its row and column integrals are p^3 , because M has constant row integral p . As $N(x, y, z) = M(x, y) + \varepsilon z(x + y)$, it follows that

$$\begin{aligned} a_p(z) &= \mathbb{E}_{x,y} N(x, y, z) K_3(x, y) \\ &= t(B, U_r) + \varepsilon z p^3 (\mathbb{E}x + \mathbb{E}y) = t(B, U_r). \end{aligned}$$

This calculation also applies when $r = 3$, when the selected edge has no other private vertices to integrate.

Conditioning on the common vertex in a one-point amalgamation makes the remaining variables in its two atoms independent. Consequently

$$\begin{aligned} t(H_c, U_r) - t(H_p, U_r) &= \mathbb{E}a_c(Z)^2 - (t(B, U_r))^2 \\ &= \text{Var}(a_c(Z)) = \varepsilon^8 s^6 \text{Var}(Z^2) = \frac{128}{6561} \varepsilon^8 > 0. \end{aligned}$$

Thus the location of a joint inside an atom carries density information that its multiplicity and the atom list do not record.

Transfer from proper-subgraph limits. The preceding estimates do not give the exact Sidorenko inequality. For a fixed uniformity the next lemma transfers a strict deficit, provided the witness family has sufficiently controlled proper edge-subgraph densities.

There is also a finite error budget behind the limit argument. Let J be a finite simple graph with $m \geq 1$ edges, write $J_A = (V(J), A)$, and let V be a finite symmetric nonnegative kernel with constant row integral one. Suppose $0 < \lambda \leq 1$, $t(J, V) \leq 1 - \delta$ with $\delta > 0$, and $|t(J_A, V) - 1| \leq \varepsilon$ for every proper edge subset. The expansion used in the next proof gives

$$t(J, 1 + \lambda(V - 1)) \leq 1 - \delta\lambda^m + \varepsilon(1 - \lambda^m). \quad (20)$$

Indeed, the coefficients of the proper-subgraph terms are nonnegative and sum to $1 - \lambda^m$. Thus $\varepsilon(1 - \lambda^m) < \delta\lambda^m$ suffices for a strict deficit. At the universal coefficient λ_r , the retained signal has order r^{-m} for a fixed full-graph deficit. This explains why the proper-subgraph approximation must be chosen after the uniformity, and why the limit statement is not a uniform bound on the density or size of a finite witnessing host.

Lemma 13.14 (Transfer from proper-subgraph limits). *Let J be a finite graph with $m \geq 1$ edges. Suppose there are finite symmetric nonnegative kernels V_q with constant row integral one such that*

$$t(J_A, V_q) \longrightarrow 1 \quad (A \subsetneq E(J)), \quad t(J, V_q) \longrightarrow a < 1,$$

where $J_A = (V(J), A)$. Then, for every integer $r \geq 2$, there is a symmetric $[0, 1]$ -valued r -kernel W with $0 < p(W) < 1$ and

$$t(J^{[r]}, W) < p(W)^m.$$

There is also a finite simple r -uniform host with the same strict inequality.

Proof. For any fixed $0 < \lambda \leq 1$, expansion over subsets of $E(J)$ gives

$$\begin{aligned} t(J, 1 + \lambda(V_q - 1)) &= \sum_{A \subseteq E(J)} \lambda^{|A|} (1 - \lambda)^{m - |A|} t(J_A, V_q) \\ &\longrightarrow 1 + \lambda^m (a - 1) < 1. \end{aligned}$$

Apply Lemma 13.2 with $\lambda = \lambda_r$. By (3), the resulting U_r has mean one and $t(J^{[r]}, U_r) < 1$ for all sufficiently large q . Here r is fixed before q is chosen; the limiting deficit is $(1 - a)\lambda_r^m$, so the required index may depend on r . Choose such a q , let $L = \max U_r$, and put $W = U_r/L$. The strict inequality implies that

U_r is nonconstant; its mean is one, so $L > 1$. Homogeneity gives the required inequality with $p(W) = 1/L \in (0, 1)$.

For the finite-host assertion, sample n independent labels from the finite probability space, and include each r -set independently, conditional on those labels, with probability given by W . Let G_n be the resulting simple r -uniform hypergraph and write $F = J^{[r]}$, $v = |V(F)|$, $p = p(W)$, and $\tau = t(F, W) < p^m$. An injective vertex map from F uses distinct target edges, whence

$$\mathbb{E}t(F, G_n) \leq \tau + \binom{v}{2}/n, \quad \mathbb{E}p_r(G_n) = \frac{\binom{n}{r}}{n^r} p.$$

Consequently, for sufficiently large n ,

$$\mathbb{E}t(F, G_n) < (\mathbb{E}p_r(G_n))^m \leq \mathbb{E}[p_r(G_n)^m],$$

the final inequality being Jensen's inequality. Some realization therefore has $t(F, G_n) < p_r(G_n)^m$. $\square$

The source pattern and its proper-subgraph limits. Let $\mathcal{T}$ be the following 22 triples on $P = \{0, \dots, 12\}$, in the order used by [6]:

$$\begin{array}{cccc} \{0, 1, 3\} & \{0, 1, 9\} & \{0, 2, 3\} & \{0, 2, 9\} \\ \{1, 3, 10\} & \{1, 7, 10\} & \{1, 7, 12\} & \{1, 9, 12\} \\ \{2, 3, 4\} & \{2, 4, 9\} & \{3, 4, 11\} & \{3, 10, 11\} \\ \{4, 8, 9\} & \{4, 8, 11\} & \{5, 6, 7\} & \{5, 6, 11\} \\ \{5, 7, 10\} & \{5, 10, 11\} & \{6, 7, 12\} & \{6, 8, 11\} \\ \{6, 8, 12\} & \{8, 9, 12\} & & \end{array}$$

Let J be the bipartite incidence graph between P and $\mathcal{T}$. It has 35 vertices and 66 edges. Every point pair occurring in a face occurs in exactly two faces, and the graph joining faces that share a pair is connected. These are the first assertions of the finite-complex proposition in [6, Section 2], and are also directly checkable from the displayed list.

For a rectangular kernel $W : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ and $B \subseteq E(J)$, the notation $t_{P, \mathcal{T}}(J_B, W)$ means that the vertices in P are sampled from $\mathcal{X}$ and those in $\mathcal{T}$ from $\mathcal{Y}$, independently. Write $W^\top(y, x) = W(x, y)$ for the transpose.

Proposition 13.15 (The source supplies the transfer hypotheses). *Assume the rank-layer limit stated in [6, Section 3]. Then the graph J admits kernels V_q satisfying all hypotheses of Lemma 13.14.*

Proof. We keep the source's dimension, type distributions, activation laws, thinning probabilities, and positive dilution parameter fixed before letting the odd prime q tend to infinity. Let $\mathcal{K}$ be its fixed finite coordinate set. The matrix coordinates range over the additive group of symmetric D -by- D matrices over $\mathbb{F}_q$, for a fixed sufficiently large even D . The source uses nonnegative normalized indicators K_ξ of the rank- $D/2$ layers with determinant sign ξ . Its rectangular

kernel has the form

$$W_q((a, x), (b, y)) = \mathbb{E}_{S, \xi|a, b} \prod_{\gamma \in S} K_{\xi_\gamma}(x^\gamma - y^\gamma),$$

where every matrix coordinate is uniform and independent, and $\mathbb{E}K_\xi = 1$. The laws depend on the types but not on the matrices. For each fixed x^γ , the difference $x^\gamma - Y^\gamma$ is uniform; the same is true of $X^\gamma - y^\gamma$ for fixed y^γ . Integrating the product coordinate by coordinate gives one for every pair of types. Averaging over the opposite type shows that W_q has row and column integrals identically one, including after thinning and dilution.

We next prove the proper-subgraph assertion, which is not a stated conclusion of the source's coordinate-expansion lemma. Write $\mathcal{C} = E(J)$ for its source incidences and $\mathcal{P}$ for the 33 point pairs occurring in the faces. At a source corner (i, j) there are two pairs $e, f \subset j$ containing i . For an activation law on a coordinate set S and signs ξ , set

$$\phi_\alpha = \mathbf{1}_{\{\alpha \subseteq S\}} \prod_{\gamma \in \alpha} \xi_\gamma, \quad A(\alpha, \beta) = \mathbb{E}(\phi_\alpha \phi_\beta).$$

Here $\alpha, \beta \subseteq \mathcal{K}$, and ϕ_α is defined to be zero when $\alpha \not\subseteq S$. The source's laws have $\mathbb{E}\phi_\alpha = 0$ for every nonempty α , so $A(\alpha, \emptyset) = A(\emptyset, \alpha) = 0$ for nonempty α .

Fix $B \subsetneq \mathcal{C}$ and a type assignment. In the proof of the coordinate-expansion lemma, force the activation at every omitted source incidence in $\mathcal{C} \setminus B$ to be $S = \emptyset$, and retain the original laws at the other incidences. These modified laws may depend on the individual source incidence, even when two incidences have the same type pair; we therefore repeat the proof, rather than invoke its type-pair statement verbatim. After independently sampling the law at each incidence, each coordinate has some active incidence set to which the source's rank-layer lemma applies. That lemma is uniform over every subset of $\mathcal{C}$ and every sign choice. There are finitely many configurations, so its $o(1)$ error survives the finite products and averages exactly as in the original proof. The resulting limiting expression is

$$\sum_{\substack{\alpha_e \subseteq \mathcal{K} \\ (e \in \mathcal{P})}} \prod_{(i, j) \in \mathcal{C}} A_{i, j}(\alpha_{e(i, j)}, \alpha_{f(i, j)}).$$

At an omitted corner the factor is zero unless both adjacent pair labels are empty. At every other corner an empty label and a nonempty label also give zero by the first-moment identity. Hence, in any nonzero term, emptiness propagates to all three pair labels of a face and then across every shared-pair adjacency. There is an omitted corner, and the face-neighbor graph is connected, so every pair label must be empty. The all-empty term equals one. After averaging over the finitely many type assignments, this proves

$$t_{P, \mathcal{T}}(J_B, W_q) \longrightarrow 1.$$

The reversed-difference version of the rank-layer lemma gives the same conclusion for the transposed rectangular kernel.

For the full incidence set, the source's signed coefficient and dilution construction instead gives

$$t_{P,\mathcal{T}}(J, W_q) \longrightarrow 1 + \varepsilon^{13}t_0, \quad t_{P,\mathcal{T}}(J, W_q^\top) \longrightarrow 1 + \varepsilon^{22}t_1,$$

where $t_0 < 0$, and $\varepsilon > 0$ is fixed small enough that the product of these two limits is some $a < 1$. These are the dilution limits and strict product inequality proved in [6, Section 3].

On the product probability space of the two rectangular sides, put

$$V_q((x, y), (x', y')) = W_q(x, y')W_q(x', y).$$

This is symmetric and nonnegative. Its row integral is one by the exact row and column identities for W_q . For any $B \subseteq \mathcal{C}$, bipartiteness separates the variables in the two factors, giving

$$t(J_B, V_q) = t_{P,\mathcal{T}}(J_B, W_q) t_{P,\mathcal{T}}(J_B, W_q^\top).$$

Thus every proper B has limit one, and the full pattern has limit $a < 1$, as required. $\square$

Obligatory atoms without a Sidorenko expansion.

Theorem 13.16 (An obligatory atom below the random-density bound). *Assume the rank-layer limit in [6, Section 3]. For every integer $r \geq 3$, the hypergraph $A_r = J^{[r]}$ is an obligatory canonical cyclic atom with*

$$|E(A_r)| = 66, \quad |V(A_r)| = 66r - 97, \quad \beta(A_r) = 32.$$

Here β is the cycle rank of the connected Levi graph. Nevertheless, there is a finite simple r -uniform host G with

$$t(A_r, G) < p_r(G)^{66}.$$

Here obligatoriness concerns injective non-induced containment in every uncountably chromatic host, whereas t counts all homomorphisms in a finite host.

Proof. First J is 2-connected. After deleting a point-node, each pair of neighboring faces still shares at least the other point of its common pair. Connectivity of the face-neighbor graph therefore connects all remaining face nodes, and every remaining point attaches to one of them. After deleting a face-node $j = \{a, b, c\}$, the other face on each of $\{a, b\}$, $\{a, c\}$ and $\{b, c\}$ provides a two-edge incidence path connecting that pair of points while avoiding j . Thus the three neighbors of j remain in one component. Replacing every passage through j by these paths shows that $J - j$ remains connected. Consequently no vertex of J is a cut vertex.

The graph is bipartite, so its private expansion is obligatory by Reiher's bipartite-expansion theorem [4, Theorem 1.2]. Since the core J is 2-connected, Corollary 11.7 identifies A_r as one cyclic atom. Equivalently, its cyclic incidence

block is the subdivision of J , and all its other incidence edges go to the private degree-one vertices. The counts follow from the expansion:

$$|V(A_r)| = 35 + 66(r - 2), \quad \beta(A_r) = (r - 1)66 - |V(A_r)| + 1 = 32.$$

Finally apply Proposition 13.15 and Lemma 13.14. $\square$

Corollary 13.17 (Failure of the expansion conjecture). *Under the rank-layer hypothesis of Theorem 13.16, there is a bipartite graph J such that $J^{[r]}$ is not Sidorenko for every integer $r \geq 2$. Consequently Conjecture 5.1 of Nie–Spiro [7] is false.*

Proof. Use the same 35-vertex graph J . For $r \geq 3$, apply Theorem 13.16. The case $r = 2$ follows from Lemma 13.14, or directly from the source counterexample, since $J^{[2]} = J$. The private expansion used by Nie–Spiro is exactly $J^{[r]}$ here. Their Conjecture 5.1 asserts that every bipartite graph has a Sidorenko expansion at some rank $r \geq 2$; this one graph has none. $\square$

This corollary does not settle the general converse question posed in Nie–Spiro’s Question 5.2. The proof uses the additional proper-subgraph limits of this particular graph’s witness family, not a transfer theorem for every non-Sidorenko graph.

Equality at the random benchmark does not force other densities.

The density deficit has a further consequence. Interpolation adapts the graph argument in [6, Section 8] to the actual hypergraph kernels constructed above. We state the consequence as three explicit limits.

Corollary 13.18 (Two densities do not determine the four-cycle density). *Under the rank-layer hypothesis of Theorem 13.16, for every $r \geq 3$ there are $p \in (0, 1)$ and finite simple r -uniform hosts G_n of orders tending to infinity such that*

$$p_r(G_n) \longrightarrow p, \quad t(A_r, G_n) \longrightarrow p^{66},$$

but

$$t(C_4^{[r]}, G_n) \longrightarrow q > p^4.$$

Thus matching the edge and A_r densities of a constant kernel does not determine even the density of the obligatory expansion $C_4^{[r]}$.

Proof. Write $Q_U = M_U$ for the pair marginal in (3). Take the negative kernel U_- of mean $p \in (0, 1)$ from Lemma 13.14, and let $Q_- = Q_{U_-}$. It has mean p and is nonconstant, since otherwise $t(A_r, U_-) = t(J, Q_-) = p^{66}$.

On the balanced sign space $\{-1, 1\}$ choose

$$0 < \delta \leq \frac{\min(p, 1 - p)}{\binom{r}{2}}, \quad U_+(z_1, \dots, z_r) = p + \delta \sum_{1 \leq a < b \leq r} z_a z_b.$$

This kernel is symmetric, takes values in $[0, 1]$, and has mean p . Its pair marginal is $Q_+(a, b) = p + \delta ab$: every other summand contains an independently integrated

sign to its first power. Expansion and sign averaging therefore give

$$t(A_r, U_+) = t(J, Q_+) = \sum_{\substack{A \subseteq E(J) \\ d_A(v) \text{ even for every } v}} p^{66-|A|} \delta^{|A|} > p^{66}.$$

The empty set contributes p^{66} ; all terms are nonnegative, and the edge set of a cycle contributes positively. The graph J has a cycle because it has 66 edges and only 35 vertices.

Lift U_- and U_+ to the independent coordinates of the product of their finite probability spaces and put

$$U_s = (1-s)U_- + sU_+, \quad 0 \leq s \leq 1.$$

Every U_s is symmetric, $[0, 1]$ -valued, and has mean p . The function $s \mapsto t(A_r, U_s)$ is a polynomial with endpoint values strictly below and above p^{66} . Thus some $s_* \in (0, 1)$ gives equality. Its pair marginal is

$$Q_s((x, a), (y, b)) = (1-s)Q_-(x, y) + s(p + \delta ab).$$

The centered summands depend on independent coordinate pairs and both have mean zero. Orthogonality gives

$$\|Q_s - p\|_2^2 = (1-s)^2\|Q_- - p\|_2^2 + s^2\delta^2 > 0 \quad (0 \leq s \leq 1).$$

In particular $Q = Q_{s_*}$ is nonconstant.

For completeness, every finite symmetric nonnegative kernel Q of mean p satisfies $t(C_4, Q) \geq p^4$, with equality only when Q is constant. Indeed, put $d(x) = \mathbb{E}_z Q(x, z)$ and $c(x, y) = \mathbb{E}_z Q(x, z)Q(y, z)$. Two applications of Cauchy–Schwarz give

$$t(C_4, Q) = \mathbb{E}_{x,y} c(x, y)^2 \geq (\mathbb{E}_z d(z)^2)^2 \geq p^4.$$

Equality forces $d(x) = p$ and $c(x, y) = p^2$ on every positive-mass atom and atom pair. Diagonal pairs then give

$$\mathbb{E}_z (Q(x, z) - p)^2 = c(x, x) - 2p d(x) + p^2 = 0.$$

Thus equality forces constancy. Applied to the present Q , this shows

$$t(C_4^{[r]}, U_{s_*}) = t(C_4, Q) > p^4.$$

It remains to pass to finite simple hosts with all three limits. Sample n independent vertex labels and then, conditionally independently, include each r -set with probability U_{s_*} of its labels. For every fixed finite simple r -uniform A , injective maps give the kernel density, and the fraction of vertex maps having a collision is $O_A(n^{-1})$. Consequently

$$\mathbb{E}t(A, G_n) = t(A, U_{s_*}) + O_A(n^{-1}).$$

The contributions from two vertex maps are independent whenever their images are disjoint. Overlapping pairs form an $O_A(n^{-1})$ fraction, so $\text{Var}(t(A, G_n)) = O_A(n^{-1})$. Chebyshev's inequality with tolerance $n^{-1/4}$, followed by a union bound for the single edge, A_r , and $C_4^{[r]}$, gives positive probability that all three desired approximations hold for each sufficiently large n . Choose one

such realization for each n . These deterministic hosts have the three asserted limits. $\square$

The density p in this corollary can depend on r . Both test patterns are obligatory, yet the equality for one leaves a strict excess for the other. This compares their finite homomorphism densities with the constant-kernel benchmark.

The earlier linear non-Sidorenko examples of Conlon et al. [36, Theorem 3.1] include systems of odd girth. The present A_r has only even Berge cycles, satisfies the bridge criterion, and is one canonical atom. Its obstruction lies within the obligatory class. The graph counterexample and its finite-field construction are due to OpenAI [6]; the additional conclusions here come from the proper-subgraph limits, matching transfer, and canonical identification.

SCOPE OF FORMAL VERIFICATION

The triple-system classification and specified finite structural extensions have verified Lean 4 implementations [42, 43]. The accepted results include the canonical atoms, finite parameter and connected atom-count spectra, classical-block identification, local decomposition products, actual-piece accounting and supported-cover increments. They use the original carriers and retain their exact finiteness and structural hypotheses.

The higher-uniformity results of Section 11, including the iterated trace lemma, classification, canonical atoms and finite transfer, and the exact chromatic spectra, reservoir calibration and finite-family results of Section 12, are proved in the manuscript and have not been formalized in Lean. They are outside the coverage of every existing export listed in Appendix A.

The results in Section 13 have not been formalized here. Its universal marginal-realization theorem and quantitative density estimates have independent manuscript proofs; they do not use the rank-layer hypothesis. The conclusions about the explicit Sidorenko counterexample, including Theorem D, do use the rank-layer limit of OpenAI [6] as an explicit external hypothesis. The source repository describes a Lean formalization of its finite graph counterexample; that description does not verify the additional proper-subgraph extraction or the hypergraph transfer in this manuscript. No replay of that external formalization is claimed.

Verification of the extended manuscript is not complete. In particular, fixed-atom profile realization, the full lattice spectra, maximal-chain and height endpoints, the complete phase/equality interfaces and the new coefficient bounds remain distinct obligations. The arithmetic deficit theorem supports the cyclic-budget calculation, but does not by itself identify every remaining geometric or lattice endpoint. The proofs in the main text establish the manuscript conclusions; successful typechecking of other statements cannot replace them. Implementation details, exact exported types, validation chronology and the remaining crosswalk are separated into Appendix A.

APPENDIX A. FORMALIZATION INTERFACES AND REPRODUCIBILITY

The implementation uses Lean 4 [42] and the mathematical library Mathlib [43]. These references describe the proof assistant and library, not the verification status of this manuscript. Formal verification checks a precisely stated theorem; its correspondence with a prose formulation still requires mathematical review [44]. Verified typechecking addresses a distinct trust question [45]; no use of that external checker is claimed here.

The self-contained Lean 4 source is available in the public `Erdos593` repository. The repository contains a classification implementation in commit `6fd00a7`, whose Git timestamp is 15 July 2026. A dated public validation record is the successful GitHub Actions run of 19 July 2026, at commit `88ce701`. A Git timestamp alone does not establish the date of public disclosure or priority. Li’s arXiv v2, submitted on 23 July 2026, reports a separate formal verification of the broader results in his paper. These records do not date either author’s private work, and no shared code or derivation between the two Lean developments is asserted. For every finite triple system F , the exported theorems establish

$$\begin{aligned} \text{F.IsObligatory} &\iff \text{F.isolatedReduction.Intrinsic}, \\ \text{F.IsObligatory} &\iff \text{Constructible F.isolatedReduction}. \end{aligned}$$

These exported theorems verify both directions of the classification, including the isolated-vertex reduction. Here “finite” describes the system F being classified. The host triple systems are not assumed finite; they are quantified within the documented ambient-universe convention of the formalisation.

Further accepted Lean results cover the canonical-atom normal form and minimal-generator classification, the finite parameter spectra, the connected atom-count spectrum and its maximisers, the unrestricted classical-block equivalence, the forward atomic cycle and even-theta boundary forms, and the supported decomposition order isomorphism with local partition products. A separately validated extension has passed canonical and standalone replay and final source reconciliation for the actual-piece/local-partition block-count identity; these results retain their stated finiteness, structural and feasible-parameter hypotheses. The block-count checkpoint is now included in the synchronized public and private repositories. The abstract capacity-two incidence-forest construction has also been validated, but it does not yet formalise assembly of an arbitrary fixed list of actual atoms.

The supported-subtype cover criterion, the unique local coordinate changed by a cover, and the additive piece-deficit identity have now passed pinned canonical and standalone Lean replay and final live source reconciliation. This extension also verifies that coarsening along a cover reduces the actual piece count by one and increases the deficit by one. These results use actual original-edge pieces and local quotient carriers. The extension is included in the public repository through the reviewed cover/grading integration. Maximal-chain lengths and height endpoints remain separate obligations.

Verification of the extended manuscript is therefore partial. Remaining interfaces include the maximal-chain grading and height endpoints, fixed-atom profile realization and canonical-label fidelity after assembly, the componentwise atom-count and full factorization-lattice spectra, the all-maximal-block running-order statement, and sorted theta parameter normalization. The full indecomposability phase diagram, decomposable spectrum and decomposable equality classification also remain separate obligations. The characteristic-polynomial recovery of the attachment profile is not claimed as an additional Lean endpoint. Exact statement-by-statement coverage is tracked in the manuscript crosswalk.

The accepted atom-deficit theorem supplies the arithmetic identity and positive-atom bound behind Proposition 10.18, under its exact structural, positive-rank and additive-count hypotheses. Its identification with lattice rank still uses the manuscript’s grading argument. The generating polynomial, coefficient bounds and equality cases in Corollary 10.19 are proved here but are not yet additional Lean endpoints. Counting by actual pieces uses the accepted product and block-count interfaces, not an assumption about covers in an ambient partition lattice; the fixed-atom sharpness assertion retains the separate assembly-realization obligation.

The Lean development does not reproduce every intermediate prose lemma verbatim. In particular, its negative direction uses a shift-graph host in place of the paper’s high-odd-girth input, and its closure construction uses a singular-safe layering rather than the displayed cofinality-indexed chain. These are distinct proof routes to the classification. Kernel validation supplements rather than replaces mathematical review; a full external audit of the extended development remains pending.

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AI ASSISTANCE

OpenAI’s GPT-5.6 Pro through ChatGPT and Aristotle [46] were used in proof development, adversarial checking, editorial restructuring, and the Lean formalisation. In particular, ChatGPT reviews informed the checking and presentation of the local forest-counting and capacity arguments in Section 10; Aristotle supplied the abstract incidence-forest construction distinguished from fixed-atom realization in the verification section. ChatGPT was also used to develop and check the higher-uniformity argument in Section 11 and the exact chromatic and finite-family arguments in Section 12. ChatGPT was used to identify the external density construction and to develop and check the transfer arguments, universal marginal theorem, quantitative density estimates for linear patterns, and attachment examples in Section 13. The author takes full responsibility for the arguments, citations, and final manuscript; newly

incorporated material in this review draft remains subject to the author's review.

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